

Long Distance Dependencies in Turkana

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Abstract

This dissertation investigates long distance movement and long distance dependencies in Turkana, an extremely under-researched Eastern Nilotic language spoken in Kenya. All relevant Turkana data come from original fieldwork.

Turkana is a VSO language (Dimmendaal 1983; [Finer 2013](#)) with an optional pre-verbal focus position. This position has not been discussed in detail in the literature before. Wh-elements can either move to this position or remain in-situ. I demonstrate that wh in-situ does not undergo covert movement but it is truly in-situ. In contrast, wh-ex-situ is a result of movement and not because the wh-element is externally merged high in the syntactic structure. I show that the pre-verbal wh-position is not located in the specifier of CP, as usually assumed, but in the specifier of TP.

Ex-situ and in-situ wh-elements exhibit the same locality patterns: They can move or be licensed via crossing a single CP but never more than one CP. This is a challenge for current theories of phasality. To account for this intricate data, I propose that only CPs function as phases in Turkana, and that phasality obeys PIC₂ leading to a relatively weak version of phasality. I further adapt Keine's ([2019](#); [2020b](#)) selected opacity approach making it more restrictive by allowing for very small horizons. Specifically, I propose that T and v in Turkana can carry probes with unrestricted horizons, while C carries an edge feature whose horizon is limited to V. This horizon mimics the locality restrictions of vP acting as a phase, but without the absolute locality constraints typically associated with full phasehood.

This suggests that not only horizons, but also the very architecture of phasal domains, are subject to cross-linguistic variation. Turkana thus offers compelling evidence that the phase system is less rigid and more diverse than traditionally assumed. These findings highlight the importance of investigating a broader range of languages beyond the Indo-European languages to get a full understanding of

locality restrictions.

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Few would disagree that no one completes a PhD unscathed, but beginning one in the midst of a global pandemic added a particular flavour of chaos to the process. When I started this project, I was stuck abroad, unable to move to Leipzig for another six months, and I did not meet most of my peers or colleagues until a full year into the programme. This thesis exists not because the circumstances were favourable (they were not) but in spite of them. I owe enormous thanks to the people who helped me keep going when things felt stuck, unclear, or simply too much.

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List of Glossing Abbreviations

Glosses

1	1 st person
2	2 nd person
3	3 rd person
APPL	Applicative
C	Complementiser
CAUS	Causative
DAT	Dative
DEFL	Default case
F	Feminine
GEN	Genitive
INS	Instrumental
IPFV	Imperfective
LOC	Locative
M	Masculine
NOM	Nominative
N	Neuter
PL	Plural
POSS	Possessive
PREP	Preposition
PRS	Present
PRF	Perfect
PST	Past
SG	Singular
TAM	Tense, aspect, mood

Other abbreviations

ATR	Advanced tongue root
PIC	Phase impenetrability condition
WCO	Weak cross over

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Chapter 1

Introduction

According to [Eberhard et al. \(2024\)](#), there are currently 7159 languages spoken worldwide. Yet much of linguistic research focuses on a relatively small number of well-documented languages, primarily Indo-European languages. The idea behind generative grammar is that all languages share a universal grammar ([Chomsky 1957](#)), but if our conclusions are based on only a small subset of languages, we risk missing important counterexamples. Broader language data are essential for making reliable generalisations about how language works.

Research on less well-documented languages can reveal grammatical patterns that challenge established assumptions. Such data broaden our understanding of cross-linguistic variation and offer critical testing grounds for theoretical claims developed on limited language samples. For this reason, research on understudied languages and language families is not only valuable but necessary for the field as a whole.

One of these under-studied language families is Nilo-Saharan. The Nilo-Saharan language family consists of 210 languages ([Eberhard et al. 2024](#)) which are mostly very under-studied. The Nilotic branch has recently received some attention (cf. [van Urk 2015](#) for Dinka, Western Nilotic, [Barasa 2017](#) for Ateso, Eastern Nilotic or [Kouneli 2019](#); [Bossi 2023b](#) for Kipsigis, Southern Nilotic) but the whole Nilo-Saharan language branch belongs to one of the least researched language families in Africa ([Kouneli 2019](#)). This dissertation contributes to giving Nilotic languages more attention, specifically Turkana, an Eastern Nilotic language. Turkana exhibits unusual wh-movement patterns which have not been systematically described in the literature yet. These patterns raise important

questions about locality.

The aim of this dissertation is twofold: (i) to provide a detailed empirical account of wh-dependencies in Turkana, and (ii) to examine the implications of these patterns for theories of locality and movement.

1.1 Major Themes

Long distance movement and licensing of wh - and focal elements in Turkana do not align with typical assumptions about locality. Ex-situ and in-situ wh-elements pattern together with regard to locality: They can move/be licensed via crossing one CP but never more than one CP:

- (1) a. $[_{CP} \text{ḡáé}] \quad \text{é-jén-ì} \quad \text{ákirù} \quad [_{CP} \text{átàmàr}]$
 $\text{who.DEFL} \quad \text{3-know.PRS-TAM Akiru.NOM} \quad \text{that}$
 $\text{é-mín-á} \quad \text{ákàyì} \quad \text{]]?}$
 $\text{3-love.PRS-TAM Akai.NOM}$
 ‘Who does Akiru know that Akai loves?’
- b. $*[_{CP} \text{ḡáé}] \quad \text{é-jén-ì} \quad \text{ákirù} \quad [_{CP} \text{átàmàr}]$
 $\text{who.DEFL} \quad \text{3-know.PRS-TAM Akiru.NOM} \quad \text{that}$
 $\text{í-tám-ì} \quad \text{ákàyì} \quad [_{CP} \text{átàmàr é-mín-á}]$
 $\text{3-think-TAM Akai.NOM} \quad \text{that} \quad \text{3-love.PRS-TAM}$
 $\text{ádžìkòn} \quad \text{]]]?}$
 Adjikon.NOM
 ‘Who does Akiru know that Akai thinks that Adjikon loves?’

These findings challenge current understanding of locality following phase theory (Chomsky 2000, 2001), where vP and CP are seen as phases. Long distance wh-movement data from Turkana challenge this view. This dissertation addresses the following questions: What is the underlying structure of long-distance dependencies in Turkana? How can theories of locality adequately capture these patterns? And what implications do these data have for the universality of phasality?

To answer these questions, I provide an in-depth investigation of long-distance dependencies in Turkana based on original fieldwork data, much of which is previously undocumented. This work

significantly expands the empirical landscape regarding variation in long-distance dependencies. I describe and analyse the complex system of long-distance dependencies in Turkana and carefully examine their properties. Based on this analysis, I argue that locality constraints are less universal than commonly assumed.

1.2 Theoretical Assumptions

This section outlines the core theoretical assumptions I adopt in this dissertation. I keep the overview brief here; any assumptions that are particularly relevant to individual arguments will be introduced again in the relevant chapters.

I adopt the framework of the *Minimalist Program* (Chomsky 1993, 1995, 2000, 2001), which assumes that syntactic structure is built bottom-up through the recursive application of two structure-building operations: (external) *Merge* and *Move* (= internal Merge). Merge combines two syntactic elements into a constituent; Move re-merges a previously merged object in a higher position. Additionally, the operation *Agree* creates a syntactic dependency between a probe and a matching goal within its c-command domain.

I assume that structure is built in phases (Chomsky 2000, 2001). Each phase is sent to Spell-Out in chunks, after which the internal domain of the phase becomes inaccessible to further operations. The standard assumption is that vP and CP are phase domains. Only their edges remain accessible to higher syntactic operations. This is formalised in the Phase Impenetrability Condition PIC:

(2) *Phase Impenetrability Condition₁ (PIC₁):*

The domain of a head X of a phase XP is not accessible to operations outside XP; only X and its edge are accessible to such operations.

While I adopt the general idea of phases and the PIC in this dissertation, I will question the universal uniformity of phasehood the PIC.

1.3 Methodological Approach

Unless otherwise indicated, all Turkana data come from my own fieldwork. Some data points in this dissertation come from [Dim-mendaal \(1983\)](#) whose work is mostly focused on the phonology of Turkana.

The fieldwork data were collected on two field trips to Nairobi, Kenya in August-September 2022 and August 2023. A total of six speakers were consulted. The speakers came from South East Turkana and North East Turkana and showed some dialectal variation in word order as well as in phonology. All speakers grew up in Turkana county and spoke only Turkana at home, while they spoke Swahili and English in school. The speakers were financially compensated for their valuable work.

The fieldwork consisted of interviews between the consultants and myself. All interviews were recorded. The consultants were given sentences in English and were asked to translate them into Turkana. Some of the sentences were given with context in order to elicit specific meanings. Many of the contexts were constructed on the basis of the information structure fieldwork guide by [van der Wal \(2021\)](#). At a later stage, the speakers were also given sentences in Turkana and were asked to judge the grammaticality of the sentences. All core data were elicited/confirmed by more than one speaker.

The recorded data were later annotated and glossed. Praat was used to analyse tone, vowel length, and ATR value.

1.4 Overview of the Dissertation

Chapter [2](#) gives basic information about Turkana and a grammar sketch of the language, with a particular focus on the word order of Turkana. Turkana is a VSO language with a pre-verbal focus position. This chapter provides a descriptive overview of that position. This chapter fills a gap in theoretical literature because of the general lack of descriptive data on Turkana.

Chapter [3](#) describes and analyses the *wh*-system in Turkana. I show that Turkana has both *wh* in-situ and ex-situ constructions: *Wh*-elements in-situ do not undergo covert movement and are truly in-situ. In contrast, *wh*-elements ex-situ are a result of movement

rather than external merge high up in the structure. In sentences containing more than two CPs, at least two overt *wh*-elements are obligatory, and they can appear in either the pre-verbal or post-verbal position within each CP. This challenges current ideas of phasality.

In chapter 4, I analyse the syntax of the basic word order of Turkana and the landing sites of the pre- and post-verbal *wh*- or focal element. I argue that the subject remains in-situ in the specifier of *vP* and that the verb raises via head movement to *T*. I further argue that the *wh*/focus-probe is located on *T*. For the post-verbal position, I propose that it is an inner specifier of *v*. Furthermore, I argue that both *v* and *T* can have an optional EPP-feature, accounting for the observed optionality in movement.

Chapter 5 explores long-distance dependencies in Turkana in the light of current theories of locality. I show that standard assumptions about phasality cannot account for the Turkana data. I argue that only CPs function as phases in the language, and that phasehood follows PIC₂, resulting in a relatively weak version of phasality. However, I adopt Keine's (2019; 2020b) horizon approach and make it more restrictive. Thereby, locality restrictions in Turkana are less absolute than under a purely phasality based account.

In chapter 6, I summarise the findings and argue that locality restrictions are less universal than previously assumed.

Chapter 2

Background

2.1 Introduction

This chapter provides a brief overview of the grammar of Turkana which is relevant for understanding the patterns and terminology used throughout this dissertation. It introduces basic phonological properties, nominal features (gender, case, plural marking), verbal agreement, and word order. While parts of this chapter build on earlier descriptions by [Dimmendaal \(1983\)](#), the majority of the data presented come from my own fieldwork.

I present novel data on an applicative marker that has not been previously discussed in the literature. Additionally, I show that the basic word order in Turkana is VSO, but that the language also has an optional pre-verbal focus position. This pre-verbal position has received little attention in the literature so far. I discuss the properties of that pre-verbal position and show that only nominals are allowed in this position, that nouns lose their case and get default case, and that not more than one element is allowed in this position which has already been observed for other Nilotic languages ([van Urk 2015](#) for Dinka, [Bossi & Diercks 2019](#); [Driemel & Kouneli 2022](#) for Kipsigis). I furthermore show that this pre-verbal position has mixed A- and $\bar{A}$ -properties like Dinka ([van Urk 2015](#)).

Section [2.2](#) gives an overview of where Turkana is spoken, section [2.3](#) addresses some brief phonological characteristics of Turkana, section [2.4](#) presents nouns and their attributes, section [2.5](#) discusses verbal inflection, section [2.6](#) explores the word order, and section [2.7](#) provides data of the pre-verbal domain. Section [2.8](#) then concludes.

2.2 Overview

Turkana is spoken by around 1 million people in the North of Kenya. There are 68 languages spoken in Kenya of which 61 are indigenous (Eberhard et al. 2024). These languages can be classified into three different language families: Bantu, Cushitic, and Nilotic. The following map (figure 2.1) gives an overview of the languages spoken in Kenya.

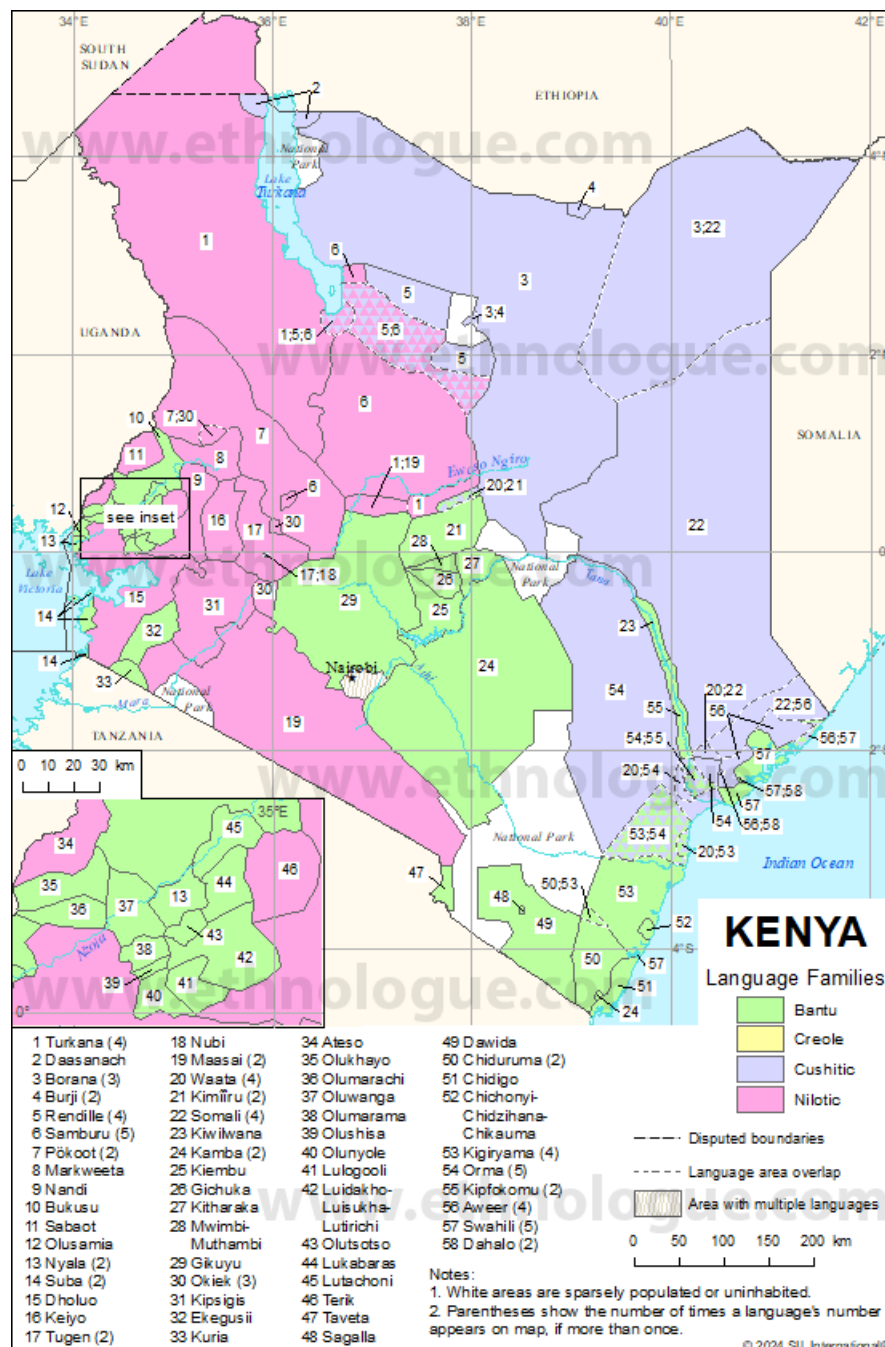


Figure 2.1: Languages spoken in Kenya (Eberhard et al. 2024)

The map in figure 2.2 shows that Turkana is spoken primarily in Kenya's northwestern region, where it borders South Sudan to the north and Uganda to the west.

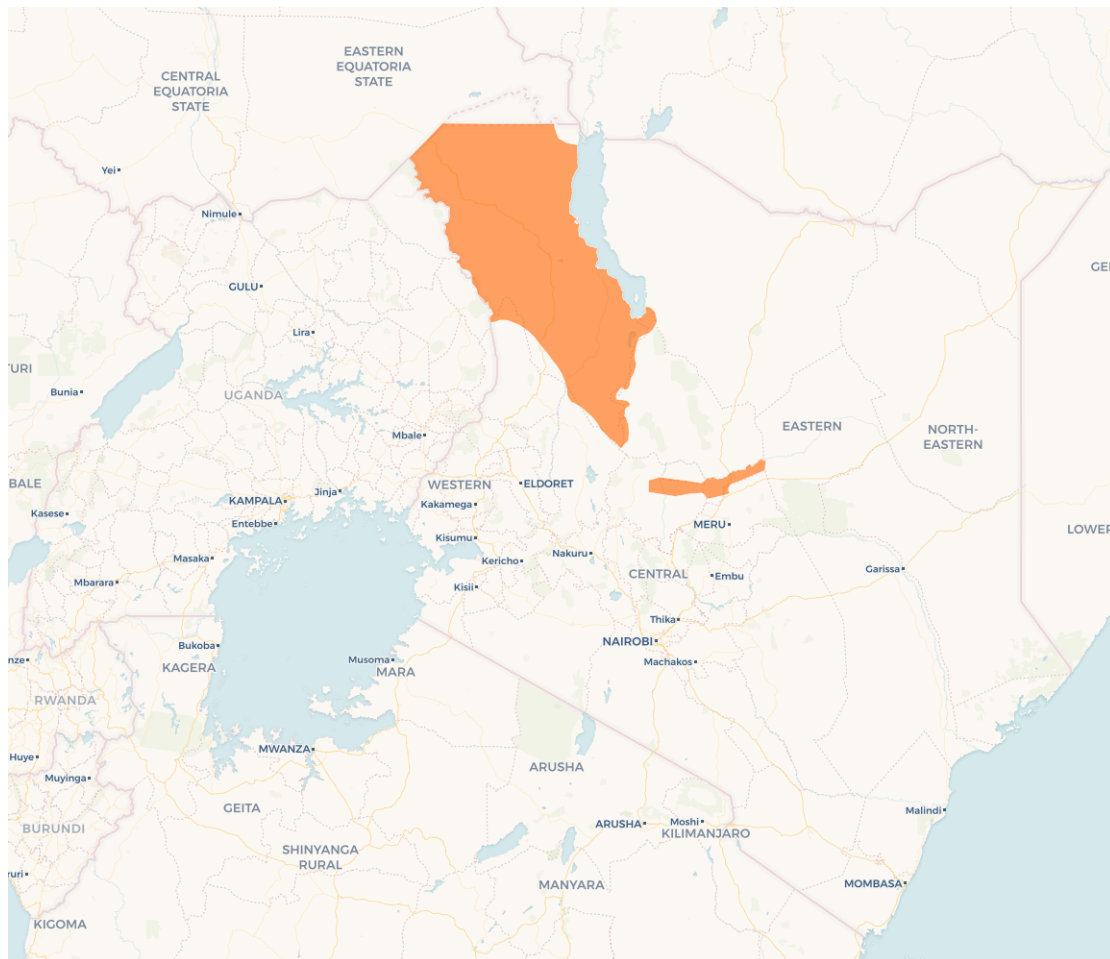


Figure 2.2: Turkana speaking area (Eberhard et al. 2024)

Turkana is an Eastern Nilotic language which belongs to the Nilo-Saharan language family. There are approximately 30 languages in the Nilotic branch (Dimmendaal to appear) which are then further divided into Southern, Western, and Eastern Nilotic. Turkana is mutually intelligible with Toposa and partially intelligible with Karimojong, Jie, and Nyangatom (Dimmendaal 1983, Eberhard et al. 2024). Eberhard et al. (2024) classifies Turkana as shown below:

- (3) Classification of Turkana:
 Nilo-Saharan, Satellite-Core, Core, Eastern-Sudanic, Southern, Nilotic, Eastern, Lotuxo-Teso (Non-Bari Group), Teso-Turkana, Turkana (Eberhard et al. 2024)

The whole Nilo-Saharan language branch belongs to one the least researched languages in Africa (Kouneli 2019) and Nilotic languages and Turkana in particular are extremely under researched. First work on Turkana is from the beginning of the 20th century (Johnston 1902, Barton 1921, Hulley 1923, Emley 1927, Anderson 1973).

First works by professional linguists are Heine & Wilhelm (1980) and Vossen (1982). The most comprehensive study of Turkana is the dissertation by Dimmendaal (1983) and his following work (Dimmendaal 1981, 1985, 1991, 1996, 2000, 2009, 2019, to appear). His dissertation gives a grammatical overview and a linguistic analysis of Turkana with a focus primarily on the phonology and morphology of the language. His collected data come from the West of Turkana county. In recent years, this work has been followed by work on phonology and morphology of Turkana by Zetterstrand (1996); Cysouw (1998); Noske (2000); Bakovic (2000); Finer (2013), and Trommer (2014).

All in all, Turkana is severely understudied and under-documented. Especially the syntactic work on Turkana is very scarce. Therefore, this dissertation aims to shed more light on the syntax of Turkana with new data. The following sections provide an overview of the grammar of Turkana, with a focus on sentence structure.

2.3 Phonology

This section gives a brief overview of some phonological characteristics of Turkana which are relevant for this dissertation. The phonological analysis in this section is based on Dimmendaal (1983) which offers a more in depth discussion of phonology in Turkana than presented here.

2.3.1 Consonants

Turkana has 22 phonemic consonants which are listed in table 2.1. I follow the standard IPA transcription throughout this dissertation. The overview of the consonant inventory is taken from Dimmendaal (1983: 7) and has been updated to the current IPA transcription standard. Furthermore, Dimmendaal notes that /s/ has /θ/ and /ð/ as allomorphs and that /c/ is optionally pronounced as /tʃ/ and

/ʃ/ as /dʒ/. These allophones have been included in the consonant inventory, as they are also attested in my collected data.

	bilabial	dental	alveolar	postalveolar	palatal	velar
stops	p, b		t, d		c, ɟ	k, g
nasals	m		n		ɲ	ŋ
trills			r			
fricatives		θ, ð	s	ʃ, ʒ		
glides					j	w
lateral approximants			l			

Table 2.1: Phonemic consonant inventory in Turkana

2.3.2 Vowels

Like all Nilotic languages (Kouneli 2019: 16), Turkana is a tonal language with advanced tongue root [ATR] and vowel length being distinctive features in the vowel inventory which will be further discussed in this subsection. Turkana has eighteen vowel phonemes which can either be [+voiced] or [-voiced]. Besides /a/ all vowels can either be [+ATR] or [-ATR] (Dimmendaal 1983: 17). The following table provides a complete overview of Turkana's vowel inventory, with all vowels shown as capable of occurring in both long and short forms.

	i	ɪ	e	ɛ	a	o	ɔ	u	ʊ	ĩ	ɪ̃	ẽ	ɛ̃	ã	õ	ɔ̃	ũ	ʊ̃
High	+	+	-	-	-	-	-	+	+	+	+	-	-	-	-	-	+	+
Low	-	-	-	-	+	-	-	-	-	-	-	-	-	+	-	-	-	-
Back	-	-	-	-	+	+	+	+	+	-	-	-	-	+	+	+	+	+
Voiced	+	+	+	+	+	+	+	+	+	-	-	-	-	-	-	-	-	-
ATR	+	-	+	-	-	+	-	+	-	+	-	+	-	-	+	-	+	-

Table 2.2: Phonemic vowel inventory in Turkana (Dimmendaal 1983: 17)

[ATR] is a distinctive feature in Turkana, as illustrated by minimal pairs that differ solely in their [ATR] value (Dimmendaal 1983). An example of such a minimal pair in verb roots is shown below:

- (4) a. -ger
 'harvest'
 b. -gɛr
 'tattoo'

(Dimmendaal 1983: 18)

According to [Dimmendaal \(1983: 23\)](#), Turkana exhibits [ATR] vowel harmony, where all vowels within a word share the same [ATR] specification, either [+ATR] or [-ATR]. This harmony is limited to the word level and does not extend across word boundaries. The root typically determines the [ATR] value, except when certain suffixes are present; in such cases, the suffix controls harmony. This type of harmony, known as dominant/recessive or strong/weak harmony, allows dominant suffixes to impose their [ATR] value on the entire word, while recessive suffixes harmonise. All prefixes in Turkana are recessive and harmonise accordingly¹.

Vowel length is also a distinctive feature in Turkana and a few minimal pairs exist:

- (5) a. jík
 'completely'
 b. jí:k
 'always' ([Dimmendaal 1983: 34](#))

Turkana is a tonal language with low and high tonemes. High tones are more frequent ([Dimmendaal 1983: 36](#)). Turkana has lexical tone with few minimal pairs which can be seen in the following example. This means that tone is lexically determined.

- (6) a. é-dí:θ
 M-banana
 'banana'
 b. é-dìθ
 M-HIV
 'HIV'

However, Turkana relies heavily on grammatical tone. Tense and case for example are marked tonally in Turkana which will be further discussed in the following sections.

2.4 Nouns

The following section gives an overview of the gender and number marking on the noun and the case system in Turkana.

¹For a more in depth analysis of vowel harmony in Turkana, see [Dimmendaal \(1983\)](#) and [Noske \(2000\)](#)

2.4.1 Gender and Number Marking

In terms of gender and number marking, Turkana distinguishes between three types of nouns. The first and largest group consists of countable and non-countable nouns, the second group includes kinship terms and proper nouns, and the third and smallest group contains exceptional nouns. These three groups differ in their marking for gender and number (Dimmendaal 1983: 209).

Gender Marking

Turkana has three grammatical genders: masculine, feminine, and neuter. Neuter often acts as a diminutive. Countable and non-countable nouns, kinship terms and proper nouns are marked for gender, while the exceptional nouns lack a gender marker in singular (Dimmendaal 1983). Gender is marked on the noun as a prefix: Masculine is marked with an *e-*, feminine with an *a-*, and neuter with an *i-*. By changing the gender marker of an animate noun like *ɲoq* ('dog'), the gender of the animate noun changes: *ɲoq* with a female gender marker (7a) refers to a female dog and *ɲoq* with a male gender marker (7b) refers to a male dog.

- (7) a. á-ɲòq
F-dog
 'female dog'
 b. é-ɲòq
M-dog
 'male dog'

The gender marker changes when the noun is in plural and becomes *ɲa-* for feminine nouns (8a) and *ɲi-* for masculine and neuter words (8b).

- (8) a. ɲá-ɲòq
PL.F-dog
 'female dogs'
 b. ɲí-ɲòq
PL.M-dog
 'male dogs'

The nominal gender markers are summarised in the following table:

	SG	PL
F	a-	ɲa-
M	e-	ɲi-
N	i-	ɲi

Table 2.3: Nominal gender markers in Turkana (Dimmendaal 1983: 210)

As it will become clear in the course of the dissertation, Turkana names always appear with the aforementioned gender markers. Typically, children are named after the place where they were born or sometimes after noticeable weather like wind or rain. For example, the author was born in a hospital and is a woman thus her name would most likely be *Asbitari* (female gender marker + *hospital*). The male equivalent for that name would be *Esbitari* (male gender marker + *hospital*).

Number Marking

Turkana has a tripartite number system which is typical for Nilotic languages (Dimmendaal 1983, 2000; Corbett 2000; Kouneli 2019). Number marking on nouns can be categorised into three groups: Some nouns have a default singular form and plural is marked with an additional plural suffix (9), some nouns have a default plural form and singular is marked with an additional singular suffix (10), the third group does not have a default form and both singular and plural are marked with an additional suffix (11).

- (9) a. á-ɲásép
F-placenta
'placenta' (Dimmendaal 1983: 224)
- b. ɲá-ɲàsèp-á
F.PL-breast-PL 'placentas' (Dimmendaal 1983: 224)
- (10) a. é-síkín-á
M-breast-SG
'breast' (Dimmendaal 1983: 224)
- b. ɲí-síkín
M.PL-breast
'breasts' (Dimmendaal 1983: 224)
- (11) a. á-kùk-út
F-chicken-SG

- ‘chicken’ (Dimmendaal 1983: 224)
- b. ṇá-kúkú-í
F.PL-chicken-PL
‘chicken (pl)’ (Dimmendaal 1983: 224)

According to Dimmendaal (1983: 224), there are various singulative and plurative suffixes in Turkana. Some suffixes are used as both singulative and plurative markers, whereas others are used exclusively to mark singulative or plurative. The typical singulative and plurative markers are shown in the table below.

Singulative markers	Plurative markers
-it	-i
-at	-a
-t	-in
-ni	-an
	-ia
	ney
	sia

Table 2.4: Nominal number markers in Turkana (Dimmendaal 1983: 258)

2.4.2 Case

Turkana has six cases which are marked tonally: default case², nominative, genitive, instrumental, locative, and vocative case.

The default case is tonally unmarked; its tones are determined purely lexically. All other cases are tonally marked and predictable. Therefore, the other cases including the nominative are derived from the unmarked default form (Dimmendaal 1983: 66).

The default case is not only morphologically but also functionally unmarked in Turkana. Default case is used as the citation form, the form for every noun in the pre-verbal position, the emphatic subject, and the nominal predicate, while nominative is only used for subjects in the post-verbal position (Dimmendaal 1983):

If the subject or any other noun for that matter moves to the pre-verbal domain for discourse reasons, it has default case. The

²The default case was previously called absolutive by Dimmendaal (1983). He called it absolutive because it is not only found with objects but it is also the citation form, the form for every noun in the pre-verbal position, the emphatic subject, and the nominal predicate. However, Turkana does not behave like an ergative-absolutive language as I will show in this section. Thus, I differ from this definition and will call this case default case since it is morphologically and functionally unmarked.

following example shows the subject in nominative case in the post-verbal position (12a) and in default case in the pre-verbal position (12b):

- (12) a. í-múdz-ì á-bèrò á-kírín.
 3.-eat.PRS-TAM F-woman. **[NOM]** F-meat.DEFL
 ‘The woman is eating meat.’
 b. á-bérú í-múdz-ì á-kírín.
 F-woman. **[DEFL]** 3-eat.PRS-TAM F-meat.DEFL
 ‘THE WOMAN is eating meat.’

The phenomenon of no overt case marking in the pre-verbal domain is not unique for Turkana. It has in fact been observed for several East-African languages, especially Nilotic languages (König 2006, 2008; Handschuh 2014). König (2008) calls this the ‘no case before the verb’ rule.

The nominal predicate is in default case and not in nominative like in marked accusative languages such as German, French etc. (13) shows the nominal predicate being in nominative case in German (13a) but in default case in Turkana (13b).

- (13) a. Ich bin schön.
 I. **[NOM]** am beautiful
 b. é-bùθ ájóη.
 1-beautiful I. **[DEFL]**
 ‘I am beautiful.’

Even though nominative is morphologically and functionally not the default case, Turkana is not an ergative-absolutive language. Instead, Turkana behaves like a nominative-accusative language for case and agreement. Nominative covers subjects in unaccusative (14), unergative (15), and transitive (16) sentences. The default case covers objects.

- (14) á-tóná ηèð/ηéðì.
 3-die he/she. **[NOM]**.SG
 ‘He is dying.’
 (15) é-pérí ηèð/ηéðì.
 3-rest he/she. **[NOM]**.SG
 ‘He is sleeping.’

- (16) á-máð-ì ájòŋ ŋá-kílé.
 1-drink.PRS-TAM I. **NOM** PL.F-milk.DEFL
 ‘I am drinking milk.’

Default case is used for subjects of some experiencer verbs, however these verbs have object marking which will be discussed more in the next section.

- (17) a. k-á-ðíí-kín-ít ájóŋ.
 K-1-being.lonely-DAT-TAM I. **DEFL**
 ‘I am lonely’
 b. k-á-búró-(nít) ájóŋ.
 K-1-being.tired-(TAM) I. **DEFL**
 ‘I am tired.’

While Barasa (2017) has argued for Ateso, a closely related Eastern-
 Nilotic language, that it has an ergative-absolutive alignment, this
 cannot be said about Turkana as shown above. Turkana is a lan-
 guage with nominative-accusative alignment and unmarked object
 case, thus a marked nominative language. While marked nomin-
 ative is the most common case system in Africa, it is topologically
 rare (König 2006). However, it is beyond the scope of this disserta-
 tion to analyse the nature of the marked nominative case system in
 Turkana³.

2.5 Verbs

This section aims to give an overview of the morphemes with a focus
 on the morphemes relevant for this dissertation. Turkana is a highly
 agglutinative language as can be seen in the following example.

- (18) pɛ-k-a-`-ìs-ìlɔt-aan-a-kin-it-o-tò-rì.
 1-2-3-4-5-6-7-8-9-10-11-12-13
 ‘That they did not force me to do the washing all the time.’
 (Dimmendaal 1983: 96)

³For a typology for marked nominative systems see König (2006), König (2008), Handschuh (2014) and for a syntactic implementation see Baker (2015), van Urk (2015), Kouneli (2024).

Negation marker	1
'Moveable k'	2
Person marker	3
Tense marker	4
Causative marker	5
Verb root	6
Extension 1	7
Epipatetic vowel	8
Extension 2	9
Aspect marker	10
Voice marker	11
Number marker	12
Subjunctive marker	13

Table 2.5: The internal structure of the conjugated verb (Dimmendaal 1983: 96)

2.5.1 Subject and Object Marking

Turkana shows verbal agreement. The person agreement marker appears on the verb as a prefix. The person agreement prefixes are shown in the table below. Turkana has two verbal classes which differ in their agreement markers in first person singular and third person.

	SG	PL
1	a- / e-	ki-
2	i-	i-
3	e- / i-	e- / i-

Table 2.6: person markers (Dimmendaal 1983: 120)

Number is not marked in singular and in first person plural. Turkana has an exclusive and inclusive first person plural which have the same person marker. First person plural inclusive and exclusive are both only marked with the prefix *ki-* while second and third person plural have a person prefix and an additional number suffix *-tV*, *-θV* or *-ðV*. The following examples give the complete verbal paradigm of the verb *dancing*.

- (19) a. á-bólíyá ájòŋ/àŋ.
1-dance I.NOM
'I am dancing.'
- b. í-bólíyá (í)jòŋ
2-dance you.NOM.SG

- ‘You are dancing’
 c. **é**-bólíà ɲèð/ɲéðì.
 3-dance s/he.NOM.SG
 ‘S/he is dancing.’
 d. **kí**-bólíyá ɲòn.
 1.PL-dance we.NOM.INCL
 ‘We (incl.) are dancing.’
 e. **kí**-bólíyá ðúà.
 1.PL-dance we.NOM.EXCL
 ‘We (excl.) are dancing.’
 f. **í**-bólíyá-**ðí** éðì.
 2-dance-**PL** you.NOM.PL
 ‘You are dancing.’
 g. **é**-bólíyá-**ðí** kètʃ.
 3-dance-**PL** they.NOM
 ‘They are dancing.’

As seen in (19), the verb agrees with the subject in intransitive sentences. The transitive verb also has a subject preference unless the object is more prominent than the subject. Turkana has a prominence hierarchy and the verb agrees with the most prominent DP. The prominence hierarchy for Turkana is the following:

- (20) Participants > Non-participants, proper names > animate nouns > inanimate nouns (Dimmendaal 1983: 88)

If the object is more prominent than the subject, the verb agrees with the object and has an additional k-marker (Dimmendaal 1985, Finer 2013, Trommer 2014). This can be seen in (21) where the object is a participant and therefore more prominent than the animate noun subject. Thus, the verb shows person agreement with the object plus the additional k-marker.

- (21) **k-á**-pám ájóɲ é-mòn.
 k-1-swallow I.DEFL M-snake.NOM
 ‘The (male) snake is swallowing me.’

The indirect object is introduced with the suffix /-kin/ and its allomorphs /-km/, /-ki/, /-kɪ/ (Dimmendaal 1983: 113, Tyrchan 2023). Dimmendaal (1983) calls this marker *dative marker* since it introduces the indirect object. I will follow his naming convention here. The

following example shows a sentence with an indirect object and the dative marker on the verb:

- (22) ε -i-na-**ki** ε -kadɔkɔ-t leonie ε diθ.
 3-EV-give-**-DAT** M-monkey-SG Leonie banana
 ‘The monkey gives Leonie a banana.’ (Tyrchan 2023)

2.5.2 Valency Changing Morphology

Applicatives

In order to promote oblique objects to arguments, Turkana has an applicative strategy where instruments, locatives, or temporal adverbs can become core arguments. The applicative is marked with the suffix *-a* or its allomorph *-o*⁴. Dimmendaal (1983) also mentions this marker and its allomorph. However, he calls it *epipatetic vowel* and states that this marker does not have any meaning now but used to mark transitive verbs at an earlier stage. I will dissent from Dimmendaal’s analysis and will show how that marker functions as an applicative marker.

Dimmendaal (1983: 203), states the *-a* suffix does not perform any grammatical function. Thus, according to his analysis, the existence of this marker should not have an influence on the argument structure of the sentence. However, this is not what we see. Instead, we see that the oblique object changes its case from instrument to default case and does not require a preposition anymore if the *-a* marker is present. This hints that this marker promotes oblique objects to core arguments as expected from an applicative. The next paragraphs will look at this in more detail.

Turkana can use prepositions or the *-a* marker to introduce instruments. There is either a preposition attached to the instrument in instrumental case (23a) or there is the *-a* marker on the verb and the instrument is in default case (23b). The sentence is ungrammatical if the verb has no *-a* marker but the instrument is in default case without a preposition (23c). The *-a* marker in combination with a preposition is also ungrammatical (23d).

⁴Dimmendaal (1983: 203) states that /o/ is a regular allomorph of /a/ in Turkana. For further discussions about allomorphy in Turkana see Dimmendaal (1983)

- (23) a. í-pír-ì ákìrù
 3-stir.PRS-TAM Akiru.NOM
 é-è-gètɕ.
 with-M-wooden.whisk. **INS**
- b. í-pír-í-**à** ákìrù
 3-stir.PRS-TAM-**APPL** Akiru.NOM
 é-gètɕ.
 M-wooden.whisk. **DEFL**
- c. *í-pír-ì ákìrù é-gètɕ.
 3-stir.PRS-TAM Akiru.NOM M-wooden.whisk. **DEFL**
- d. *í-pír-í-**à** ákìrù
 3-stir.PRS-TAM-**APPL** Akiru.NOM
 é-è-gètɕ.
 with-M-wooden.whisk. **INS**
 ‘Akiru is stirring with a wooden whisk.’

Applicatives introduce arguments and the noun being in default case without a preposition if the verb has the additional -a marker is a strong indicator that the noun is an argument (23b). Without this marker, having a bare noun as an instrument is ungrammatical (23d). Hence, I assume that this marker is an applicative marker and will call the -a marker from now on an applicative marker.

The applicative marker can also introduce locatives. There are no specific locative prepositions but if there is no applicative on the verb, the locative is missing its gender marker and has only a locative marker (24a). Thus, the locative loses part of its noun properties. If there is an applicative marker on the verb, the noun has a gender and a locative marker (24b).

- (24) a. é-pér-ì ákìrù **à-ló**-síbìtár.
 3-sleep.PRS-TAM Akiru.NOM **F-LOC-**hospital
- b. é-pér-í-**ò** ákìrù **ló**-síbìtár.
 3-sleep.PRS-TAM-**APPL** Akiru.NOM **LOC-**hospital
 ‘Akiru is sleeping in the hospital.’

Applicatives also introduce temporals, however only when they are in the pre-verbal position. When temporals are in a post-verbal position, the applicative marker is ungrammatical (25a). Temporals in a post-verbal position are only grammatical without the applicative

marker on the verb (25b). However, in the pre-verbal domain, the opposite is true. A temporal expression is only allowed in the pre-verbal domain if there is the applicative marker on the verb (25c). Without the applicative marker, the sentence is ungrammatical (25d).

- (25) a. *í-múdz-í-à ákìrù tóxóná.
 3-eat.PRS-TAM-APPL akiru.NOM now
 b. í-múdz-ì ákìrù tóxóná.
 3-eat.PRS-TAM akiru.NOM now
 c. tóxóná. í-múdz-í-à ákìrù
 now 3-eat.PRS-TAM-APPL akiru.NOM
 d. *tóxóná í-múdz-ì ákìrù.
 now 3-eat.PRS-TAM akiru.NOM
 ‘Akiru is eating now.’

Commitatives

Commitatives can also be introduced as a core argument by an applicative on the verb. However, the marker for commitatives has a different morphological realisation: *-ta*. The marker looks similar to the plural marker but it is not a plural marker as the following example shows. (26a) shows that the plural marker for *dancing* is *-te*. The *wh*-word is in nominative case since it is part of the subject and not a commitative. *ka-* functions as a conjunction and not as a preposition in both (26a) and (26b). Thus, in (26b) *ka-* is also allowed to be fronted even though prepositions are not allowed in the pre-verbal domain (this will be further discussed in 2.7.3.)

- (26) a. í-dóná-á-tè ákìrù kà-ḡáè?
 3-dance.PRS-TAM-PL Akiru.NOM and-who.NOM
 b. ákìrù kà-ḡáè í-dóná-á-tè
 Akiru.DEFL and-who.DEFL 3-dance.PRS-TAM-PL
 ‘Akiru and who are dancing?’

However, if the verb is in singular (27a), the *wh*-element is not part of the subject. It functions as a commitative and is in default case. The commitative is introduced by *ka-* which functions as a commitative preposition. The commitative can also be introduced by the *-ta* marker (27b) instead of the preposition. The reason why I assume this marker to be an applicative is the same as mentioned

above. Applicatives introduce core arguments which the marker is doing here.

- (27) a. í-dóŋá-ì ákírù **kà**-ŋáé?
 3-dance.PRS-TAM Akiru.NOM **with**-who. **DEFL**
 b. **ŋáé** í-dóŋá-á-**tà** ákírù?
who.DEFL 3-dance.PRS-TAM-**APPL** Akiru.NOM
 ‘With whom is Akiru dancing?’

Furthermore, the plural marker and the commitative marker can occur together⁵ which also indicates that they are not the same marker:

- (28) (ká)tá-ŋáé í-té-**té-tà** (kétfi) ákírù
 PL-who.DEFL 3.see.PRS-**PL-APPL** (they.NOM) Akiru.DEFL
 ‘With whom (pl) are they seeing Akiru’

Causatives

Turkana does not only have an applicative strategy to increase the valency of the verb but also a causative strategy. This causative marker has already been observed by [Dimmendaal \(1983\)](#). (29a) shows the verb *é-gáloû* (‘s/he is crying’) without a causative. Adding the *ǝi*- marker to the verb increases the valency of the verb and gives the verb a causative meaning as can be seen in (29a).

- (29) a. é-gáloû ŋéǝi.
 3-cry he/she.NOM.SG
 ‘S/he is crying.’
 b. é-**ǝi**-gáloû á-bèrù í-kókù
 3-**CAUS**-cry F-woman.NOM N-child.DEFL
 ‘The woman makes the child cry.’

2.5.3 TAM Marking

According to [Dimmendaal \(1983: 129\)](#), Turkana distinguishes between [+past] and [-past]. While [-past] is not marked (30), [+past] is

⁵*ká*- is an optional part of the plural marker of *ŋáé* and not a preposition which can be seen in the following example where (ká)táŋáé is the subject:

- (i) (ká)tá-ŋáé í-té-té ákírù
 PL-who.DEFL 3.see.PRS-PL akiru.DEFL
 ‘Who(pl) is seeing Akiru’

marked with a floating low tone (31).

- (30) á-**bú** ɲéðì.
 3.SG-come.**[PRS]** he/she.NOM.SG
 'S/he is coming.'
- (31) à-**bùn**-í ákìrù nàðá:
 3-come.**[PST]**-TAM Akiru.NOM at.a.time
 n-á-ɲár-**ɛ**-à ákàjì ɛ-pídín.
 when-3-open-**[PST]**.TAM-APPL Akai.NOM M-window.DEFL
 'Akiru came at a time when Adjikon quickly opened a window/gate/door.'

As can be seen in (31), the verb can also have additional markers, which I grouped together for simplicity and labelled as TAM marker. They include the aspect marker, the voice marker, and the mood marker. [Dimmendaal \(1983\)](#) offers a more in depth analysis of these markers. He shows that Turkana has an imperfective marker *-it* and a perfective marker *-i/-e* ([Dimmendaal 1983: 131](#)). Stative verbs are lacking this marker. Turkana has an unmarked active voice and an impersonal and middle voice which are both marked with *-i/-o* on dynamic verbs and with *-a* on stative verbs ([Dimmendaal 1983: 132](#)). Unfortunately, it is beyond the scope of this dissertation to discuss the tense, aspect, mood, and voice system of Turkana in further detail.

2.6 Basic Word Order

Turkana is a V1 language. The unmarked word order in Turkana is VSO (32a) ([Dimmendaal 1983](#)). VOS is generally ungrammatical (32b).

- (32) a. ɛ-múdɜ-í ájòŋ ákírín.
 1-eat.PRS-TAM I.NOM meat.DEFL
 (V-S-O)
- b. *ɛ-múdɜ-í ákírín ájòŋ.
 1-eat.PRS-TAM meat.DEFL I.NOM
 (V-O-S)

‘I am eating meat.’

As shown in 2.5.1, Turkana has a prominence hierarchy. This does not only influence agreement but also the word order. The word order changes if the object is more prominent than the subject (Dimendaal 1983, 1985, Finer 2013, Tyrchan 2023). While VSO is still grammatical (174a), the preferred word order is VOS (174b).

- (33) a. k-á-kóp é-mòn ájóη.
K-1-bite M-snake.NOM I.DEFL
(V-S-O)
- b. k-á-kóp ájóη é-mòn.
K-1-bite I.DEFL M-snake.NOM
(V-O-S)

‘The snake is biting me.’

In ditransitive sentences, the word order is also subject to variation. The word order can either be V-S-IO-DO (34a) or V-S-DO-IO (34b) without any specific requirements (Tyrchan 2023).

- (34) a. ε-ma-ki ε-kadəkɔ-t leonie ediθ.
3-give-DAT M-monkey-SG Leonie banana
(V-S-IO-DO)
- b. ε-ma-ki ε-kadəkɔ-t ediθ leonie.
3-give-DAT M-monkey-SG banana Leonie
(V-S-DO-IO)

‘The monkey gives Leonie a banana.’ (Tyrchan 2023: 4)

The indirect object can precede the subject if it is a participant (35a). If the indirect object is not a participant, the subject following the indirect object is ungrammatical (35b) (Tyrchan 2023).

- (35) a. k-í-nà-kín-ít ijòn líónì èdìθ.
K-2-give-DAT-TAM 2SG Leonie banana
(V-IO-S-DO)

‘Leonie gives you a banana.’

- b. *á-má-kí ékádókòt ák^wóm ákàlaliòit.
 3-give-DAT monkey Akwom akalalioit
 (V-S-IO-DO)

‘Akwom gives the monkey an akalalioit.’ (Tyrchan 2023: 4)

The direct object can never precede the subject in ditransitive sentences, independent of being higher in the prominence hierarchy or not (Tyrchan 2023).

- (36) *è-dódí-k àjòŋ ɣéθ ɣèθ.
 3-show-DAT 1SG 3SG 3SG
 (V-DO-S-IO)

‘S/he shows me to him/her.’ (Tyrchan 2023)

The position in front of the verb is usually empty but for information structural reasons, nominal elements can also occur in the pre-verbal domain. This will be discussed in more detail in the next section.

2.7 The Pre-verbal Domain

Besides the canonical word order (37a), nominals can also move to the pre-verbal position (37b).

- (37) a. é-múdʒ-í ájòŋ ákíríŋ.
 1-eat.PRS-TAM I.NOM meat.DEFL
 (V-S-O)

‘I am eating meat.’

- b. ákíríŋ é-múdʒ-í ájòŋ.
 meat.DEFL 1-eat.PRS-TAM I.NOM
 (O-V-S)

‘I am eating MEAT.’

The pre-verbal domain has some very specific properties that it shares with several other Nilotic languages such as being an information structure position, not allowing more than one element,

only allowing nominals, and not allowing overt case marking. This section presents these properties in more detail.

2.7.1 Focus Position

Nominals can appear in the pre-verbal domain for information structural reasons which has already been shown by [Dimmendaal \(1983: 407\)](#). He claimed however that the pre-verbal position is a dedicated topic position which my data do not confirm. Topics are the part of the sentence about which something is uttered ([Gundel 1988](#)). Thus, the 'tell me about X' question by ([Reinhart 1981](#)) is a standard test to detect aboutness topics. My data show that the pre-verbal domain cannot host aboutness topics in Turkana. The following example shows the 'tell me about X' question where *Akai* is what the utterance is about, meaning *Akai* is the topic of the sentence. The topic is allowed in the post-verbal domain (38a) but not in the pre-verbal domain (38b).

(38) *Context: We are talking and Akai is mentioned who I do not know.*
I ask: "Tell me about Akai."

- a. é-lòm-ìt **ákàjì** nàiròbí.
 3-live.PRS-TAM **akai.NOM** nairobi.LOC
- b. # **ákájì** é-lòm-ìt nàiròbí.
 akai.DEFL 3-live.PRS-TAM nairobi.LOC
 'Akai lives in Nairobi.'

Unlike topics, focal elements are the only elements which are allowed in the pre-verbal domain which I will show in the following paragraphs.

Wh-elements are inherently focal and can, unlike topics, occur both in the post-verbal (39a) and the pre-verbal (39b) position:

- (39) a. í-mudʒ-ì **ɲáé** á-kírín?
 3-eat.PRS-TAM **who.NOM** F-meat.DEFL
- b. **ɲáé** í-mudʒ-ì á-kírín?
 who.DEFL 3-eat.PRS-TAM F-meat.DEFL
 'Who is eating meat?'

While the prominence hierarchy influences the basic word order, this is not the case for the pre-verbal position. The wh-element can

stand in the pre-verbal domain even if it is less prominent than other elements of the sentence. It is also not of importance whether the wh-element is the subject of the sentence or not. The following example shows a sentence with an wh-object which is less prominent than the subject. It can appear both in the post-verbal (40a) and the pre-verbal position (40b).

- (40) a. í-múdǝ-ì á-bèrù **ɲó?**
 3-eat.PRS-TAM F-woman.NOM **what.DEFL**
- b. **ɲó** í-múdǝ-ì á-bèrù?
what.DEFL 3-eat.PRS-TAM F-woman.NOM
 ‘What is the woman eating?’

Wh-elements can also be in-situ or in the pre-verbal position when they do not target an argument. However, when these kind of wh-elements get fronted, the verb has to have an applicative. The following example shows that the wh-element *when* can appear both post-verbally (41a) or pre-verbally (41b).

- (41) a. é-kér-ì ákìrù **órí?**
 3-run.PRS-TAM Akiru.NOM **when**
- b. **órí** é-kér-é-à ákìrù?
when 3-run.PRS-TAM-APPL Akiru.NOM
 ‘When is Akiru running?’

The question word *ɲó* cannot only mean *what* (40) but also *why*. Unlike every other wh-element in Turkana, *ɲó* is not allowed in the post-verbal position when it has the meaning *why*. This is not unusual and *why* not being allowed in the post-verbal domain has already been observed for many other languages like Akan (Saah 1988; Torrence & Kandybowicz 2015), Chinese (Lin 1992), English (Hornstein 1995), Ikalanga (Letsholo 2011), Italian (Rizzi 2001), Japanese (Ko 2005), Kitharaka (Muriungi 2005), Korean (Ko 2005), Lubukusu (Wasike 2006), Nupe (Kandybowicz 2020), Persian (Karimi 2005), Romanian (Shlonsky & Soare 2011), and others.

- (42) a. **ɲó** é-kér-é-à ákìrù?
what 3-run.PRS-TAM-APPL Akiru.NOM
- b. *é-kér-ì ákìrù **ɲó?**
 3-run.PRS-TAM Akiru.NOM **what**

‘Why is Akiru running?’

Not only *wh*-elements are allowed in the pre-verbal domain but also other focal elements. The following examples show that new information focus (the target information of a *wh*-question) can occur in-situ (43a) or in the pre-verbal position (176b). However, DPs other than the focal element are not allowed in the pre-verbal domain (43c).

- (43) A: **ɲáé** í-múdz-ì á-kíríŋ?
who.DEFL 3-eat.PRS-TAM F-meat.DEFL
 ‘Who is eating meat?’
- a. B: í-múdz-ì **á-bèrù** á-kíríŋ.
 3.-eat.PRS-TAM **F-woman.NOM** F-meat.DEFL
- b. B: **á-bérú** í-múdz-ì á-kíríŋ.
F-woman.DEFL 3-eat.PRS-TAM F-meat.DEFL
- c. B: #á-kíríŋ í-múdz-ì **á-bèrù**.
 F-meat.DEFL 3.-eat.PRS-TAM **F-woman.NOM**
 ‘The WOMAN is eating meat.’

As seen before with *wh*-elements, neither the prominence hierarchy nor being the subject or not influences the pre-verbal domain. The same picture as in the previous example emerges with a focal element which is not the subject nor the most prominent element of the sentence. The new information focal element can occur in-situ (44a) or in the pre-verbal position (44b). However, non-focal DPs are not allowed in the pre-verbal domain (44c).

- (44) A: **ɲó** í-múdz-ì á-bèrù?
what.DEFL 3-eat.PRS-TAM F-woman.NOM
 ‘What is the woman eating?’
- a. B: í-múdz-ì á-bèrù **á-gwálí**.
 3-eat.PRS-TAM F-woman.NOM **F-ugali.DEFL**
 ‘The woman is eating meat’
- b. B: **á-gwálí** í-múdz-ì á-bèrù.
F-ugali.DEFL 3-eat.PRS-TAM F-woman.NOM
- c. B: #á-bérú í-múdz-ì **á-gwálí**.
 F-woman.DEFL 3-eat.PRS-TAM **F-ugali.DEFL**
 ‘The woman is eating UGALI.’

The same picture emerges in subordinate clauses: Focal elements can stay in-situ (45a) or can be fronted (45b). In the following example the focus *á-kítáb ná* ('this book') is marked by the focus marker *bón* ('only'):

- (45) a. é-ǒóm-ì ákìrù á-kìtáb ná
 3-read.PRS-TAM Akiru.NOM F-book.DEFL this.DEFL
 kótérè é-mín-à ɲèǒ
 because 3-like.PRS-TAM she.NOM

á-kìtáb	ná	bón
---------	----	-----

F-book.DEFL this.DEFL only

- b. é-ǒóm-ì ákìrù á-kìtáb ná
 3-read.PRS-TAM Akiru.NOM F-book.DEFL this.DEFL
 kótérè

á-kìtáb	ná	bón
---------	----	-----

 é-mín-à
 because

F-book.DEFL this.DEFL only

 3-like.PRS-TAM
 ɲèǒ.
 she.NOM
 ‘Akiru is reading this book because she likes ONLY THIS BOOK.’ (*She likes no other book*)

2.7.2 Not More Than One Element

The pre-verbal domain is a quite restricted position. As shown, it only allows focal elements. In addition, also not more than one element can precede the verb. This can be seen in (46) where both subject and object precede the verb. The order of subject and object does not influence the ungrammaticality of the sentence.

- (46)
- a. *á-bérú á-kírínj í-múdǝ-ì.
 F-woman.DEFL F-meat.DEFL 3-eat.PRS-TAM
 (S-O-V)
- b. *á-kírínj á-bérú í-múdǝ-ì.
 F-meat.DEFL F-woman.DEFL 3-eat.PRS-TAM
 (O-S-V)

‘The woman is eating meat.’

The reason that only one element is allowed in the pre-verbal domain might be due to the fact that a sentence has only one focal element. However, other Nilotic languages like Kipsigis (Southern Nilotic, [Driemel & Kouneli 2022](#); [Bossi 2023b](#)) and Dinka (Western Nilotic, [van Urk 2015](#)) with a different information structure show the same

pattern where only one element can precede the verb.

The pre-verbal domain in Kipsigis is also an information structural position like in Turkana, but, unlike in Turkana, only topics are allowed in this position (Driemel & Kouneli 2022; Bossi 2023b). Topics are marked with the obligatory topic marker *ko*. However, two elements in the pre-verbal domain are ungrammatical whether only the subject (47a), only the object (47b), or both (47c) are marked with this marker. The sentence is grammatical if only one element is preceding the verb, like the subject (47d) or the object (47e).

2.7.3 Only Nominals

The pre-verbal domain in Turkana shares more similarities with the pre-verbal domain in Kipsigis and Dinka. The pre-verbal domain is restricted to nominals in both Kipsigis (Driemel & Kouneli 2022) and Dinka (van Urk 2015). This is also the case for Turkana. The next paragraphs show this in more detail.

Prepositional phrases preceding the verb are ungrammatical in Kipsigis (Driemel & Kouneli 2022).

- (49) ***Ak kiplànàt kó** ka-Ø-tʃap-e kíbê:t amitwa:gik.

with Kiplangat TOP PST-3-make-IPFV Kibeet.NOM food
 ‘With Kiplangat, Kibeet made food.’ (Kipsigis, [Driemel & Kouneli 2022](#): 7)

In Dinka, PPs are nominalised when they move to the pre-verbal domain ([van Urk 2015](#)).

- (50) a. àyén à-càm cuḡin **nè** pǎal.
 Ayen 3SG-eat.SV food **PREP** knife
 b. pǎal à-céem-**è** áyèn cuḡin.
 knife 3SG-eat. **OBLV** Ayen.GEN food
 ‘Ayen is eating food with a knife.’ (Dinka, [van Urk 2015](#): 74)

In Turkana, PPs can also not precede the verb ([51b](#)) like in Kipsigis ([49](#)). However, bare nominals with a modification on the verb ([51c](#)) are allowed like in Dinka ([50b](#)). Without the applicative, the sentence becomes ungrammatical ([51d](#)).

- (51) a. í-pír-ì ákìrù
 3-stir.PRS-TAM Akiru.NOM
é-è-gètʃ.
with-M-wooden.whisk. **INS**
 b. ***é**-è-gètʃ í-pír-ì
with-M-wooden.whisk. **INS** 3-stir.PRS-TAM
 ákìrù.
 Akiru.NOM
 c. é-gètʃ í-pír-ì-**à**
 M-wooden.whisk.DEFL 3-stir.PRS-TAM- **APPL**
 ákìrù.
 Akiru.NOM
 d. *é-gètʃ í-pír-ì ákìrù.
 M-wooden.whisk.DEFL 3-stir.PRS-TAM Akiru.NOM
 ‘Akiru is stirring with a wooden whisk.’

Like PPs, manner adverbs are not allowed in the pre-verbal domain in Kipsigis ([52](#)) or Turkana ([53b](#)). Unlike with PPs, modifying the verb does not save the construction ([53c](#)).

- (52) ***mù:tjà** kó Ø-kèt-é kíbê:t.
slowly TOP 3-drive-IPFV Kibeet.NOM
Slowly, Kibeet drives.' (Kipsigis, [Driemel & Kouneli 2022](#): 7)

- (53) a. é-kér-ì ákìrù à:típéjì.
 3-run.PRS-TAM Akiru.NOM quickly
 b. *à:típéjì é-kér-ì ákìrù.
 quickly 3-run.PRS-TAM Akiru.NOM
 c. *à:típéjì é-kér-é-à ákìrù.
 quickly 3-run.PRS-TAM-APPL Akiru.NOM
 'Akiru is quickly running.'

Unlike manner adverbs, temporal adverbs can precede the verb in Turkana. However, if temporals are in the pre-verbal domain, an applicative is obligatory on the verb (54b). Otherwise, the sentence is ungrammatical (54c).

- (54) a. í-múdz-ì ákàjì é-mútǵélé tóxóná.
 3-eat.PRS-TAM Akai.NOM M-rice.DEFL now
 b. tóxóná í-múdz-ì-à ákàjì é-mútǵélé.
 now 3-eat.PRS-TAM-APPL Akai.NOM M-rice.DEFL
 c. *tóxóná í-múdz-ì ákàjì é-mútǵélé.
 now 3-eat.PRS-TAM Akai.NOM M-rice.DEFL
 'Akai is eating rice now.'

Furthermore, Bossi & Diercks (2019) claim for Kipsigis that temporal adverbs have nominal attributes since they can act as a subject in a copular sentence:

- (55) Koo beetuut nɛ-mie amut.
 PST day REL.SG-good.SG yesterday
 'Yesterday was a good day.' (Kipsigis, Bossi & Diercks 2019: 9)

The following data support this claim for Turkana. In Turkana temporal adverbs can be subjects in copula constructions (56) while manner adverbs cannot (57). Thus, temporal adverbs do not refute the claim that only nominals can appear in the pre-verbal domain.

- (56) èdʒók tóxóná.
 good now
 'Now is good.' (meaning: *now is a good time*)
 (57) *èdʒók átípéjì.
 good quickly

‘Quickly is good.’

Locatives are as well allowed in the pre-verbal domain but they need to be introduced by an applicative. In addition, the locative loses its gender marker, when it moves to the pre-verbal domain (58b). Otherwise, it will be ungrammatical (58c).

- (58) a. é-kér-ì ákìrù **á-nà-mòni**.
 3-run.PRS-TAM Akiru.NOM **F-LOC-forest**
- b. **nà-mòni** é-kér-é-**à** ákìrù.
 LOC-forest 3-run.PRS-TAM-**-APPL** Akiru.NOM
- c. ***á-nà-mòni** é-kér-ì ákìrù.
 F-LOC-forest 3-run.PRS-TAM Akiru.NOM
 ‘Akiru is running in the forest.’

Furthermore, the wh-element *álùwàè* (‘where’) has a preposition when it is in the post-verbal domain. This preposition is not allowed in the pre-verbal domain independently of whether the verb has an applicative or not. *álùwàè* (‘where’) is only allowed in the pre-verbal domain without the preposition and with an applicative marker on the verb. Without the applicative marker, the sentence is also ungrammatical.

- (59) a. é-kér-ì ákìrù **k**-álùwàè?
 3-run.PRS-TAM Akiru.NOM **PREP**-where
- b. ***k**-álùwàè é-kér-é-**à** ákìrù?
 PREP-where 3-run.PRS-TAM-**APPL** Akiru.NOM
- c. ***k**-álùwàè é-kér-ì ákìrù?
 PREP-where 3-run.PRS-TAM Akiru.NOM
- d. **álùwàè** é-kér-é-à ákìrù?
 where 3-run.PRS-TAM-**APPL** Akiru.NOM
- e. ***álùwàè** é-kér-ì ákìrù?
 where 3-run.PRS-TAM Akiru.NOM
 ‘Where is Akiru running?’

While the status of the locatives is a bit more unclear than the rest of the data, they still follow the trend of only nominals in the pre-verbal domain.

2.7.4 No Overt Case Marking

As stated in 2.4.2, ‘no overt case marking in the pre-verbal domain’ has been accounted for several East-African languages, especially Nilotic (König 2006, 2008; Handschuh 2014). Subjects that move to the pre-verbal domain lose their nominative case and get default case:

- (60) a. í-mùdž-ì á-bèrò á-kíríŋ.
3.-eat.PRS-TAM F-woman.NOM F-meat.DEFL
- b. á-bérú í-mùdž-ì á-kíríŋ.
F-woman.DEFL 3-eat.PRS-TAM F-meat.DEFL
'The WOMAN is eating meat.'

This has also been observed for Kipsigis (61) and Dinka (62).

- (61) a. kímpé:t kó kà-ø-ám kíbê:t.
ugali TOP PST-3-eat Kibeet.NOM
- b. kíbê:t kó kà-ø-ám kímpé:t.
Kibeet TOP PST-3-eat ugali
'Kibeet ate Ugali.' (Kipsigis, [Driemel & Kouneli 2022](#): 6-7)
- (62) a. àyén à-càm cuŋin nè pǎal.
Ayen 3SG-eat.SV food PREP knife
- b. cuŋin à-céem àyén nè pǎal.
food 3SG-eat.OV Ayen.GEN with knife
'Ayen is eating food with a knife.' (Dinka, [van Urk 2015](#): 61)

Adjuncts like PPs are not allowed in the pre-verbal domain (63a) and instead can only appear there as a bare noun in default case with an applicative on the verb (63b):

- (63) a. *á-ŋá-kàn-kèŋ í-mùdž-ì ákìrù.
with-PL.F-hand.INST-her 3-eat.PRS-TAM akiru.NOM
- b. ŋá-kàn-kèŋ í-mùdž-ì-á ákìrù.
PL.F-hand.DEFL-her 3-eat.PRS-TAM-APPL akiru.NOM
- c. *ŋá-kàn-kèŋ í-mùdž-ì ákìrù.
PL.F-hand.DEFL-her 3-eat.PRS-TAM akiru.NOM
'Akiru is eating with her hands.'

We have already seen the same pattern earlier for Dinka:

- (64) a. àyén à-càm cuŋin nè pǎal.
Ayen 3SG-eat.SV food PREP knife
- b. pǎal à-céem-è àyén cuŋin.
knife 3SG-eat.OBLV Ayen.GEN food
'Ayen is eating food with a knife.' (Dinka, [van Urk 2015](#): 74)

2.7.5 Mixed A- & $\bar{A}$ -properties

It is well established that long-distance movement such as focus and wh-movement ($\bar{A}$ -movement) and more local movement such as passivisation and raising (A-movement) have different properties (Chomsky 1977, 1981b). These properties are summarised in the following table and will be further discussed in more detail.

A-properties	$\bar{A}$ -properties
Local	Long distance
Restricted to nominals	Not restricted to nominals
No Reconstruction for Principle C	Reconstruction for Principle C
No Weak Crossover	Weak Crossover
New antecedents for anaphors	No new antecedents for anaphors
No parasitic gap licensing	Parasitic gap licensing

Table 2.7: A- & $\bar{A}$ properties (van Urk 2015: 23)

This clear distinction has only recently been challenged by van Urk (2015) who showed that movement to the pre-verbal domain in Dinka has mixed A- & $\bar{A}$ -properties. Same is true for Turkana which will be discussed in the following paragraphs. I first explore the A-properties of that position and then the $\bar{A}$ -properties⁶.

While $\bar{A}$ -movement is not restricted to nominals (65a), A-movement is (65b) as the following examples show:

- (65) a. [How proud of Mary] is Anna __?
 b. *[How proud of Mary] seems [__ to be Anna]?

I have already shown in detail in 2.7.3 that only nominals can move to the pre-verbal domain in Turkana but also in Dinka and Kipsigis. This is a typical A-property.

A- and $\bar{A}$ -movement also differ in respect to weak cross over (WCO). WCO arises when a binding relationship cannot be reconstructed after movement.

- (66) *Binding*: (Chomsky 1995: 93):
 α binds β if α c-commands β and α, β are conindexed.

Thus, a sentence becomes ungrammatical after movement when the binding relation cannot be reconstructed, resulting in WCO effects.

⁶However, parasitic gaps are hard to test, since Turkana, like Dinka (van Urk 2015: 57), does not allow them.

$\bar{A}$ -movement exhibits WCO effects (Postal 1971; Wasow 1972), meaning there is no reconstruction for binding after movement (67a). On the other hand A-movement shows no WCO effects (67b), therefore there is reconstruction for binding relations such as variable binding. While $\bar{A}$ -movement cannot establish new binding relations after movement, A-movement still allows binding after movement (Ruys 2004; Safir 2019).

- (67) a. ***Which woman_i** does **her_i** mother believe ___ is smart? ($*\forall > \exists$)
 b. **Which woman_i** seemed to **her_i** mother ___ to be smart? ($\forall > \exists$)

Turkana can also show WCO effects with quantifier raising. (68) shows that the universal quantifier *dà:ŋ* (every) can bind pronouns.

- (68) a. é-nú-ì **á-bèrù** **dà:ŋ_i** í-kókú
 3-hug-TAM **F-woman-NOM every.NOM** N-child-DEFL
kéŋ_{i/k}.
3.POSS.DEFL
 ‘Every woman is hugging her child.’ ($\exists > \forall / \forall > \exists$)
 b. é-nú-ì **á-bèrù** **dà:ŋ_i** í-kókú
 3-hug-TAM **F-woman-NOM every.NOM** N-child-DEFL
ání_{i/k}?
which..DEFL
 ‘Every woman is hugging which child?’ ($\exists > \forall / \forall > \exists$)

Moving the possessive pronoun leads to WCO effects. The quantifier cannot bind the pronoun once the pronoun moved as shown in the following example.

- (69) a. í-kókú **kéŋ_{*i/k}** é-nú-ì
 N-child-DEFL **3.POSS.DEFL** 3-hug-TAM
á-bèrù **dà:ŋ_i** ____.
F-woman-NOM every.NOM
 ‘Her child, every woman is hugging.’ ($\exists > \forall / * \forall > \exists$)
 b. í-kókú **ání_{*i/k}** é-nú-ì
 N-child-DEFL **which..DEFL** 3-hug-TAM
á-bèrù **dà:ŋ_i** ____?
F-woman-NOM every.NOM

‘Which child is every woman hugging?’ ($\exists > \forall$ / $*\forall > \exists$)

However, unlike in English (67a), WCO effects can be obviated in Turkana by fronting the binder. There are no WCO effects when the binder is moved to the pre-verbal domain in Turkana as shown in (70a) for focus movement and in (70b) for wh-movement. Thus, Turkana can reconstruct for variable binding which is a typical characteristic of A-movement.

- (70) a.

á-bérú	dá:ŋ_i
F-woman-DEFL	every.DEFL

é-nú-ì	í-kòkù
3-hug-TAM	N-child-NOM

kèŋ_{i/k}	—.
3.POSS.NOM	

‘Every woman, her child is hugging.’ ($\exists > \forall$ / $\forall > \exists$)
- b.

á-bérú	ání_i
F-woman-DEFL	which.DEFL

é-nú-ì	í-kòkù
3-hug-TAM	N-child-NOM

kèŋ_{i/k}	—?
3.POSS.NOM	

‘Which woman, her child is hugging?’ ($\exists > \forall$ / $\forall > \exists$)

The same picture emerges in Dinka. The universal quantifier in Dinka can bind pronouns (71a) but fails to bind it when the pronoun is raised (71b):

- (71) a.

dhùk ébén_i	à-cé	thòk-dè_{i/k}	t̩iŋ.
boy every	3S-PRF.SV	goat.CS-	SG.3SG

see.NF
‘Every boy has seen his goat.’ ($\exists > \forall$ / $\forall > \exists$)
- b.

thòk-dè_{i/k}	à-cé	dhùk ébén_i	kâac.
goat.CS-	SG.3SG	3S-PRF.SV	boy every

bite.NF
‘His goat has bitten every boy.’ ($\exists > \forall$ / $*\forall > \exists$)
(Dinka, van Urk 2015: 110)

Like in Turkana, the universal quantifier (72a) or the wh-element (72b) can still bind the pronoun after being moved to the pre-verbal domain. Thus, no WCO effects occur.

- (72) a.

dhùk ébén_i	à-c̩i	thòk-dè_{i/k}	—	kâac.
boy every	3S-PRF.OV	goat.CS-	SG.3SG	bite.NF

‘Every boy, his goat bit.’ ($\exists > \forall$ / $\forall > \exists$)

- b. yè **dhùŋ-ó_i** c̣ií thók-**dè_{i/k}** — kâac?
 be **boy.SC-which** PRF.OV goat.CS-**SG.3SG** bite.NF
 ‘Which boy did his goat bite?’ (∃ > ∀ / ∀ > ∃)
 (Dinka, [van Urk 2015: 110](#))

The pre-verbal domain in Turkana shows the following typical attributes for A-movement: only nominals are allowed in the pre-verbal position in Turkana and Turkana allows reconstruction for variable binding or in other words, there are now WCO effects when an element raises to the pre-verbal position. This aligns with the observations about the pre-verbal position in Dinka ([van Urk 2015](#)).

However, the pre-verbal domain does not only have A-properties but also $\bar{A}$ -properties. While A-movement is more local and does not allow movement across several phrases (73a), $\bar{A}$ -movement permits long distance movement across several phrases (73b).

- (73) a. **Who** did Kim say [that Alex thinks [that Sam likes __]]?
 b. ***Those people** were said [that Alex thinks [that Sam likes __]].
 ([van Urk 2015: 27](#))

Turkana allows movement across several phrases to the pre-verbal domain as demonstrated in the following example:

- (74) [_{CP} **ŋáé** é-jén-ì ákìrù [_{CP} (átàmàr)
who.DEFL 3-know.PRS-TAM Akiru.NOM that
 é-mín-á ákàjì __]]?
 3-love.PRS-TAM Akai.NOM
 ‘Who does Akiru know that Akai loves?’

This is also the case in Dinka. Movement to the pre-verbal domain can cross clause boundaries:

- (75) **cũin** à-yàa tàak [_{CP} kẹ cém áyèn __].
food 3S-HAB.1.SG thinkNF C eat.OV Ayen.GEN
 ‘The food, I think Ayen is eating.’ (Dinka, [van Urk 2015: 95](#))

Another property of $\bar{A}$ -movement is reconstruction for Principle C which states that R-expressions must be free and cannot be bound. The following example is ungrammatical because the R-expression *Anna* is bound by the pronoun *she* which is a violation of Principle C.

(76) *She_i saw Anna_i in the mirror.

Certain types of movement are able to reconstruct principle C, meaning that the sentence remains ungrammatical after movement because the Principle C violation persists. While A-movement does not reconstruct principle C (77a), $\bar{A}$ -movement does (77b):

- (77) a. This side of **Anna_i** seems to **her_i** to be hidden.
 b. *Which friend of **Anna_i** is **she_i** calling?

To determine whether this test applies to Turkana, we first need to establish whether Turkana exhibits Principle C effects in general. The following example demonstrates that it does. *Akiru* cannot be bound:

- (78) *é-pár-ìt **nèð_i** [ná-páç kénj á
 3-call-TAM **she.NOM** F-friend.DEFL 3.POSS.DEFL PREP
àkírù_i]?
Akiru.GEN
 ‘She_i is calling a friend of Akiru_i.’

Turkana shows reconstruction for principle C. The sentence remains ungrammatical even after the R-expression has been moved.

- (79) *[ná-páç kénj ání á **àkírù_i**]
 F-friend.DEFL 3.POSS.DEFL which.DEFL PREP **Akiru.GEN**
 é-pár-ìt **nèð_i**?
 3-call-TAM **she.NOM**
 ‘Which friend of Akiru_i is she_i calling?’

Dinka differs in that respect as it does not reconstruct for principle C. Like Turkana, Dinka shows Principle C violations:

- (80) ***yêen_i** à-cé mánh è **màyén_i** côl.
He 3S-PRE.SV brother.CS P **Mayen** call.NF
 ‘He_i has called a brother of Mayen_i.’ (Danika, [van Urk 2015](#): 113)

But unlike in Turkana, Dinka does not reconstruct for Principle C after movement and therefore Principle C is not violated after movement:

- (81) [_{DP} mánh è máyèn kù àyén_i] cìikè_i —
 brother.CS P Mayen.GEN and Ayen PRF.3PL
 t̩iŋ.
 seeNF
 ‘The brother of Mayen and Ayen_i, they_i have seen.’

Thus, the pre-verbal domain in Turkana shows the following typical properties of $\bar{A}$ -movement: Long distance movement is allowed and Principle C is reconstructed.

To summarise, the pre-verbal position is not a clear A- or $\bar{A}$ -position. Instead, it has properties of both. The following table outlines the A-properties and $\bar{A}$ -properties of the pre-verbal position.

A-properties	$\bar{A}$ -properties
Restricted to nominals	Long distance movement
No WCO effects	Reconstruction for Principle C

Table 2.8: A- & $\bar{A}$ properties of the pre-verbal position in Turkana

2.8 Conclusion

This chapter has provided a comprehensive overview of key grammatical properties of Turkana. I demonstrated that Turkana exhibits canonical VSO word order with a dedicated pre-verbal focus position. This focus position is restricted to nominals, which consequently lose their case marking when occupying this position. Furthermore, the position exhibits both properties of an A- and an $\bar{A}$ -position.

Additionally, I showed that this language is severely understudied, especially in the theoretical literature. I discovered an applicative marker which has not been mentioned before in the literature. By presenting this data, I aim to address this gap and contribute to the integration of Turkana into broader theoretical linguistic debates.

Chapter 3

Wh-Movement in Turkana

3.1 Introduction

In this chapter, I examine the nature of the position of wh-elements in Turkana. Turkana permits wh-elements to appear either in-situ (283a) or ex-situ (283b) but not simultaneously at the same time in a simple clause (82c).

- (82) a. é-mín-à ákàjì ŋáé ?
 3-like-TAM Akai.NOM who.DEFL
- b. ŋáé é-mín-à ákàjì ?
 who.DEFL 3-like-TAM Akai.NOM
- c. *ŋáé é-mín-à ákàjì ŋáé ?
 who.DEFL 3-like-TAM Akai.NOM who.DEFL
 ‘Who does Akai like?’

I show that wh in-situ is true wh in-situ and does not have any covert movement. In contrast, wh ex-situ results from movement rather than the wh-element being externally merged in a higher position.

Furthermore, I discuss the limits of long-distance licensing and long-distance movement of focal and of wh-elements in Turkana. Long-distance licensing of wh-elements in-situ is allowed if only one CP is crossed. However, it becomes ungrammatical if long-distance licensing crosses two CPs. Wh-elements can move to the next higher CP without leaving a copy. However, it becomes ungrammatical to move via more than two CPs without leaving a copy. This is a challenge for the standard view on locality and phasality (Chomsky 2000, 2001) Under a traditional phasal approach, either wh in-situ with one CP should already be ungrammatical because Turkana has

true wh in-situ and vP is a phase or there should be no restriction at all.

The chapter is structured as follows: In 3.2, I examine the state of wh in-situ in Turkana. In 3.3 show that wh ex-situ involves movement in Turkana. In 3.4, I explore the limits of long-distance licensing and long-distance movement in Turkana and show how this is a challenge for standard phase theory. In 3.5, I discuss the possible shapes of copies. 3.6 concludes the chapter.

3.2 Wh In-Situ

The phenomenon of wh in-situ languages has been the subject of extensive discussion in the literature. Languages which display wh in-situ can be categorised into two types: those with true wh in-situ and those exhibiting covert wh-movement (cf. Pesetsky 1987; Bošković 2000; Polinsky & Potsdam 2001; Cheng 2003; İşsever 2009; Polinsky 2015; Demirok 2017; Aravind 2018; Amaechi & Georgi 2021). In true wh in-situ languages, wh-elements remain in their original position and do not undergo movement. Conversely, in languages with covert wh-movement, the wh-element is syntactically displaced but not phonetically realised at the surface level. Although these two types may appear similar, their syntactic implications are quite distinct.

To determine whether a language features covert movement, we must investigate the potential movement of the wh-element. A standard test for movement involves examining island sensitivity, as islands typically prohibit extraction (cf. Ross 1967; Huang 1982; Kayne 1983; Cinque 1990; Rizzi 1990; Boeckx 2012).

(83) *What_i did you wonder [what Anna ate _i] ?

If the wh-element genuinely remains in-situ without moving out of an island, the sentence should be grammatical. However, if the wh-element does move out of the island, the sentence becomes ungrammatical. Thus, if an in-situ wh-element is sensitive to island violations, this suggests that it has covertly moved and, consequently, violated an island constraint.

This is observed in languages like Tsez (Polinsky & Potsdam 2001; Polinsky 2015) and Igbo (Amaechi & Georgi 2021) both of which

permit wh-elements to remain in-situ:

- (84) a. uʒ-ä kid-be-r **jebi** teʎ-ä?
boy-ERG girl-OS-LT **what.ABS.IV** give-PST.WIT.INTERR
'What did the boy give to the girl?' (Tsez, [Polinsky 2015](#):
183)

- b. Òbí hù-rù **ònyé** n'-áhíá?
 Obi saw **who** PREP-market
 Lit.: 'Who saw Obi at the market?' (Igbo, Amaechi & Georgi 2021: 300)

However, in both languages, the in-situ wh-elements are sensitive to island constraints. The following examples illustrate for Tsez (85a) and Igbo (85b) that it is ungrammatical to have a wh-element in-situ within an adjunct clause, indicating that the wh-element must have covertly moved out of the adjunct clause, thereby violating an island constraint:

- (85) a. *[beɬi-λ' **jebi** b-äk'-äsi yäɬ-zay]
 chase.SUPER.ESS **who** IPL-GO-RES be.PRS-WHILE
 ɬirbabi bayä?
 guests came
 Int: 'The guest arrived when who were away hunting?'
 (Tsez, Polinsky 2015: 192)
- b. *àdá hù-rù Òbí [túpú ọ zúrú **íní** n'-áhíá?]
 Ada saw Obi before she buy **what** PREP-market
 Lit.: 'Ada saw Obi before she bought what at the market?'
 (Igbo, Amaechi & Georgi 2021: 312)

In contrast, certain languages like Turkish do not exhibit island violations with wh-elements in-situ within island constructions. Turkish allows wh-elements to remain in-situ, as illustrated in the following example:

- (86) sen-i **kim** sec-iyor?
 you-ACC **who.NOM** love-PROG.3.SG
 'Who loves you?' (Turkish, İşsever 2009: 106)

Unlike in Tsez and Igbo, Turkish permits wh-elements in-situ even within in adjunct clauses. This suggests that no covert movement occurs and that the wh-element genuinely remains in-situ.

- (87) Misafir-ler [**kim** avlanmaya git-miş-ken] hel-di?
 guest-PL **who** hunt.NMLZ go.PRF-WHILE come-PST
 'The guests arrived when who was away hunting?' (Turkish, Demirok 2017: 5)

To demonstrate that adjunct clauses are indeed islands, Demirok (2017) shows that generally movement out of adjunct clauses is prohibited since it violates island constraints:

- (88) *_{[RC} Misafir-ler-in [*e_i* avalnmaya git-miş-ken]
 guest-PL-GEN *e* hunt.NMLZ go.PRF-WHILE
 geldikleri] *adam_i*.
 come.NMLZ *man*
 Int: ‘The man_i such that the guests arrived when he_i was
 away hunting.’ (Turkish, Demirok 2017: 6)

Turkana is another language which also allows wh-elements to remain in-situ. When the wh-elements stay in-situ, the VSO word order discussed in 2.6 is followed and does not change the prominence hierarchy. If the wh-element is the subject of the sentence it will follow the verb (89a). If the wh-element is the object of the sentence it will follow the subject (89b).

- (89) a. í-mudʒ-ì *ɲàè* á-kírín?
 3-eat.PRS-TAM *who.NOM* F-meat.DEFL
 ‘Who is eating meat?’
 b. í-múdʒ-ì ákìrù *ɲó?*
 3-eat.PRS-TAM Akiru.NOM *what.DEFL*
 ‘What is Akiru eating?’

The same can be seen in d-linked wh-questions. If the wh-element is the subject of the sentence it will follow the verb (90a). If the wh-element is the object of the sentence it will follow the subject (90b).

- (90) a. í-múdʒ-é-tè *ɲí-tùɲà* àlù
 3-eat.PRS-TAM-PL *PL-people.NOM which.PL.NOM*
 á-kírín?
 meat.DEFL
 ‘Which people are eating meat?’
 b. í-té ákìrù *ɲí-tùɲá* álù?
 3-see.PRS Akiru.NOM *PL-people.DEFL which.DEFL.PL*
 ‘Which people is Akiru seeing?’

Turkana pairs with Turkish in the respect that it shows true wh in-situ. Turkana does not exhibit island violations with wh-elements

in-situ. Movement out of islands like adjunct clauses is prohibited in Turkana as shown in the following example.

- (91) *_{[CP} **é-pídín** à-bùn-í ákìrù nàðá:
M-window.DEFL 3-come.PST-TAM Akiru.NOM at.a.time
_{[CP} n-á-ḡár-è-à ákàjì]].
 when-3-open-TAM-APPL Akai.NOM
 Int: 'Akiru came at a time when Akai opened a window / gate / door.'

However, wh-elements are permitted in-situ in adjunct clauses which indicates that no covert wh-movement takes place:

- (92) _{[CP} à-bùn-í ákìrù nàðá: _{[CP}
 3-come.PST-TAM Akiru.NOM at.a.time
 n-á-ḡár-è-à ákàjì **nó**]].
 when-3-open-ITV-TAM akai.NOM **what.DEFL**
 'Akiru came at a time when Akai opened what?'

Wh in-situ languages can be classified into two types: those with true wh in-situ and those exhibiting covert wh-movement. By finding signs of movement, it can be determined into which category a language falls. In this section, I have shown that Turkana demonstrates true wh in-situ with wh-elements remaining genuinely in-situ and not undergoing covert movement.

3.3 Wh Ex-Situ

In the previous section, I demonstrated that wh in-situ in Turkana does not involve covert movement. In this section, I will turn to the properties of wh-elements in ex-situ positions, showing that Turkana wh ex-situ constructions are not concealed pseudo-clefts but a result of movement.

Turkana allows wh-elements to appear not only in-situ but also in the pre-verbal domain. As discussed in 2.7.4, when a wh-element is the subject of a sentence and occupies the pre-verbal domain, it loses its nominative case marking and instead appears in default case:

- (93) **ḡáé** í-mudḡ-ì á-kírín?
who.DEFL 3-eat.PRS-TAM F-meat.DEFL
 'Who is eating meat?'

3.3.1 Pseudo-Cleft

van Urk (2018) suggests for the Western-Nilotic language Dinka that the pre-verbal wh-construction is a wh-cleft. Unlike Turkana, Dinka has an overt copula in the pre-verbal domain:

- (94)

yè	nà	[_{CP} cīi	Bôl	tīiŋ	] ?
be.3.SG	who	PRF.OV	Bol.GEN	see.NF	

 ‘Who has Bol seen?’ (Dinka, van Urk 2018: 945)

However, Turkana does not have an overt copular verb (95).

- (95) é-bùθ ájóŋ.
 1-beautiful I.DEFL
 ‘I am beautiful.’

Thus, Turkana may exhibit a concealed pseudo-cleft structure with a silent copular verb, a phenomenon previously identified in several Austronesian languages like Malagasy (Paul 2001; Potsdam 2006), Marshallese (Willson-Sturman 2014), Malay (Aman, Cole & Hermon 2009) or Tuvaluan (Besnier 2002; Potsdam & Polinsky 2011). Similar to Turkana, these languages display both wh-in-situ (96a, 97a) and ex-situ constructions (96b, 97b), as seen in Malagasy and Marshallese.

- (96) a. nividy cary

 Rasoa?
 buy rice

 Rasoa
 b.

 no nividy cary Rasoa?

 PRT buy rice Rasoa
 ‘Where did Rasoa buy rice’ (Malagasy, Potsdam 2006: 2155)

- (97) a. Kidu ko re=kar køkure

?
 dog the.PL.NH 3PL.ARG=PST destroy

 b.

 eo kidu ko re=kar køkure?

 the.SG dog the.PL.NH 3PL.ARG=PST destroy
 ‘What did the dogs destroy?’ (Marshallese, Willson-Sturman 2014: 13)

It has been argued for these languages that the position of the initial wh-element is not a result of movement but a concealed pseudo-cleft. The initial wh-element is a wh-predicate and the rest of sentence is a headless relative clause in subject position. Thus, the wh-element

does not move to the pre-verbal domain but is base-generated there. The following structure has been proposed for (97b):

- (98) [FP [ta_i]_{predicate} F [IP [DP/headless RC eo kidu ko
what the.SG dog the.PL.NH
re=kar køkure pro]_{subject} I t_i]] ?
3PL.ARG=PST destroy
‘What did the dogs destroy?’ (Marshallese, Willson-Sturman
2014: 15)

Whether constructions with an initial *wh*-element are pseudo-clefts can be tested even in the absence of an overt copula. Since the *wh*-element does not undergo movement, there should be no evidence of movement, such as island effects. This pattern is observed in Austronesian languages, where relative clauses are islands, and extraction from them results in an island violation. However, extracting a *wh*-element out of an island is grammatical in Marshallese, suggesting that no movement has occurred.

- (99) ta_i eo [leddik eo me e=j kōṇake
what the.SG girl the.SG that 3SG.ARG=PRS weat-OBJ
 —_i] e=j iukkure nabōj?
 3SG.ARG=PRS play outside
 ‘The girl that is wearing what is playing outside?’ (Mar-
 shallese, Willson-Sturman 2014: 19)

Another piece of evidence against movement is the presence of resumptive pronouns. If resumptive pronouns are regular pronouns following McCloskey (2002) and not spelled-out copies, they should not be able to be base generated at the same position as the wh-element. Willson-Sturman (2014) argues for Marshallese, that resumptive pronouns are regular pronouns. These pronouns can co-occur with wh-elements, which indicates that the wh-element is not base-generated low in the clause, as this position is already occupied by the resumptive pronoun.

- (100) **wōn** eo Libokmeto e=j iukurre ñae **e**?
who the.SG Libokmeto 3.SG=PRS play against **3.SG**
 ‘Who is Libokmeto playing against?’ (Marshallese,
 Willson-Sturman 2014: 18)

While these are clear indications, that in Austronesian, constructions with an initial *wh*-element are pseudo-clefts, I will show that this is not the case in Turkana. These constructions behave the opposite in Turkana and are thus not concealed pseudo-clefts but a result of movement.

3.3.2 Movement

I will begin by testing aforementioned island constraints in constructions with an initial *wh*-element in Turkana. Violations on island constraints are an indicator for movement as shown before. Testing island violations in Turkana with only one CP is somewhat challenging. While extracting the *wh*-element results in ungrammaticality (101b, 101c), it cannot be ruled out that this ungrammaticality arises for independent reasons. (101b) is also ungrammatical because Turkana permit only nominals in the pre-verbal domain and prohibits prepositions in this position (see 2.7.3). (101c) shows that preposition stranding is not allowed in Turkana.

- (101) a. í-té ákìrù [DP à-pàè ká **ḡàè**]?
 3-see.PRS Akiru.NOM F-friend.DEFL of who.GEN
- b. ***ká ḡàè_i** í-té ákìrù [DP à-pàè
of who.GEN 3-see.PRS Akiru.NOM F-friend.DEFL
 —_i]?
- c. ***ḡàè_i** í-té ákìrù [DP à-pàè **ká**
who.DEFL 3-see.PRS Akiru.NOM F-friend.DEFL **of**
 —_i]?

‘Akiru is seeing a friend of who?’

However, the picture becomes clearer upon looking at different kind of islands. As shown in (91), moving out of adjunct clauses leads to ungrammaticality because of an island violation. This also holds, when a wh-element moves out of the adjunct clause.

- (102) *_{[CP} **ḡó** à-bùn-í ákìrù nàḡá: _{[CP}
what.DEFL 3-come.PST-TAM Akiru.NOM at.a.time
 n-á-ḡár-è-à ákàjì]]?
 when-3-open-TAM-APPL Akai.NOM
 Int: ‘Akiru came at a time when Akai opened what?’

This holds not only for temporal adjunct clauses but for all different kind of adjunct clauses. Wh-elements in the pre-verbal domain are also prohibited when they are extracted from a causal adjunct clause:

- (103) a. _{[CP} é-kíyán-á ákìrù _{[CP} kòtèré
 3-laugh.PRS-TAM Akiru.NOM because
 é-límó-kín-ít ákàjì ḡéḡí **ḡó**]]?
 3-tell-DAT-TAM Akai.NOM 3.SG.DEFL **what.DEFL**
- b. *_{[CP} **ḡó** é-kíyán-á ákìrù _{[CP}
what.DEFL 3-laugh.PRS-TAM Akiru.NOM
 kòtèré é-límó-kín-ít ákàjì ḡéḡí]]?
 because 3-tell-DAT-TAM Akai.NOM 3.SG.DEFL
 ‘Akiru is laughing because Akai is telling her what?’

These data indicate that the wh-element moves to the pre-verbal position and is not base generated there like in pseudo-cleft constructions

which are not sensitive to islands.

Another data point is resumption. Unlike Austronesian languages, Turkana does not allow resumptive pronouns which indicates that the *wh*-element was base generated in the subject position which can therefore not also be filled with a pronoun.

- (104) ***ḡáé** í-mudʒ-ì **ḡéðì** á-kírínʒ?
who.DEFL 3-eat.PRS-TAM **3.SG.NOM** F-meat.DEFL
 'Who is eating meat?'

Furthermore, the island data from above can also not be saved by a resumptive pronoun:

- (105) a. *_[CP] **ḡó** à-bùn-í ákìrù nàðá:
what.DEFL 3-come.PST-TAM Akiru.NOM at.a.time
_[CP] n-á-ḡár-è-à ákàjì
 when-3-open-TAM-APPL Akai.NOM
ḡó / ḡéðì]?
what.DEFL / 3.SG.DEFL
 Int: 'Akiru came at a time when Akai opened what?'
 b. *_[CP] **ḡó** é-kíyán-á ákìrù _[CP]
what.DEFL 3-laugh.PRS-TAM Akiru.NOM
 kótèré é-límó-kín-ít ákàjì ḡéðì
 because 3-tell-DAT-TAM Akai.NOM 3.SG.DEFL
ḡó / ḡéðì]?
what.DEFL / 3.SG.DEFL
 Int: 'Akiru is laughing because Akai is telling her what?'

Another piece of evidence suggesting that fronting elements involves movement comes from fronted adjuncts. Before discussing the complex movement data, I will explain the basic pattern of adjunct movement. As shown in 2.7.3, Turkana only allows nominals in the pre-verbal domain. When adjuncts are fronted, an applicative strategy takes place as prepositions are prohibited in the pre-verbal domain. Locatives are allowed in the pre-verbal domain but they lose their gender marker and the verb shows the obligatory applicative marker:

- (106) a. é-kér-ì ákìrù **á-nà-mòni**.
 3-run.PRS-TAM Akiru.NOM **F-LOC-forest**

- b. ***á-nà-mòni** é-kér-ì-(a) ákìrù.
F-LOC-forest 3-run.PRS-TAM-(APPL) Akiru.NOM
- c. **nà-mòni** é-kér-é-*(à) ákìrù.
LOC-forest 3-run.PRS-TAM-**APPL** Akiru.NOM
 'Akiru is running in the forest.'

This also holds for the locative question word *álùwàe* (*where*). If the question words appears in front of the verb, the applicative marker is obligatory:

- (107) **álùwàe** í-páp-*(à) ákìrù é-ká:ð?
where 3-work.PRS.TAM-**APPL** Akiru.NOM job.DEFL
 'Where is Akiru working?'

The applicative strategy is optional for the post-verbal domain for locatives. If the applicative is present, the locative in the post-verbal domain also loses its gender-marker (108).

- (108) é-pér-í-ò ákìrù **ló-síbìtár**.
 3-sleep.PRS-TAM-**APPL** Akiru.NOM **LOC-**hospital
 'Akiru is sleeping in the hospital.'

However, the applicative strategy is prohibited for locative questions in the post-verbal domain (109).

- (109) *í-páp-à ákìrù é-ká:ð **álùwàe**?
 3-work.PRS.TAM-**APPL** Akiru.NOM job.DEFL **where**
 'Where is Akiru working?'

For the locative question word to appear in the post-verbal domain, a preposition is required. The verb must stand without the applicative (110a). This preposition is, like all prepositions, not allowed in the pre-verbal position (110b).

- (110) a. í-pán-ì-(*a) ákìrù é-ká:ð **k-álùwàe**?
 3-work.PRS-TAM Akiru.NOM job.DEFL **PREP-where**
- b. ***k-álùwàe** í-pán-ì ákìrù é-ká:ð?
PREP-where 3-work.PRS-TAM Akiru.NOM job.DEFL
 'Where is Akiru working?'

Having presented the simple adjunct data, I turn now to the data that

shows that fronting adjuncts involves movement. In sentences with more than one CP, the following picture emerges: The applicative always appears on the verb of the clause from which the element has moved. Meaning, if the adjunct is base generated in the subordinate clause and fronted, the verb of the subordinate clause carries the applicative marker. Conversely, if the adjunct is base generated in the main clause and fronted, the applicative appears on the verb of the main clause. This indicates that the adjunct is not base-generated in the pre-verbal domain, when it appears there overtly. Let's examine this in greater detail.

Adjuncts provide an excellent testing ground for movement because they allow us to determine to which action the adjuncts refers. For instance, the question *Where does Anna think that Sarah is sleeping* can be interpreted in two ways: it can ask where Anna herself is located while she is thinking, or it can inquire about the location where Anna believes Sarah is sleeping. This distinction helps to test the syntactic movement of locative elements.

Locatives that refer to the action of the verb in the lower clause can be extracted from that CP, and only the verb in the lower CP carries the applicative marker. This indicates that the locative must have been in the lower clause at one point of the derivation and then moved:

- (111) [CP **álùwàè** í-tám-ì ádzìkòn [CP átàmàr
where 3-think.PRS-TAM Adjikon.NOM that
 í-páp-**à** ákìrù é-ká:ǝ]] ?
 3-work.PRS.TAM-**APPL** Akiru.NOM job.DEFL
 'Where does Adjikon think that Akiru is working?' (Mean-
 ing: *where is the working happening*)

On the other hand, if the fronted locative refers to the action verb of the main clause, the applicative marker is on the verb of the main clause:

- (112) [CP **álùwàè** í-tám-í-**à** ádzìkòn [CP
where 3-think.PRS-TAM-**APPL** Adjikon.NOM
 átàmàr í-páp-ì ákìrù é-ká:ǝ]] ?
 that 3-work.PRS-TAM Akiru.NOM job.DEFL
 'Where does Adjikon think that Akiru is working?' (Mean-

(113) [CP í-tám-ì ádʒìkòn [CP átàmàr
3-think.PRS-TAM-APPL Adjikon.NOM that
áìùwàè í-páp-à ákìrù é-ká:ǝ]] ?
where 3-work.PRS-TAM-APPL Akiru.NOM job.DEFL
'Where does Adjikon think that Akiru is working?' (Mean-
ing: *where is the working happening*)

⁷I assume that *where* is not allowed in the postverbal domain without a preposition for independent reasons.

- (114) a. *_{[CP í-tám-ì ádzìkòn **álùwàè** _{[CP átàmàr}}
 3-think.PRS-TAM Adjikon.NOM **where** that
 í-páp- **à** ákìrù é-ká:ǝ]] ?
 3-work.PRS-TAM- **APPL** Akiru.NOM job.DEFL
 ‘Where does Adjikon think that Akiru is working?’ (Mean-
 ing: *where is the working happening*)
- b. _{[CP í-tám-ì ádzìkòn}
 3-think.PRS-TAM Adjikon.NOM
 á-kájí **ání** _{[CP átàmàr}
 F-house.DEFL which.F-DEFL that
 é-pér-í- **ò** ákìrù]] ?
 3-sleep.PSR-TAM- **APPL** Akiru.NOM
 ‘In which house does Adjikon thinks that Akiru is sleep-
 ing?’ ⁸ (Meaning: *where is the sleeping happening*)

On the other hand, if the locative is base generated in this position. There is no applicative present and the locative appears with a preposition:

- (115) _{[CP í-tám-ì ádzìkòn **k-álùwàè** _{[CP}}
 3-think.PRS-TAM-APPL Adjikon.NOM **PREP-where**
 átàmàr í-pán-ì ákìrù é-ká:ǝ]] ?
 that 3-work.PRS-TAM Akiru.NOM job.DEFL
 ‘Where does Adjikon think that Akiru is working?’ (Mean-
 ing: *where is the thinking happening*)

A similar pattern can be observed for temporals. Temporals in the pre-verbal domain require an applicative ⁹:

- (116) a. **órí** í-páp- **à** ákìrù é-ká:ǝ?
 when 3-work.PRS-TAM- **APPL** Akiru.NOM job.DEFL
- b. ***órí** í-pán-ì ákìrù é-ká:ǝ?
 when 3-work.PRS-TAM Akiru.NOM job.DEFL
 ‘When is Akiru working?’

However, in the postverbal domain, temporals differ from locatives:

⁸We have seen Akai before as a name. This is due to the fact that the Turkana naming convention is the place of birth plus the respective gender marker.

⁹I follow (Bossi & Diercks 2019) who claim for Kipsigis (Southern Nilotic) that temporals have a nominal character and are therefore allowed in the pre-verbal domain (cf. 2.7.3).

The applicative is always prohibited but there is also no additional preposition.

- (117) a. í-pán-ì ákìrù é-ká:ǎ **órí?**
 3-work.PRS-TAM Akiru.NOM job.DEFL **when**
 b. *í-páp-**à** ákìrù é-ká:ǎ **órí?**
 3-work.PRS.TAM-**APPL** Akiru.NOM job.DEFL **when**
 ‘When is Akiru working?’

Unlike locatives, temporals base generated in the lower clause seem not be able to be moved to the pre-verbal domain of the higher clause:

- (118) *_{[CP} **órí** í-tám-ì ádžìkòn _{[CP} átàmàr
when 3-think.PRS-TAM Adjikon.NOM that
 í-páp-**à** ákìrù é-ká:ǎ] ?
 3-work.PRS.TAM-**APPL** Akiru.NOM job.DEFL
 ‘When does Adjikon think that Akiru is working?’ (Meaning
when is the working happening)

The only possibility for the temporal to be at the front of sentence is if it is referring to the action of the higher verb and the higher verb carries an applicative marker:

- (119) _{[CP} **órí** í-tám-ì-**à** ádžìkòn _{[CP}
when 3-think.PRS-TAM-**APPL** Adjikon.NOM
 átàmàr í-pán-ì ákìrù é-ká:ǎ] ?
 that 3-work.PRS-TAM Akiru.NOM job.DEFL
 ‘When does Adjikon think that Akiru is working?’ (Meaning:
when is the thinking happening)

To express the meaning that the temporal refers to the lower verb, the temporal must stay in the lower CP, either in-situ (120a) or in the pre-verbal domain of the lower CP with the applicative only on the lower verb (120b).

- (120) a. _{[CP} í-tám-ì ádžìkòn _{[CP} átàmàr
 3-think.PRS-TAM Adjikon.NOM that
 í-pán-ì ákìrù é-ká:ǎ **órí]]** ?
 3-work.PRS.TAM Akiru.NOM job.DEFL **when**
 b. _{[CP} í-tám-ì ádžìkòn _{[CP} átàmàr **órí**
 3-think.PRS-TAM Adjikon.NOM that **when**

í-páp-à ákìrù é-ká:ǝ]] ?
 3-work.PRS.TAM-APPL Akiru.NOM job.DEFL
 ‘When does Adjikon think that Akiru is working?’ (Mean-
 ing: *when is the working happening*)

Another way to get that reading, is by moving the temporal to the higher CP in the post-verbal position. The verb of the lower clause carries an applicative marker indicating that the temporal originated in the lower clause:

(121) [_{CP} í-tám-ì ádzìkòn órí [_{CP} átàmàr
 3-think.PRS-TAM Adjikon.NOM when that
 í-páp-à ákìrù é-ká:ǝ]]
 3-work.PRS.TAM-APPL Akiru.NOM job.DEFL
 ‘When does Adjikon think that Akiru is working?’ (Meaning:
when is the working happening)

Without the applicative on the lower verb, the temporal refers to the action of the higher verb which indicates that the temporal did not originate in the lower clause:

(122) [_{CP} í-tám-ì ádzìkòn órí [_{CP} átàmàr
 3-think.PRS-TAM-APPL Adjikon.NOM when that
 í-pán-ì ákìrù é-ká:ǝ]]
 3-work.PRS-TAM Akiru.NOM job.DEFL
 ‘When does Adjikon think that Akiru is working?’ (Meaning:
when is the thinking happening)

Thus, the applicatives in the adjunct data show us if the adjunct has moved and where it was originally base generated.

To conclude, based on island, resumption, and adjunct data, I have shown that the pre-verbal wh-element moves to the pre-verbal position and is not base generated high up in a clause like it would be the case in pseudo-clefts. I will now turn to long-distance movement in Turkana and its limits.

3.4 Long-Distance Movement

I have established in the previous section that wh-elements which appear in-situ have no covert movement and wh-elements ex-situ

moved to this position. Furthermore, the wh-elements cannot occur at the same time.

- (123) a. é-mín-à ákàjì ɲáé?
 3-like-TAM Akai.NOM who.DEFL
- b. ɲáé é-mín-à ákàjì?
who.DEFL 3-like-TAM Akai.NOM
- c. *ɲáé é-mín-à ákàjì ɲáé?
who.DEFL 3-like-TAM Akai.NOM who.DEFL
 ‘Who does Akai like?’

3.4.1 Constructions with 2 CPs

This pattern also holds in longer constructions. With two CPs, the wh-element is allowed in-situ (124a) or ex-situ in the pre-verbal position in the lower (124b) or in the higher CP (124c). If the wh-element is in the pre-verbal domain of the lower CP, it appears after the complementiser *atamar* (‘that’) which is optional.

- (124) a. [_{CP} é-jén-ì ákìrù [_{CP} (átàmàr)
 3-know.PRS-TAM Akiru.NOM that
 é-mín-à ákàjì ɲáé]]?
 3-love.PRS-TAM Akai.NOM who.DEFL
- b. [_{CP} é-jén-ì ákìrù [_{CP} (átàmàr)
 3-know.PRS-TAM Akiru.NOM that
 ɲáé é-mín-à ákàjì]]?
 who.DEFL 3-love.PRS-TAM Akai.NOM
- c. [_{CP} ɲáé é-jén-ì ákìrù [_{CP}
 who.DEFL 3-know.PRS-TAM Akiru.NOM
 (átàmàr) é-mín-à ákàjì]]?
 that 3-love.PRS-TAM Akai.NOM
 ‘Who does Akiru know that Akai loves?’

While the wh-element is allowed in the post-verbal position of the lower CP, the wh-element is not allowed in the post-verbal domain of the higher CP¹⁰:

¹⁰I will discuss in the next chapter why assume that this is the post-verbal position of the higher CP and not another pre-verbal position of the lower CP.

- (125) *_{[CP} é-jén-ì ákìrù ḡáé _{[CP} (átàmàr)
 3-know.PRS-TAM Akiru.NOM who.DEFL that
 é-mín-à ákàjì]]?
 3-love.PRS-TAM Akai.NOM
 ‘Who does Akiru know that Akai loves?’

However, this is not a general observation that elements can not move to the post-verbal domain. I have shown before that there seems to be a discrepancy between what can land in that position and what cannot. This is repeated in the following examples. While the *wh*-word *álùwàḡ* (‘where’) cannot land in this position, the d-linked element *á-kájí ání* (‘which house’) can:

- (126) a. *<sub>[CP í-tám-ì ádzìkòn **álùwàè** _{[CP átàmàr}
 3-think.PRS-TAM Adjikon.NOM **where** that
 í-páp- **à** ákìrù é-ká:ǝ]] ?
 3-work.PRS-TAM- **APPL** Akiru.NOM job.DEFL
 ‘Where does Adjikon think that Akiru is working?’ (Mean-
 ing: *where is the working happening*)</sub>
- b. _{[CP í-tám-ì ádzìkòn}
 3-think.PRS-TAM Adjikon.NOM
á-kájí **ání** _{[CP átàmàr}
F-house.DEFL which.F-DEFL that
 é-pér-í- **ò** ákìrù]] ?
 3-sleep.PSR-TAM- **APPL** Akiru.NOM
 ‘In which house does Adjikon thinks that Akiru is sleep-
 ing?’ ¹¹ (Meaning: *where is the sleeping happening*)

Furthermore, an additional overt copy is allowed if none of the overt copies appear in-situ (127a). If one of the overt copies appears in-situ, the construction is ungrammatical (127b).

- (127) a. <sub>[CP **ɲáé** é-jén-ì ádzìkòn _{[CP}
who.DEFL 3-know.PRS-TAM Adjikon.NOM
 (átàmàr) **ɲáé** é-mín-à ákàjì]]?
 that **who.DEFL** 3-love.PRS-TAM Akai.NOM</sub>
- b. *<sub>[CP **ɲáé** é-jén-ì ákìrù _{[CP}
who.DEFL 3-know.PRS-TAM Akiru.NOM
 (átàmàr) é-mín-à ákàjì **ɲáé**]]?
 that 3-love.PRS-TAM Akai.NOM **who.DEFL**
 ‘Who does Akiru know that Akai loves?’</sub>

Having two overt copies in the same CP is also ungrammatical. The following examples demonstrate that with two overt copies in the lower CP (128a) and with two overt copies in the higher CP (128b).

¹¹We have seen Akai before as a name. This is due to the fact that the Turkana naming convention is the place of birth plus the respective gender marker.

- (128) a. *_{[CP} é-jén-ì ákìrù _{[CP} (átàmàr) **ḡáé**
 3-know.PRS-TAM Akiru.NOM that who.DEFL
 é-mín-à ákàjì **ḡáé**]]?
 3-love.PRS-TAM Akai.NOM **who.DEFL**
 b. *_{[CP} **ḡáé** é-jén-ì ákìrù
 who.DEFL 3-know.PRS-TAM Akiru.NOM
 ḡáé _{[CP} (átàmàr) é-mín-à ákàjì]]?
 who.DEFL that 3-love.PRS-TAM Akai.NOM
 ‘Who does Akiru know that Akai loves?’

The pattern remains the same when the wh-element is the subject. The wh-element can either stay in-situ in the lower CP with the difference that the wh-element carries nominative case. Furthermore, the in-situ wh-element is located in front of the object, following the VSO word order of Turkana (129a). The wh-element can also move to the pre-verbal domain of either the lower (129b) or the higher CP (129c). When the subject wh-element undergoes movement, it receives default case (cf. 2.7.4). Movement to the post-verbal domain of the higher CP is ungrammatical.

3.4.2 Constructions with 3 CPs

When a third CP is introduced, several differences arise in comparison to constructions with two CPs. In sentences with two CPs, *átàmàr* ('that') was optional. However, in sentences with three CPs, at least one instance of *átàmàr* ('that') must be present:

- (131) a. [_{CP} ... [_{CP} *átàmàr* ... [_{CP} *átàmàr* ...]]]
 b. [_{CP} ... [_{CP} *átàmàr* ... [_{CP} ...]]]
 c. [_{CP} ... [_{CP} ... [_{CP} *átàmàr* ...]]]
 d. * [_{CP} ... [_{CP} ... [_{CP} ...]]]

With two CPs, an additional *wh*-element is optional. However, with three CPs, at least two *wh*-elements are required. A single *wh*-element is ungrammatical, regardless of whether it is in-situ (132a) or ex-situ (132b). I tested all possible pre- and post-verbal positions and every sentence with three CPs and containing only one *wh*-element was always ungrammatical. The complete overview is provided in (appendix, 288).

- (132) a. *[_{CP} é-jén-ì ákirù [_{CP} átàmàr
3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàjì [_{CP} átàmàr é-mín-à
3-think-TAM Akai.NOM that 3-love.PRS-TAM
ádžìkòn **ɲáé**]]]?
Adjikon.NOM **who.DEFL**
- b. *[_{CP} **ɲáé** é-jén-ì ákirù [_{CP} átàmàr
who.DEFL 3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàjì [_{CP} átàmàr é-mín-à
3-think-TAM Akai.NOM that 3-love.PRS-TAM
ádžìkòn]]]?
Adjikon.NOM
'Who does Akiru know that Akai thinks that Adjikon
loves?'

At least two *wh*-elements are obligatory to render the sentence grammatical. Unlike with 2 CPs, where two *wh*-elements are not allowed when one of them stays in-situ (127b), with 3 CPs one of the *wh*-elements can remain in-situ without making the sentence ungrammatical:

- (133) $[_{CP} \boxed{\eta\acute{a}é} \quad \acute{e}\text{-jén-ì} \quad \acute{a}kìrù \quad [_{CP} \acute{à}tàmàr$
 $\boxed{\text{who.DEFL}} \quad 3\text{-know.PRS-TAM} \quad \text{Akiru.NOM} \quad \text{that}$
 $\acute{í}\text{-tám-ì} \quad \acute{à}kàjì \quad [_{CP} \acute{à}tàmàr \acute{é}\text{-mín-à}$
 $3\text{-think-TAM} \quad \text{Akai.NOM} \quad \text{that} \quad 3\text{-love.PRS-TAM}$
 $\acute{à}džìkòn \quad \boxed{\eta\acute{a}é} \quad]]]?$
 $\text{Adjikon.NOM} \quad \boxed{\text{who.DEFL}}$
 ‘Who does Akiru know that Akai thinks that Adjikon loves?’

In constructions with two CPs, wh-elements can only appear in the pre-verbal position. With three CPs, however, they may also occur post-verbally in any CP.

- (134) [CP é-jén-ì ákìrù [CP átàmàr í-tám-ì
 3-know.PRS-TAM Akiru.NOM that 3-think-TAM
 ákàjì **ḡáé** [CP átàmàr é-mín-à
 Akai.NOM **who.DEFL** that 3-love.PRS-TAM
 ádzìkòn **ḡáé**]]]?
 Adjikon.NOM **who.DEFL**
 'Who does Akiru know that Akai thinks that Adjikon loves?'

As with two CPs, two wh-elements within the same CP result in ungrammaticality.. The example below illustrates this with the lowest CP containing two wh-elements. This applies to all cases where two wh-elements appear in the same CP (see appendix, 291).

- (135) *[CP é-jén-ì ákìrù [CP átàmàr í-tám-ì
 3-know.PRS-TAM Akiru.NOM that 3-think-TAM
 ákàjì [CP átàmàr **ḡáé** é-mín-à
 Akai.NOM that **who.DEFL** 3-love.PRS-TAM
 ádzìkòn **ḡáé**]]]?
 Adjikon.NOM **who.DEFL**
 'Who does Akiru know that Akai thinks that Adjikon loves?'

All other combinations in which two wh-elements are present, whether in the pre- or post-verbal position, are grammatical as long as the wh-elements are not within the same CP. The following example shows all the possible positions of the second obligatory wh-element when the other wh-element is in the pre-verbal position of the highest CP:

- (136) [CP **ḡáé** é-jén-ì ákìrù [CP átàmàr ☒
who.DEFL 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì ☒ [CP átàmàr ☒ é-mín-à
 3-think-TAM Akai.NOM that 3-love.PRS-TAM
 ádzìkòn ☒]]]?
 Adjikon.NOM
 'Who does Akiru know that Akai thinks that Adjikon loves?'

A complete overview of all grammatical possible constructions with 3 CPs is provided in the appendix: (appendix, 289) lists all grammatical configurations where one wh-element remains in-situ, while (appendix, 290) presents all grammatical cases where both wh-elements are ex-situ.

Furthermore, constructions with three CPs and three wh-elements are also grammatical, as long as the wh-elements are not located in the same CP, meaning each CP contains exactly one copy. As observed in (134), wh-elements can appear in both pre- and post-verbal positions within each CP. Additionally, one of the copies can remain in-situ. The following examples shows a construction with three CPs, each containing one wh-element, with the lowest wh-element remaining in-situ. The complete overview of all constructions of three CPs containing three wh-elements can be found in (appendix, 292).

- (137)

[_{CP} é-jén-ì	ákìrù	ɲáé	[_{CP} átàmàr
3-know.PRS-TAM	Akiru	who.DEFL	that
í-tám-ì	ákàjì	ɲáé	[_{CP} átàmàr
3-think.PRS-TAM	Akai.NOM	who.DEFL	that
é-mín-à	ádžìkòn	ɲáé	]]]?]
3-love.PRS-TAM	Adjikon.NOM	who.DEFL	

‘Who does Akiru know that Akai thinks that Adjikon loves?’

So far, I have only presented data with wh-fronting. However, as shown in the previous chapter, the pre-verbal domain serves a focus position, meaning it also allows focal elements. The behaviour of focal elements in constructions with one or two CPs is the same as that of wh-elements. The element can either stay in-situ (138a) or move to the pre-verbal position of each CP (138b, 138c). Landing in the post-verbal position of the higher CP is ungrammatical (138d).

- (138) a.

[_{CP} í-tám-ì	ádžìkòn	[_{CP} átàmàr
3-think-TAM	Adjikon.NOM	that
í-múdz-ì	ákìrù	á-kíríŋ
3-eat.PRS-TAM	Akiru.NOM	meat.DEFL

b.

[_{CP} í-tám-ì	ádžìkòn	[_{CP} átàmàr	á-kíríŋ
3-think-TAM	Adjikon.NOM	that	meat.DEFL
í-múdz-ì	ákìrù	]]	
3-eat.PRS-TAM	Akiru.NOM		

c.

[_{CP} á-kíríŋ	í-tám-ì	ádžìkòn	[_{CP} átàmàr
meat.DEFL	3-think-TAM	Adjikon.NOM	that
í-múdz-ì	ákìrù		
3-eat.PRS-TAM	Akiru.NOM		

- d. *_[CP í-tám-ì ádzìkòn á-kírìŋ] _[CP átàmàr]
 3-think-TAM Adjikon.NOM meat.DEFL that
 í-múdz-ì ákìrù]].
 3-eat.PRS-TAM Akiru.NOM
 ‘Adjikon thinks that Akiru is eating the meat.’ (answer
 to the question: What is Akiru eating)

The crucial difference between in-situ wh-elements in constructions with several CPs and focal elements is that the latter do not require licensing. As a result, a construction with three CPs and a focal element in-situ without additional copies is grammatical, whereas a construction with three CPs and only one wh-element in-situ is ungrammatical.

- (139) [CP é-jén-ì ákìrù [CP átàmàr
 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ádzìkòn [CP átàmàr é-mín-à
 3-think.PRS-TAM Adjikon.NOM that 3-love.PRS-TAM
 ákàjì **é-kílè ló**]]].
 Akai.NOM **M-man.DEFL this.M.DEFL**]]].

However, if this focal element is fronted without leaving a copy behind, the sentence becomes ungrammatical. As with *wh*-elements, the position of the focal element does not matter; if there is no copy, the sentence is ungrammatical. A complete overview of this can be found in the appendix.

- (140) *[CP **é-kílè ló** é-jén-ì
M-man.DEFL this.M.DEFL 3-know.PRS-TAM
 ákìrù [CP átàmàr í-tám-ì ádzìkòn [CP
 Akiru.NOM that 3-think.PRS-TAM Adjikon.NOM
 átàmàr é-mín-à ákàjì]]].
 that 3-love.PRS-TAM Akai.NOM
 ‘Akai knows that Adjikon thinks that Akai loves this man.’

Like with *wh*-elements the sentence becomes grammatical when there are two copies. Those two copies can be in-situ or in any pre- or post-verbal position of the three CPs. Just like with *wh*-elements, the two copies are not allowed to be in the same CP. Unlike with *wh*-elements, the copy is a pronoun (*ɲéǎí*).

- (141) a. [CP **é-kílè ló** é-jén-ì
M-man.DEFL this.M.DEFL 3-know.PRS-TAM
 ákàjì [CP átàmàr í-tám-ì ákìrù [CP
 Akai.NOM that 3-think.PRS-TAM Akiru.NOM
 átàmàr é-mín-à ádzìkòn **néǎí**]]].
 that 3-love.PRS-TAM Adjikon.NOM **3.SG.DEFL**]]].
- b. [CP **é-kílè ló** é-jén-ì
M-man.DEFL this.M.DEFL 3-know.PRS-TAM
 ákìrù [CP átàmàr **ɲéǎí** í-tám-ì
 Akiru.NOM that **3.SG.DEFL** 3-think.PRS-TAM
 ádzìkòn [CP átàmàr é-mín-à ákàjì]]].
 Adjikon.NOM that 3-love.PRS-TAM Akai.NOM]]].

‘Akai knows that Adjikon thinks that Akai loves this man.’

The pronoun is not allowed to stand in front of the focal element as shown in the following example:

- (142) *_[CP] **ɲéðí** é-jén-ì ákirù _[CP] átàmàr
 3.SG.DEF 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ádzìkòn _[CP] átàmàr é-mín-à
 3-think.PRS-TAM Adjikon.NOM that 3-love.PRS-TAM
 ákàjì **é-kílè** **lís**]]].
 Akai.NOM **M-man.DEFL**
 ‘Akai knows that Adjikon thinks that Akai loves this man.’

In summary, long-distance licensing of wh-elements in-situ is allowed only if one CP is crossed. However, it becomes ungrammatical if long-distance licensing crosses two CPs. Wh-elements can move to the next higher CP without leaving a copy but moving via more than two CPs without a copy is ungrammatical. An extra copy makes the sentence grammatical. Focal elements behave like wh-elements with regard to movement. The only difference is that their copy is a pronoun. Section 3.5 will discuss the form of copies in Turkana in more detail.

3.4.3 A Challenge for Phasality

Chapter 5 will discuss the locality constraints of long distance movement in Turkana in detail. However, I already want to briefly show here that the Turkana data poses a challenge for current analyses of phasality. In standard phase theory (Chomsky 2000, Chomsky 2001), spell out happens in chunks known as phases. Once a phase is sent to spell out, it becomes inaccessible for further syntactic operations. There is an ongoing debate about what constitutes a phase, but traditionally vP and CP are seen as phases (Chomsky 2000, Chomsky 2001). Only the edge of a phase is accessible to operations (Chomsky 2000, Chomsky 2001):

- (143) *Phase Impenetrability Condition₁ (PIC₁):*

- (146) *siitaa-ne soc-aa [_{CP} ki ravii-ne **kis-ko**
 Sita-ERG think-PERF-M-SG that Ram-ERG **who-ACC**
 dekh-aa]?
 see-PERF-M-SG
 ‘Who did Sita think ravi saw?’ (Hindi-Urdu, Mahajan 2000:
 319, retrieved from Keine 2017)

Following phasal theory, this should also be expected for Turkana but this is not the case and the sentence is grammatical, as already shown in 3.4.1:

- (147) [_{CP} é-jén-ì ákìrù [_{CP} átàmàr é-mín-à
 3-know.PRS-TAM Akiru.NOM that 3-love.PRS-TAM
 ákàjì **ḡáé**]]?
 Akai.NOM **who.DEFL**
 ‘Who does Akiru know that Akai loves?’

Wh in-situ only becomes ungrammatical if there is an additional CP, as already shown in 3.4.2:

- (148) *_{[CP é-jén-ì ákìrù _{[CP átàmàr í-tám-ì}}
 3-know.PRS-TAM Akiru.NOM that 3-think-TAM
 ákàjì _{[CP átàmàr é-mín-à ádzìkòn}
 Akai.NOM that 3-love.PRS-TAM Adjikon.NOM
 _{ηáé}]]]?
 _{who.DEFL}
 ‘Who does Akiru know that Akai thinks that Adjikon loves?’

The sentence remains ungrammatical if the *wh*-element moves to the pre-verbal domain. This is unexpected because according to phase theory, the *wh*-element should be able to move successively through the edge of every phase and therefore not violate any locality constraints.

- (149) *_{[CP _{ηáé} é-jén-ì ákìrù _{[CP átàmàr}}
 _{who.DEFL} 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì _{[CP átàmàr é-mín-à}
 3-think-TAM Akai.NOM that 3-love.PRS-TAM
 ádzìkòn]]]?
 Adjikon.NOM
 ‘Who does Akiru know that Akai thinks that Adjikon loves?’

In order to be grammatical, the *wh*-element needs to move and needs to leave at least one copy behind which cannot be explained by standard locality constraints.

To summarise, in both *ex-situ* and *in-situ* contexts, *wh*-elements behave alike with respect to locality: they can move or be licensed across a single CP boundary, but not across more than one CP. This suggests that only every other CP acts as a phase in Turkana, unlike languages such as Hindi-Urdu, where every CP acts as a phase boundary (Keine 2017, 2020b). This subsection has only sketched the challenge; Chapter 5 will return to this issue in depth, situating the Turkana pattern within broader debates on phasehood, locality, and selective opacity.

3.5 The Form of Copies

Before, I will offer an analysis to this data under a phasal approach, I want to discuss which forms the copies can have. As shown in

the previous sections, both focal and wh long-distance movement sometimes allow and sometimes require copies. In this section, I examine the possible forms these copies can take and how they differ between wh-movement, focal movement, and d-linked wh-movement. While wh-elements require their copies to also be wh-elements, focal elements only allow pronouns as copies. D-linked wh-elements permit both wh-elements and pronouns. Both focal elements and d-linked wh-elements do not allow copies which look like the focal or d-linked element itself.

3.5.1 Copies in Wh-Movement

In the previous section, I described in detail how long-distance licensing of wh-elements often allows or even requires copies. For example:

- (150) [CP **ḡáé** é-jén-ì ádzìkòn [CP (átàmàr)
who.DEFL 3-know.PRS-TAM Adjikon.NOM that
ḡáé é-mín-à ákàjì]]
who.DEFL 3-love.PRS-TAM Akai.NOM
 ‘Who does Akiru know that Akai loves?’

These copies cannot be replaced by a pronoun, as shown in the following example:

- (151) *[CP **ḡáé** é-bál-à ákìrù [CP í-tám-ì
who.DEFL 3-say-TAM Akiru.NOM 3-think-TAM
 ádzìkòn [CP átàmàr é-mín-à ákàjì **ḡéǎí**
 Adjikon.NOM that 3-like-TAM Akai.NOM **s/he.DEFL**
]]]?

‘Who does Akiru say that Adjikon thinks that Akai loves?’

However, other moved elements, such as focal elements or d-linked wh-elements, can have copies which do not look the identical.

3.5.2 Copies in Focal Movement

Unlike wh-elements, moved focal DPs do not allow identical copies:

- (152) *_{[CP} é-kílè ló é-jén-ì ákàjì
M-man.DEFL this.M.DEFL 3-know.PRS-TAM Akai.NOM
_{[CP} átàmàr í-tám-ì ákírù _{[CP} átàmàr
that 3-think.PRS-TAM Akiru.NOM that
é-mín-à ádjikòn é-kílè ló
3-love.PRS-TAM Adjikon.NOM M-man.DEFL this.M.DEFL
]]].

‘Akai knows that Akiru thinks that Adjikon loves this man.’

However, they allow pronouns as copies:

- (153) _{[CP} é-kílè ló é-jén-ì ákàjì
M-man.DEFL this.M.DEFL 3-know.PRS-TAM Akai.NOM
_{[CP} átàmàr í-tám-ì ákírù _{[CP} átàmàr
that 3-think.PRS-TAM Akiru.NOM that
é-mín-à ádjikòn néđí]]].
3-love.PRS-TAM Adjikon.NOM 3.SG.DEFL
‘Akai knows that Akiru thinks that Adjikon loves this man.’

As shown in (142), the pronoun is not allowed to be in front of the focal DP.

Focal DPs behave in the opposite way to *wh*-elements: whereas *wh*-elements require another *wh*-element as a copy and do not allow pronouns, focal DPs require pronouns as copies and do not permit another identical copy of the focal DP.

3.5.3 Copies in D-linked Wh-Movement

D-linked *wh*-elements behave like focal elements in that they do not allow an identical copy of the whole DP:

- (154) *_{[CP} é-kílè álí é-jén-ì
M-man.DEFL which.M.DEFL 3-know.PRS-TAM
ákírù _{[CP} átàmàr í-tám-ì ádjikòn _{[CP}
Akiru.NOM that 3-think.PRS-TAM Adjikon.NOM
átàmàr é-mín-à ákàjì
that 3-love.PRS-TAM Akai.NOM
é-kílè álí]]]?
M-man.DEFL which.M.DEFL

‘Which man does Akiru know that Adjikon think that Akai loves?’

Like focal elements, they do allow pronouns as copies:

(155) [CP

álú-túḡá	/ ḡí-túḡá	álú
which.PL-people.DEFL / PL-people.DEFL which.PL.DEFL		
é-jén-ì	ákàjì	[CP átàmàr í-tám-ì
3-know.PRS-TAM Akai.NOM	that	3-think.PRS-TAM
ádžikòn	[CP átàmàr	kétfí í-múdž-é-tè
Adjikon.NOM	that	3.PL.DEFL 3-eat.PRS-TAM-PL
á-kíríḡ	]]]	
meat.DEFL		
‘Which people does Akai know that Adjikon thinks are eating meat?’		

Furthermore, wh-elements are allowed as copies:

(156) [CP

álú-túḡá	/ ḡí-túḡá	álú
which.PL-people.DEFL / PL-people.DEFL which.PL.DEFL		
é-jén-ì	ákàjì	[CP átàmàr í-tám-ì
3-know.PRS-TAM Akai.NOM	that	3-think.PRS-TAM
ádžikòn	[CP átàmàr	tá-ḡáé í-múdž-é-tè
Adjikon.NOM	that	PL-who.DEFL 3-eat.PRS-TAM-PL
á-kíríḡ	]]]	
meat.DEFL		
‘Which people does Akai know that Adjikon thinks are eating meat?’		

Interestingly, unlike with focal elements, the reduced copy is allowed to stand in front of the complete d-linked wh-phrase:

(157) [CP **tá-ḡáé** é-jén-ì ákàjì [CP átàmàr
PL-who.DEFL 3-know.PRS-TAM Akai.NOM that
 í-tám-ì ádžikòn [CP átàmàr
 3-think.PRS-TAM Adjikon.NOM that

álú-túná	/ ɲí-túná	álú
which.PL-people.DEFL / PL-people.DEFL which.PL.DEFL		
í-múdʒ-é-tè	á-kírɪŋ	]]]]?
3-eat.PRS-TAM-PL meat.DEFL		
‘Which people does Akai know that Adjikon thinks are eating meat?’		

In summary, *wh*-elements require another *wh*-element as a copy. The copy cannot be a pronoun. Focal DPs, on the other hand, require pronouns as copies and do not permit identical copies. D-linked *wh*-elements are more flexible, allowing both *wh*-elements and pronouns as copies, but they also do not allow exact copy of the moved element.

3.6 Conclusion

In this chapter, I have examined the properties of *in*- and *ex-situ* *wh*-elements. I have shown that *wh*-elements that remain *in-situ* do not undergo covert movement but truly remain *in-situ*. In contrast, *wh*-elements that appear *ex-situ* undergo movement rather than being base-generated in a high position within the clause.

Furthermore, I explored long-distance licensing and movement of focal and *wh*-elements. The following pattern emerged: In single CP constructions, copies are not allowed; in two CP constructions, copies are optional; and in three CP constructions, copies become obligatory. This pattern challenges the standard view on phasality because it seems like only every other CP acts as a phase.

While this chapter described the pattern of long-distance movement and licensing, the next two chapters will provide an analysis to this intriguing phenomenon. The following chapter will focus on the landing sites of moved elements.

Chapter 4

Landing Sites

4.1 Introduction

In this chapter, I investigate the syntax of focus and wh-movement in Turkana, with particular attention to their optionality and landing sites. I begin with examining the derivation of the V1 word order, following [Finer's \(2013\)](#) proposal, that the verb undergoes head movement to T.

The core focus of the chapter is on the landing sites of focal and wh-movement. As discussed in the previous chapter, Turkana allows wh-elements to appear both in-situ and ex-situ, the latter surfacing either pre-verbally or post-verbally in a higher clause. The following pair illustrates this alternation: The wh-element can either move to a pre- (158b) or to a post-verbal position (158a) in a higher CP.

- (158) a. $[_{CP}$ é-jén-ì ákìrù $[_{CP}$ átàmàr
3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàjì ḡáé $[_{CP}$ átàmàr
3-think-TAM Akai.NOM who.DEFL that
é-mín-à ádzìkòn ḡáé]]]?
3-love.PRS-TAM Adjikon.NOM who.DEFL
b. $[_{CP}$ é-jén-ì ákìrù $[_{CP}$ átàmàr ḡáé
3-know.PRS-TAM Akiru.NOM that who.DEFL
í-tám-ì ákàjì $[_{CP}$ átàmàr é-mín-à
3-think-TAM Akai.NOM that 3-love.PRS-TAM
ádzìkòn ḡáé]]]?
Adjikon.NOM who.DEFL
'Who does Akiru know that Akai thinks that Adjikon loves?'

I argue that the pre-verbal position of the *wh*-element is not the specifier of CP, as typically assumed for $\bar{A}$ -movement (Chomsky 1977, 1995). Instead, I propose that it is the specifier of TP, building on feature-driven accounts of movement.

As for the post-verbal position, I show that the *wh*-element does not land in the embedded CP, contrary to what the surface order might suggest (which is already indicated by the bracket notion in the example above). It rather surfaces in a higher clause, specifically, in the inner specifier of the vP domain of that clause.

The chapter is structured as follows: In 4.2, I analyse the basic word order in Turkana, 4.3 examines the pre-verbal focus position, in 4.4, I discuss the post-verbal landing sites of long distance movement, 4.6 concludes.

4.2 Structure of the Basic Word Order

In this section, I discuss the structure of the basic word order in Turkana. This provides a necessary foundation for understanding the landing sites of *wh*-movement. As shown in 2.6, the basic word order in Turkana is VSO:

- (159) é-múdz-í ájòŋ ákírín.
 1-eat.PRS-TAM I.NOM meat.DEFL
 ‘I am eating meat.’

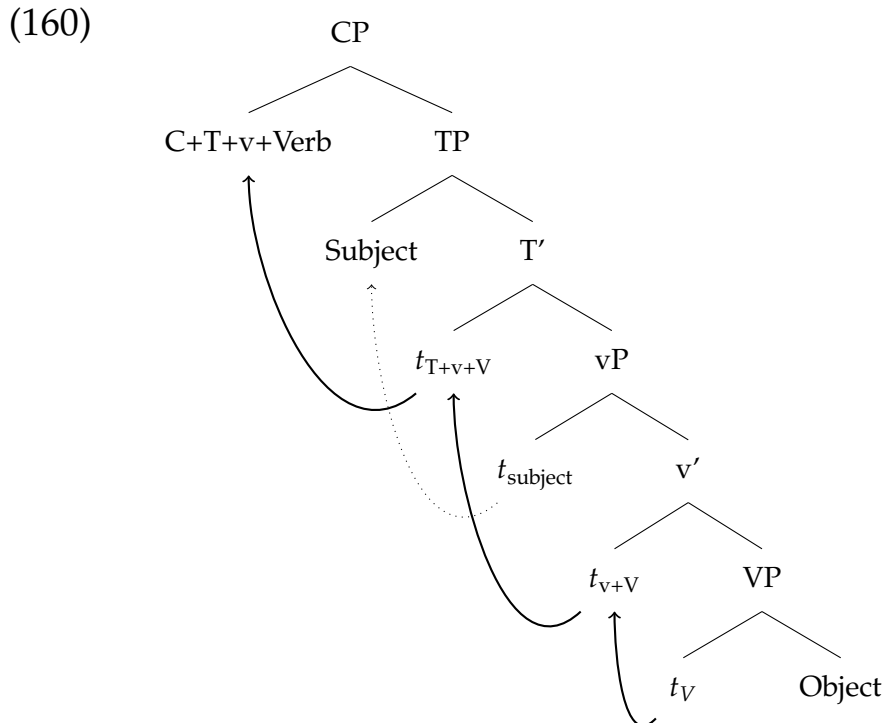
There has been extensive debate about the structure of VSO languages, with particular focus on Celtic and Austronesian languages. However, there has been some work on Nilotic (Creider 1977; Creider & Creider 1983; Bossi & Diercks 2019 for Kipsigis, Finer 2013 for Turkana). Traditionally, VSO order is derived by raising the verb via head movement past the subject (Clemens & Polinsky 2017; see also Guilfoyle 1990; McCloskey 1991, 1996, 1997, 2001, 2005 among others for Irish, Sproat 1985; Sadler 1988; Roberts 2005 among others for Welsh, Guilfoyle, Hung & Travis 1992 for Cebuano, Woolford 1991 for Chamorro and Niuean, Pearce 2002 for Māori, Richards 2000; Rackowski 2002; Aldridge 2004; Rackowski & Richards 2005 for Tagalog, Otsuka 2000, 2005a,b; Custis 2004 for Tongan, Ouhalla 1994 among others for Arabic, Creider 1977; Creider & Creider 1983; Bossi & Diercks 2019 for Kipsigis, Finer 2013 for Turkana).

Early approaches like [Emonds \(1980\)](#) argue that the verb raises to C while the subject moves to the specifier of TP. This account is also called weak V2-approach ([Clemens & Polinsky 2017](#)). [Otsuka \(2005b\)](#) supports this for Tongan. In contrast, [Sproat \(1985\)](#); [McCloskey \(1996\)](#); [Richards \(2000\)](#) among others argue that verb only moves to T while the subject remains in-situ. A third account, proposed by [Bossi & Diercks \(2019\)](#) for Kipsigis (a Western Nilotic language), argues that V1 order results from head movement of the verb to a functional projection located between TP and CP.

In this section, I examine these verb raising approaches for Turkana. I argue that in Turkana the subject stays in-situ while the verb raises to T.

4.2.1 V to C Movement

The weak V2-approach assumes that an EPP feature on T triggers movement of the subject to the specifier of TP (cf. [Miyagawa 2001, 2003](#)). In this account, the verb raises to C via head movement. The tree below illustrates this derivation:



An example for such an approach comes from [Otsuka \(2002, 2005a,b\)](#) for Tongan. Tongan is an Austronesian language that exhibits a VSO/VOS alternation, with VSO being the unmarked word order

(Otsuka 2000, 2002, 2005a,b):

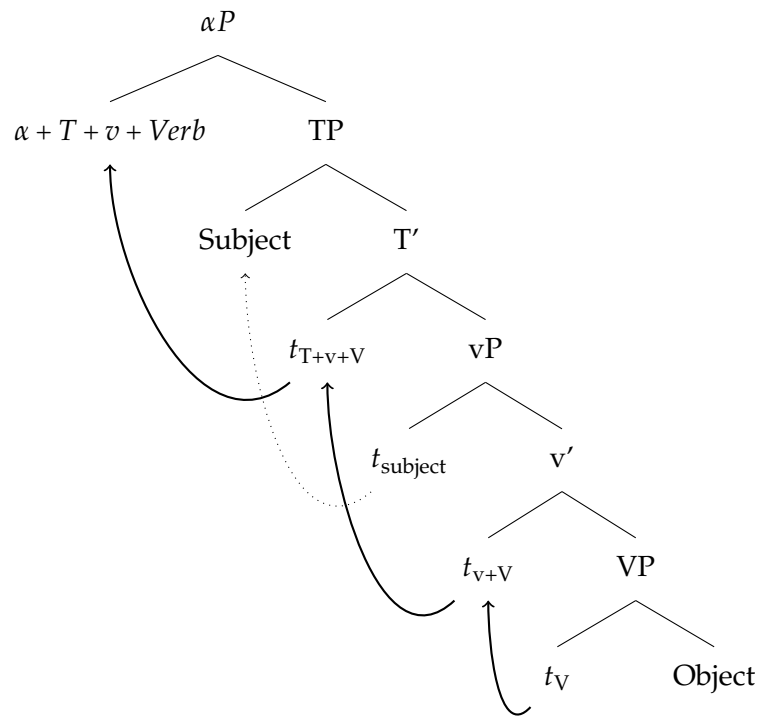
- (161) a. Na'e fili 'e Sione 'a Pila. (VSO)
PST choose ERG Sione ABS Pila
b. Na'e fili 'a Pila 'e Sione. (VOS)
PST choose ABS Pila ERG Sione
'Sione chose Pila.' (Tongan, Otsuka 2005b: 246)

Otsuka (2002, 2005a,b) argues that the position immediately following the verb is a discourse marked position that attracts focal elements. In this view, T carries an optional discourse-based feature in addition to its EPP feature. Thus, the focal element which can both be subject or object can move to the specifier of T. This accounts for the VSO/VOS alternation. Following the standard VP-internal subject hypothesis (cf. Koopman & Sportiche 1982; Kratzer 1996), the subject originates in the specifier of vP. This, the verb must move to a position higher than the subject. Once the verb reaches T, both subject and object become equidistant from T (cf. Chomsky 1993), allowing either to move to the specifier of T without violating the minimal link condition. To derive V1 order, the verb subsequently raises from T to C.

4.2.2 V to α Movement

A related account to the verb raising to C is proposed by Richards (1996) for Tagalog and Bossi & Diercks (2019) for Kipsigis. They assume that the subject moves to the specifier of TP, as in the V-to-C approach. However, rather than raising all the way to C, the verb undergoes head movement to a functional projection located immediately above TP. This is illustrated in the tree below:

(162)



Kipsigis is a V1 language with VSO as the canonical word order (Creider 1977; Creider & Creider 1983; Kouneli 2019; Bossi & Diercks 2019). However, Kipsigis shows, similar to Tongan, a VSO/VOS alternation in which the most prominent DP appears immediately after the verb:

- (163) a. kii-Ø-geer džiita tɛɛta. (V-S-O)
 PST-3SG-see person cow
 b. kii-Ø-geer tɛɛta džiita. (V-O-S)
 PST-3SG-see cow person.
 ‘A person saw a cow (long ago).’ (Kipsigis, Bossi & Diercks 2019: 2)

According to Bossi & Diercks (2019), this alternation can be explained by T having an optional discourse-based feature, in addition to its EPP feature. This discourse-feature attracts the most prominent DP, whether subject or object. This approach adapts the analysis of Otsuka (2005b) for Tongan, presented above.

Bossi & Diercks (2019) further show that manner adverbs, unlike temporal adverbs, cannot appear immediately following the verb in Kipsigis (164). Manner adverbs are usually assumed to be vP level adverbs which can be adjoined to either side of vP (cf. Bowers 1993; Cinque 1999; Diercks 2010). This restriction suggests that the position immediately following the verb is located above vP, within the TP domain:

- (164) a. *Koo-Ø-min **komie** laagook baandeek.
 PST-3PL-plant **well** children maize
 b. *Koo-Ø-min **komie** baandeek laagook.
 PST-3PL-plant **well** maize children
 ‘The children planted the maize well.’ (Kipsigis, Bossi & Diercks 2019: 9)

However, Bossi & Diercks (2019) deviate from Otsuka’s (2005b) analysis regarding the position of the verb. They argue that the VSO/VOS alternation in Kipsigis is independent of CP-level phenomena such as various types of embedded structures. The following example shows that the VSO/VOS alternation persists under declarative embedding complementisers which are usually located in C.

- (165) a. koo-Ø-ger kiproono kɔɛ koo-Ø-min laagook
 PST3-SG-say Kiproono that PST-3PL-plant children
 baandeek komie.
 maize well

- b. koo-Ø-ger kiproono kɔɛ koo-Ø-min baandeek
 PST3-SG-say Kiproono that PST-3PL-plant maize
 laagook komie.
 children well
 ‘Kiproono said that the children planted the maize well.’
 (Kipsigis, Bossi & Diercks 2019: 6)

Bossi & Diercks (2019) take this as evidence that the verb raises to a position below the CP domain because C is filled by the complementiser. However, since the verb must raise to a position above the most prominent DP located in the specifier of TP, they conclude that it must occupy a projection between T and C in order to derive the V1 order, which they label α P.

However, Driemel & Kouneli (2025) argue that complementisers in Kipsigis are verbs (see also Bossi 2023a). They say that the complementiser consists of the root of the verb *le* (‘say’) and a prefix with person and number agreement:

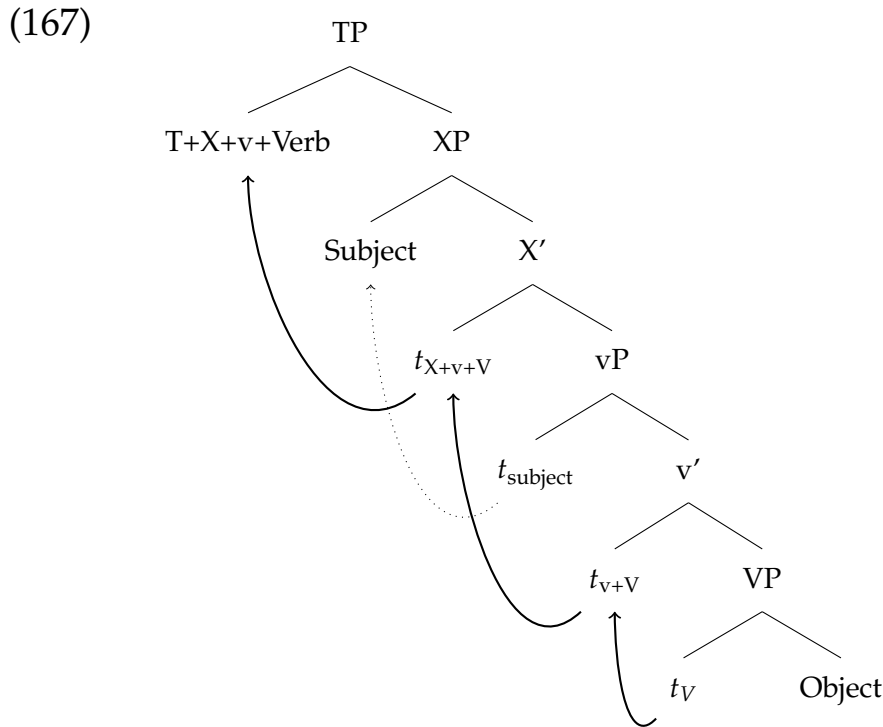
- (166) a. â:-ŋgɛn **â:-lé** Ø-rú-è kíbê:t.
 1SG-know **1SG-LE** 3-sleep-IPFV Kibeet.NOM
 ‘I know that Kibeet is sleeping.’
 b. kà-ó-mwá **ò:-lè** Ø-rú-è kíbê:t.
 PST.CURR-2PL-say **2PL-LE** 3-sleep-IPFV Kibeet.NOM
 ‘You (pl) said that Kibeet is sleeping.’ (Kipsigis, Driemel & Kouneli 2025: 9)

Driemel & Kouneli (2025) argue that an embedded clause with *le* in Kipsigis consists of two clauses: A TP with the verb *le* which takes a CP as a complement. This weakens Bossi & Dierck’s (2019) argument that the verb cannot move to C because C is filled with the complementiser, since there is in fact no complementiser.

4.2.3 V to T movement

While the previous approaches assume that the subject moves to the specifier of T because of the EPP feature, the analysis presented here rejects that assumption (McCloskey 1996, 1997; Roberts 2005; Finer 2013; Tyrchan 2023). McCloskey (1996) in particular questions the universal existence of an EPP-feature on T. Instead, in these approaches, the verb moves via head movement to T and the subject

moves to a position between vP and TP. This is illustrated in the following tree.

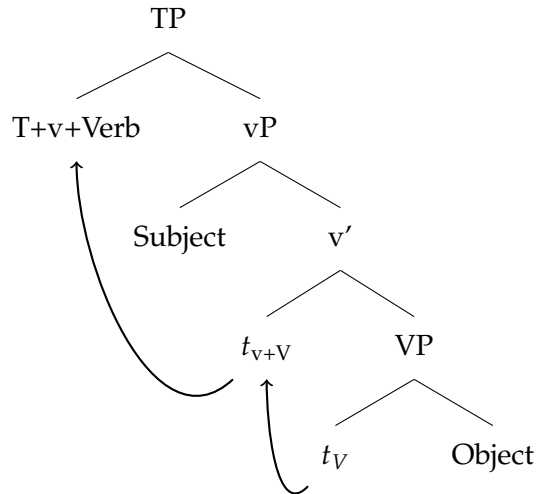


McCloskey (1996, 1997); Roberts (2005) argue for Irish and Welsh, respectively, that the subject moves from the specifier of vP to a projection between vP and TP. As in Kipsigis (see 4.2.2), vP-level adverbs cannot appear immediately after the verb. However, they interpret these findings differently and argue that the subject moves to a phrase above vP. *Finer* (2013) adopts this for Turkana. However, he notes that his data are inconclusive, as speaker judgments varied on whether adverbs can appear between the verb and subject.

In contrast, my own data consistently show that manner adverbs are permitted immediately after the verb:

- (168)
- | | | | |
|----|-----------------------------|----------------|-----------------|
| a. | á-kér-ì | átùpéjì | ájòṅ. |
| | 1-run.PRS-TAM | quickly | 1.SG.NOM |
| | 'I am quickly running.' | | |
| b. | á-kér-ì | lógé | ájòṅ. |
| | 1-run.PRS-TAM | slowly | 1.SG.NOM |
| | 'I am slowly running.' | | |
| c. | é-múdz-í | átùpéjì | ájòṅ ákírín. |
| | 1-eat.PRS-TAM | quickly | I.NOM meat.DEFL |
| | 'I am quickly eating meat.' | | |

(171)



This approach aligns with analyses of subject inversion in Bantu languages (cf. Demuth 1990; Demuth & Mmusi 1997; Letsholo 2004; Carstens 2005; Baker 2003; Zerbian 2006; Zeller 2008, 2013; Diercks 2010, 2011). While Bantu languages typically exhibit SVO order (172a), they also allow subject inversion, where the external argument appears in a postverbal position. Bantu languages have noun classes that behave like grammatical gender and number. The verb agrees with the respectable DP in that noun class. This can be seen in the following examples in Lubukusu, a Bantu language spoken in Kenya (Diercks 2010, 2011). In subject inversions, the subject marker can either agree with the post-verbal logical subject (172b) or with the fronted locative (172c). However, the external argument is always seen as the logical subject, independently of whether the verb agrees with it or not.

- (172) a. **Kú**-mú-saala **kw**-á-kwá mu-mu-siiru.
 3-3-tree 3SM-PST-fall 18-3-forest
 'A tree fell in the forest.'
- b. Mú-mú-siirú **kw**-á-kwá-mó **kú**-mú-saala.
 18-3-forest 3SM-PST-fall-18L 3-3-tree
 'In the forest fell a tree.'
- c. **Mú**-mú-siirú **mw**-á-kwá-mó kú-mú-saala.
 18-3-forest 18SM-PST-fall-18L 3-3-tree
 'In the forest fell a tree.' (Lubukusu, Diercks 2011: 2-3)

In order to test the position of the subject, Diercks (2010, 2011) uses manner adverbs as a testing ground. Manner adverbs in Lubukusu can appear on either side of the logical subject, but not in pre-verbal

position, as the following examples show:

- (173) a. Mú-mú-siiru o-mwó ma-á-kwá-mó kú-mú-saala
 18-3-forest DEM-18 3SM-PST-fall 3-3-forest
bwangu.
quickly
- b. Mú-mú-siiru o-mwó ma-á-kwá-mó bwangu
 18-3-forest DEM-18 3SM-PST-fall quickly
 kú-mú-saala.
 3-3-forest
- c. *Mú-mú-siiru o-mwó bwangu ma-á-kwá-mó
 18-3-forest DEM-18 quickly 3SM-PST-fall
 kú-mú-saala.
 3-3-forest
 ‘In the forest fell a tree quickly.’ (Lubukusu, [Diercks 2011](#): 30)

[Diercks \(2010, 2011\)](#) takes this as evidence that the verb moves to T while the logical subject stays in-situ in the specifier of vP. The adverb can either left or right adjoin in the vP which explains why it can precede or follow the subject. The adverb cannot occur pre-verbally, because it does not adjoin to T.

Diercks concludes that in subject inversion, the verb moves to T while the logical subject remains in-situ. Manner adverbs can adjoin within the vP but not in pre-verbal position, since they do not adjoin to T.

Nonetheless, the prominence data presented in [2.6](#) suggest an additional projection between vP and TP where the object can move to in VOS constructions:

- (174) a. k-á-kójn é-mòn ájóη.
 K-1-bite M-snake.NOM I.DEFL
 (V-S-O)
- b. k-á-kójn ájóη é-mòn.
 K-1-bite I.DEFL M-snake.NOM
 (V-O-S)

‘The snake is biting me.’

To capture this VSO/VOS alternation, [Finer \(2013\)](#) and [Tyrchan \(2023\)](#) suggest the aforementioned extra projection between vP and TP which serves as a landing site for either the subject or the object. [Finer \(2013\)](#) suggests, given his unclear data, that manner adverbs may also be able to attach to this suggested extra projection between vP and TP. Whether such a projection is necessary remains open. The crucial generalisation is that the verb raises to T, but not to C.

Further evidence that the verb does not raise to C comes from embedded contexts. In embedded clauses, the V1 order persists even in the presence of an overt complementiser. This suggests that the verb does not target C, since that position is already occupied by the complementiser. Instead, the verb must raise only as far as T:

- (175) í-tám-ì ádzìkòn [CP átàmàr í-múdz-ì
 3-think-TAM Adjikon.NOM that 3-eat.PRS-TAM
 ákìrù á-kírìŋ].
 Akiru.NOM meat.DEFL
 ‘Adjikon thinks that Akiru is eating the meat.’

To summarise, this section has explored the derivation of V1 word order in Turkana. I have argued that the verb moves to T via head movement. Regarding the subject, two possibilities were considered: either the subject remains in-situ or moves to a position between vP and TP. While my own data support a subject in-situ position, I remain agnostic about whether an additional projection is necessary. Crucially, the data are incompatible with a model in which the subject moves to the specifier of T and the verb raises further to C.

4.3 Pre-Verbal Focus Position

In the previous section, I investigated the structure of the canonical VSO word order in Turkana. I argued that the verb moves via head movement to T, while the subject remains in a position below T. I now turn to the pre-verbal focus position. As previously shown in [2.7.1](#), Turkana allows focal elements to move to the pre-verbal domain. This is illustrated again in the following examples. (176a) shows a *wh*-element, which are inherently focal, in the pre-verbal

domain and (176b) shows the answer to the question, also known as new information focus, in the pre-verbal domain.

- (176) a. ḡáé í-múdz-ì á-kírín?
who.DEFL 3-eat.PRS-TAM F-meat.DEFL
 ‘Who is eating meat?’
 b. á-bérú í-múdz-ì á-kírín.
F-woman.DEFL 3-eat.PRS-TAM F-meat.DEFL
 ‘The WOMAN is eating meat.’

In this section, I argue that the focus position in Turkana is in the specifier of T even though focus movement is typically considered a type of $\bar{A}$ -movement which is usually associated with the CP domain. However, the pre-verbal position in Turkana displays mixed A- and $\bar{A}$ -properties as shown in 2.7.5. Before turning to this issue, I first discuss the standard analysis of focus movement to the specifier of CP.

4.3.1 Focus Movement to the Specifier of C

Under the standard analysis, focus and wh-movement are classified as $\bar{A}$ -movement (Chomsky 1977, 1981b). $\bar{A}$ -movement differs in their properties from A-movement i.e. $\bar{A}$ -movement allows long distance movement or weak cross over (for a detailed overview, see 2.7.5)

In this view, $\bar{A}$ -movement, such as wh-movement, topicalisation or focalisation, a designated $\bar{A}$ -position in the left periphery (cf. Chomsky 1977, 1995; Rizzi 1997; Kiss 1998 among others). This is illustrated for German in the following examples (taken from Fanselow & Lenertová 2011): (177a) shows that the wh-phrase moves to the specifier of CP and (177b) shows that the topic moves to the specifier of CP. In both examples, the verb moves to C.

- (177) a. [_{CP} [Wen]_i [_C hat] [_{TP} der Peter t_i eingeladen t_V
 who.ACC has the.NOM Peter invited
]]?
 ‘Who has Peter invited?’
 b. [_{CP} [den Josef]_i [_C mag] [_{TP} jeder t_i t_V]].
 the.ACC Josef likes everyone
 ‘Everyone likes Josef.’ (German, Fanselow & Lenertová

2011: 171)

Rizzi (1997) follows the analysis that $\bar{A}$ -movement targets a designated landing site in the left periphery. However, he observes that topics and foci can occur simultaneously in the left periphery. Furthermore, the foci and topic move to a position below the complementiser located in C, meaning that the fronted element must move to be a position below C. Both observations are illustrated for Italian in the following example:

- (178) Credo che a Gianni, QUESTO, domani, gli dovremmo dire
 C TOP FOC TOP IP
 ‘I believe that to Gianni, THIS we should say.’ (Italian, Rizzi 1997: 295)

To account for the simultaneous occurrence of topics and foci, Rizzi (1997) argues that the left periphery is split into several functional projections. He presents the structure as follows:

- (179) [Force [Top* [Foc [Top* [Fin]]]]] ¹²

While Rizzi (1997) posits that $\bar{A}$ -movement lands in fixed positions in the left periphery, van Urk (2015) & van Urk & Richards (2015) challenge the idea of fixed A- and $\bar{A}$ -positions. They argue that topics and foci in Dinka land in the left periphery, consistent with the aforementioned approaches. However, Dinka, like Turkana, shows mixed A- & $\bar{A}$ -properties (see 2.7.5 for a detailed discussion). Hence, they suggest that movement is driven by features rather than by inherent A- and $\bar{A}$ -positions. Given that Turkana exhibits similar mixed A- & $\bar{A}$ -properties, van Urk’s feature-driven approach is particularly relevant for understanding the Turkana data.

Dinka is a V2 language in which both topics and foci can appear pre-verbally as shown in the following examples: (180) shows a wh-element in the pre-verbal domain of the question and the topic in the pre-verbal in the corresponding answer. (181) shows the focus in the pre-verbal domain in the answer.

- (180) a. A: yè nà cè cuñin câam?
 be who PRE.SV food eat.NF
 ‘Who has eaten the food?’

¹²The asteriks are used to notate the recursion of topics.

- b. B:

cuñin
food

 à-cïi bôl câam.
3S-PRF.OV Bol eat.NF
'The food, Bol has eaten.' (Dinka, [van Urk 2015](#): 96)

- (181) a. A: yè **ṇó** cǐi bôl câam?
 be **what** PRF.OV Bol eat.NF
 ‘What has Bol eaten?’
 b. B: **cuṇin** à-cǐi bôl câam.
 food 3S-PRF.OV Bol eat.NF
 ‘The food, Bol has eaten.’ (Dinka, *van Urk 2015*: 96)

van Urk (2015) & *van Urk & Richards (2015)* assume the standard V2 analysis for Dinka in which the verb moves to C and the fronted constituent moves to Spec,CP, as shown above for German (177): The verb moves to C and the fronted element moves to the specifier of CP. This is supported by the observation that some complementisers intervene with the V2 structure (182a) and only the verb initial order is allowed (182b). This means that the verb and the complementiser compete for the same position.

- (182) a. majók à-gàii [mèn cǐ dēṇ lô
 Majok 3SG-wonder whether PRF Deng.GEN go
 bâai].
 home.
 b. *majók à-gàii [mèn dēṇ (à)-cǐi lô bâai].
 Majok 3SG-wonder whether Deng 3SG-PRF go home.

‘Majok wonders whether Deng went home.’ (Dinka, *van Urk & Richards 2015*: 119)

However, a few complementisers permit V2 structure:

- (183) à-cà tâak [ké cǎn bǐ wít tiáam].
 3SG-PRF.1SG think C Can FUT wrestling win.TR
 ‘I thought that Can will win wrestling.’

Thus, *van Urk (2015)* & *van Urk & Richards (2015)* adapt the split CP approach by *Rizzi (1997)* and suggest that C has at least two heads with the lower head triggering V2 and functioning as a phase head.

Nonetheless, as discussed in 2.7.5, focalisation and topicalisation in Dinka display mixed A- and $\bar{A}$ -properties. Hence, *van Urk (2015)* & *van Urk & Richards (2015)* argue that the distinction between CP and TP properties is not always clear-cut in Dinka. They question the

existence of designated A- & $\bar{A}$ -positions and argue that movement is better understood as being feature driven: All fronting operations result from uninterpretable features [uF] on the probe, with the fronted constituent carrying a matching interpretable feature [F]. In this view, there is no syntactic distinction between A- & $\bar{A}$ -movement. They are both phrasal movement and different movement types only differ in the features involved.

As such, fronted elements like topics and foci are not required to move to a fixed $\bar{A}$ -position. Instead they move because they are attracted by a syntactic feature. They could very well occupy the specifier of TP if T carried the relevant uninterpretable feature. In fact, [van Urk \(2015\)](#) briefly entertains the possibility that Dinka's V2 structure is set at the TP level where functions are traditionally associated with C would be carried out by T.

Given that Turkana, like Dinka, shows mixed A- & $\bar{A}$ -properties, I will adopt the proposal by [van Urk \(2015\)](#) & [van Urk & Richards \(2015\)](#) that movement is feature driven and that a designated $\bar{A}$ -position does not exist. However, I argue that in Turkana, focus movement targets the specifier of TP rather than the specifier of CP in the following subsection.

4.3.2 Focus Movement to the Specifier of T

In the previous subsection, I discussed approaches that assume focus targets the specifier of CP which is the standard view of focus movement. However, alternative approaches suggest that foci or other information structural material may instead land in the specifier of TP.

In 4.2.1, I presented Otsuka's (2002; 2005a; 2005b) analysis of word order in Tongan. To summarise briefly: Tongan exhibits a VSO/VOS alternation, with VSO being the unmarked word order. Otsuka argues that the verb raises to C and that the position immediately following the verb, the specifier of TP, is a discourse marked position that attracts focal elements. According to [Otsuka \(2005a\)](#), focus movement in Tongan is A-movement as it displays the typical properties for A-movement. Specifically, focus movement in Tongan is obligatorily local and not long distance; it is not subject to WCO effects; it does not reconstruct for principle C. [Otsuka \(2005a\)](#) takes this as an argument that focus elements land in the designated position

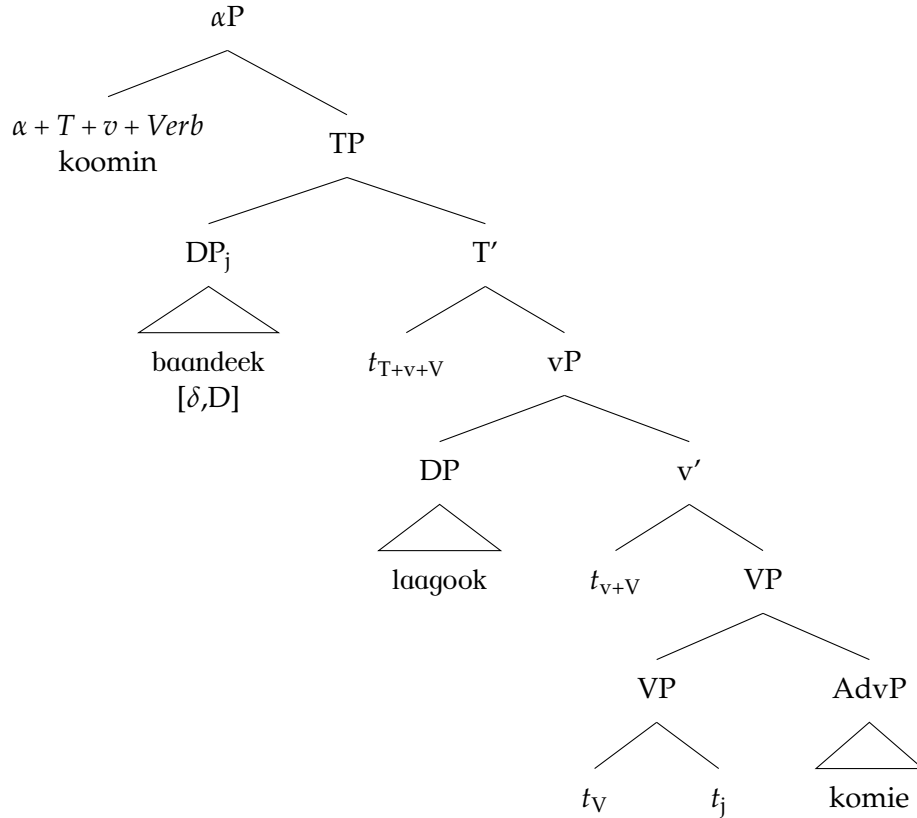
for A-movement namely the specifier of TP. Otsuka (2005a) follows Mayagawa's (2001; 2003) analysis for A-scrambling in Japanese and proposes that focus movement is driven by an EPP feature on T. Additionally, Otsuka adopts the proposal by Zubizarreta (1998) for Spanish in which T carries an optional discourse-based feature in addition to its EPP feature. Thus, Otsuka assumes that T carries a discourse feature which attracts foci to the specifier of T.

This analysis has also been extended to Nilotic, specifically Kipsigis. As discussed in 4.2.2, Bossi & Diercks (2019) argue that discourse prominent elements in Kipsigis likewise land in the specifier of TP. They adopt Otsuka's (2005a) analysis that TP has a discourse feature (δ) which attracts the most prominent DP. Consider the following example where the object is the most prominent element.

- (184) koo- $\emptyset$ -min baandeek laagook komie.
PST-3PL-plant maize children well
'The children plant the maize well.' (Kipsigis, Bossi & Diercks 2019: 16)

The verb moves via head movement to α P (cf. 4.2.2). T bears an unvalued D feature and an unvalued prominence feature which attracts the most prominent DP, in this case the object. The derivation is illustrated in the following tree (taken from Bossi & Diercks 2019: 17).

(185)



Unlike Otsuka (2005a), Bossi & Diercks (2019) do not categorise the discourse-prominent position in Kipsigis as a dedicated A-position. Instead, they argue that this position exhibits mixed A- and $\bar{A}$ -properties. On one hand, the position behaves like an $\bar{A}$ -position with respect to quantifier binding and reconstruction. For instance, the moved elements reconstruct for quantifier-binding. In the following example, the quantifier DP subject can bind the pronoun regardless of the surface word order:

- (186) a. koo- $\emptyset$ -sus η ookta age t $\acute{u}$ g $\acute{u}$ l_k ξ iita-iin_{k/i}.
 PST-3SG-bite dog every person-POSS
 b. koo- $\emptyset$ -sus ξ iita-iin_{k/i} η ookta age t $\acute{u}$ g $\acute{u}$ l_k.
 PST-3SG-bite person-POSS dog every
 ‘Every dog_k bit its_{k/i} person.’ (Kipsigis, Bossi & Diercks 2019: 34)

On the other hand, the same position behaves like an A-position in that it does not reconstruct for Principle C. Principle C blocks binding into R-expressions, but movement to the discourse-prominent position in Kipsigis does not induce such violations. In the following example, a fronted R-expression can corefer with a pronoun in its

base position, indicating that the moved subject does not reconstruct for Principle C.

- (187) koo-Ø-geer tʃoorweet-aap kiproono_k mɛɛndɛt_{k/i}.
 PST-3SG-see friend.POSS Kiproono him
 ‘Kiproono’s_k friend saw him_{k/i}.’

Bossi and Diercks conclude that the discourse prominent position cannot be categorically classified as either A- or $\bar{A}$. However, they do not offer an analysis for the mixed A/ $\bar{A}$ -properties.

Turkana aligns with the aforementioned Nilotic languages Dinka and Kipsigis in having a discourse position with mixed A/ $\bar{A}$ -properties (see 2.7.5 for a detailed discussion). Thus, I will adopt van Urk’s (2015) which rejects the existence of dedicated A- & $\bar{A}$ positions. Instead, movement is driven by syntactic features rather than distinct designated positions. Furthermore, van Urk (2015) shows for Dinka that the properties of the CP and TP domains in Dinka are not as segregated as in other languages. I have already shown in 2.7.5 that this is also holds for Turkana.

Since the theoretical framework adopted here does not assume designated A- and $\bar{A}$ -positions, focus movement is not inherently linked to the CP domain. I assume that the focus position in Turkana is in the specifier of TP and not in the specifier of CP as suggested by van Urk (2015) for Dinka.

The primary motivation for positing that focus movement targets the specifier of TP is structural: This position is available, as subjects remain in-situ in the specifier of vP. Additional evidence against a CP-related focus position arises from embedded clause data. In such contexts, the focal element follows complementisers like *that* which are usually assumed to be located in C:

- (188) í-tám-ì ádzìkòn [CP átàmàr **á-kírín**
 3-think-TAM Adjikon.NOM that **meat.DEFL**
 í-módz-ì ákìrù].
 3-eat.PRS-TAM Akiru.NOM
 ‘Adjikon thinks that Akiru is eating MEAT.’

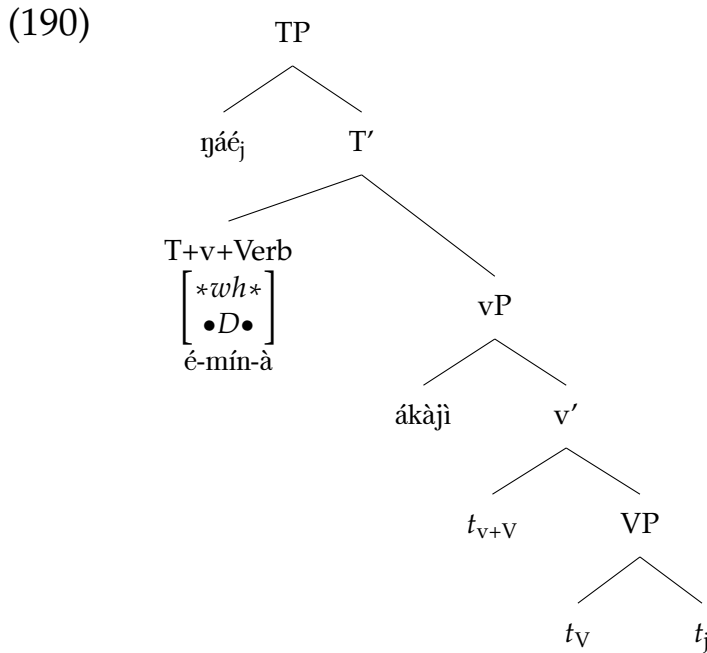
Although one could consider a split CP analysis in the sense of Rizzi (1997) like van Urk (2015) does for Dinka, there is currently no empirical evidence from Turkana that motivates the postulation of

such a fine-grained CP structure.

Thus, I assume that T carries syntactic features such as [focus] or [wh]. In addition, T can also carry an additional EPP feature to attract these focal or wh-elements. Since this movement is optional, I argue that this EPP feature itself is optional. I follow Heck & Müller (2007); Müller (2010) and notate probes that are valued via Agree without movement as $[*\pi*]$ and probes that trigger movement as $[\bullet\pi\bullet]$. Thus, the EPP-feature is notated as $[\bullet D \bullet]$. The optionality of movement will be further discussed in 4.5. To demonstrate the derivation, take the following sentence:

- (189) ɲáé é-mín-à ákàjì?
 who.DEFL 3-like-TAM Akai.NOM
 ‘Who does Akai like?’

The wh-element is the object of the sentence. It is attracted by the wh-feature on T. The subject stays in-situ in vP:



I have adopted van Urk’s (2015) approach which rejects the existence of dedicated A- and $\bar{A}$ -positions, arguing instead that movement is feature-driven. Further, I have argued that in Turkana, both focus and wh-movement target the specifier of TP, rather than the specifier of CP.

4.4 Post-verbal Focus Position

In this section, I discuss the landing site of the post-verbal copy shown in the following example:

- (191) $[_{CP} \text{ é-jén-ì} \quad \text{ákìrù} \quad [_{CP} \text{ átàmàr í-tám-ì}$
 3-know.PRS-TAM Akiru.NOM that 3-think-TAM
 $\text{ákàjì} \quad \boxed{\text{ɲáé}} \quad [_{CP} \text{ átàmàr é-mín-à}$
 Akai.NOM $\boxed{\text{who.DEFL}}$ that 3-love.PRS-TAM
 $\text{ádžìkòn} \quad \text{ɲáé} \quad]]]?$
 Adjikon.NOM who.DEFL
 ‘Who does Akiru know that Akai thinks that Adjikon loves?’

I argue that the post-verbal wh-element lands in the specifier of the vP of the higher CP, not in the specifier of the CP of the embedded clause, as suggested by the bracket notation. This subsection further explores the constraints and structure of this post-verbal position.

4.4.1 The Post-Verbal Domain

In this subsection, I show that the post-verbal wh-element lands in the vP domain of the higher clause, rather than the specifier of CP of the embedded clause. This claim is supported by adverbial data. The surface position of the wh-element in the following sentence does not make its landing site immediately clear:

- (192) $\text{é-jén-ì} \quad \text{ákìrù} \quad \text{átàmàr í-tám-ì}$
 3-know.PRS-TAM Akiru.NOM that 3-think-TAM
 $\text{ákàjì} \quad \boxed{\text{ɲáé}} \quad \text{átàmàr é-mín-à}$
 Akai.NOM $\boxed{\text{who.DEFL}}$ that 3-love.PRS-TAM
 $\text{ádžìkòn} \quad \text{ɲáé?}$
 Adjikon.NOM who.DEFL
 ‘Who does Akiru know that Akai thinks that Adjikon loves?’

Structurally, the wh-element could either be in the specifier of CP of the lower clause (193a) or in the post-verbal position of the higher clause (193b):

- (193) a. $[_{CP} \dots [_{CP} \dots [_{CP} \text{ ɲáé átàmàr } \dots]]]$
 b. $[_{CP} \dots [_{CP} \dots \text{ ɲáé } [_{CP} \text{ átàmàr } \dots]]]$

- (196) a. [_{CP} í-tám-ì ákìrù [_{CP} átàmàr
 3-think.PRS-TAM Akiru.NOM that
 é-jén-ì ákàjì **nóí** ɲáé [_{CP} átàmàr
 3-know.PRS-TAM Akai.NOM **well** who.DEFL that
 é-mín-á ádzìkòn ɲáé]]]?
 3-love.PRS-TAM Adjikon.NOM who.DEFL
 b. [_{CP} í-tám-ì ákìrù [_{CP} átàmàr
 3-think.PRS-TAM Akiru.NOM that
 é-jén-ì ákàjì ɲáé **nóí** [_{CP} átàmàr
 3-know.PRS-TAM Akai.NOM who.DEFL **well** that
 é-mín-á ádzìkòn ɲáé]]]?
 3-love.PRS-TAM Adjikon.NOM who.DEFL
 ‘Who does Akiru think that Akai knows well that Adjikon
 loves?’

Since the adverb cannot appear in the pre-verbal domain, the *wh*-element cannot be in the specifier of the CP of the embedded clause. Instead, it must occupy the post-verbal domain of the higher CP.

Additional evidence comes from the ungrammaticality of multiple *wh*-elements within a single CP, even when the two elements are almost next to each other. (197a) is ungrammatical because the two *wh*-elements are located in the same CP. However, (197b) is grammatical.

- (197) a. *[_{CP} é-jén-ì ákìrù [_{CP} átàmàr
 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì [_{CP} átàmàr ɲáé
 3-think-TAM Akai.NOM that **who.DEFL**
 é-mín-à ádzìkòn ɲáé]]]?
 3-love.PRS-TAM Adjikon.NOM **who.DEFL**
 b. [_{CP} é-jén-ì ákìrù [_{CP} átàmàr
 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì ɲáé [_{CP} átàmàr
 3-think-TAM Akai.NOM **who.DEFL** that
 ɲáé é-mín-á ádzìkòn]]]?
who.DEFL 3-love.PRS-TAM Adjikon.NOM
 ‘Who does Akiru know that Akai thinks that Adjikon
 loves?’

This pattern supports the claim that the *wh*-element does not occupy the specifier of the embedded CP, as this would result in an illicit configuration of two *wh*-elements being in the same CP.

In summary, the *wh*-element surfaces in a post-verbal position in the higher clause and does not move to the specifier of the CP of the embedded clause. This conclusion is supported by adverbial diagnostics and the constraint on multiple *wh*-elements within a single CP.

4.4.2 Focus Movement to the Specifier of *v*

Having established that *wh*-elements land in a post-verbal position in Turkana, I now address their precise syntactic landing site. I argue that these elements optionally land in the inner specifier of *vP*, motivated by an optional EPP feature on *v*.

Before introducing this analysis for Turkana, I discuss relevant evidence from Dinka where *wh*-elements supposedly also move through the (outer) specifier of *vP*. Although in Dinka constituents do not surface post-verbally, movement via these positions is detectable through morphological reflexes. [van Urk \(2015, 2018\)](#); [van Urk & Richards \(2015\)](#) argue that long-distance movement in Dinka proceeds cyclically through every specifier of *vP* and CP. They argue that movement through the specifier of *vP* and CP is motivated by phasality.

The first piece of evidence for successive-cyclic movement through *vP* comes from extraction data. The focus here will be restricted to the *vP* domain in this section. As shown in [4.3.1](#), Dinka is a V2 language where the verb raises to C. In contexts where the verb does not raise to C, such as in the presence of an auxiliary, only one constituent can precede the verb. [van Urk \(2015\)](#); [van Urk & Richards \(2015\)](#) identify this position as the specifier of *vP*. The following examples show that having only the indirect ([198a](#)) or only the direct object ([198b](#)) in the specifier of *vP* is grammatical; the position cannot be empty independent of word order following the verb ([198c, 198d](#)); the position cannot be filled with more than one element independent of the word order ([198e, 198f](#)).

- (198) a. *yêɛn cé ayén yiɛn kítáp.*
 I PRF Ayen give book

- 'I gave Ayen a book.'
- b. yêen cê kítáp yiên ayén.
I PRF book give Ayen
'I gave a book to Ayen.'
- c. *yêen cê — yiên kítáp ayén.
I PRF give book Ayen
- d. *yêen cê — yiên ayén. kítáp
I PRF give Ayen book
- e. *yêen cê kítáp ayén yiên.
I PRF book Ayen give
- f. *yêen cê ayén kítáp yiên.
I PRF Ayen book give (Dinka,

van Urk & Richards 2015: 122)

However, when a constituent is extracted from a vP, the specifier of that vP must be empty (van Urk 2015; van Urk & Richards 2015). While this might initially appear to contradict the previous generalisation illustrated in (198), van Urk argues that the moved element leaves a gap in the specifier of vP, accounting for the absence of any overt material in that position. The following example illustrates this point:

- (199) a. yejà cǐ môc — yiên kítáp.
who PRF.NS man.GEN give book
b. *yejà cǐ môc kítáp yiên.
who PRF.NS man.GEN book give
'Who did the man give the book to?' (Dinka, van Urk & Richards 2015: 125)

Additional support for intermediate movement through the specifier vP comes from the plural morpheme *ké* which surfaces when a plural DP is extracted. *ké* appears in the specifier of every vP that the extracted DP crosses showing that the DP passes through these positions:

- (200) yeyínà cǐ bôl ké tǐŋ?
who.PL PRF.NS Bol.GEN PL see
'Who all did Bol see?' (Dinka, van Urk & Richards 2015: 127)

- (204) a. é-mín-à ákàjì ɲáé?
 3-like-TAM Akai.NOM who.DEFL
 b. ɲáé é-mín-à ákàjì?
who.DEFL 3-like-TAM Akai.NOM
 ‘Who does Akai like?’

Furthermore, I showed that wh-elements can also move to the pre- (205a) or post-verbal (205b) position of a higher CP as in:

- (205) a. [_{CP} é-jén-ì ákìrù [_{CP} átàmàr ɲáé
 3-know.PRS-TAM Akiru.NOM that who.DEFL
 í-tám-ì ákàjì [_{CP} átàmàr é-mín-à
 3-think-TAM Akai.NOM that 3-love.PRS-TAM
 ádzìkòn ɲáé]]]?
 Adjikon.NOM who.DEFL
 b. [_{CP} é-jén-ì ákìrù [_{CP} átàmàr
 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì ɲáé [_{CP} átàmàr
 3-think-TAM Akai.NOM who.DEFL that
 é-mín-à ádzìkòn ɲáé]]]?
 3-love.PRS-TAM Adjikon.NOM who.DEFL
 ‘Who does Akiru know that Akai thinks that Adjikon
 loves?’

In previous sections, I argued that v and T can have an optional EPP-feature. Movement to the specifier of v or T occurs only if the EPP feature is present. In this section, I elaborate on the underlying structure that accounts for this optionality and draw cross-linguistic parallels from Dholuo.

The optional nature of the EPP feature has previously been observed in another Nilotic language, Dholuo, by Cable (2012). Dholuo, a Western Nilotic language with SVO word order:

- (206) Ochieng’ ne ok oneno Onyango.
 Ochieng’ PAST NEG saw Onyango
 ‘Ochieng didn’t see Onyango.’ (Dholuo, Cable 2012: 655)

Cable (2012) proposes the following clause structure for Dholuo:

- (207) [_{CP} C [_{TP} TENSE-PARTICLES [_{NegP} NEG [_{AspP} PERF/IMPERF [_{VP}

SUBJ ... V ... OBJ]]]]]

While subjects typically move to the specifier to T, Cable (2012) observes that they may also appear after the tense particle tense particle (208a) or the negation (208b):

- (208) a. ne Ochieng' ok oneno Onyango.
 PAST Ochieng' NEG saw Onyango
 b. ne ok Ochieng' oneno Onyango.
 PAST NEG Ochieng' saw Onyango
 'Ochieng didn't see Onyango.' (Dholuo, Cable 2012: 657)

Wh-movement in Dholuo is likewise optional:

- (209) a. Ng'a ma Achieng' enono?
 who C Achieng' oneno
 b. Achieng' oneno ng'a?
 Achieng' saw who
 'Who did Achieng' see?'

Cable (2012) argues that these instances of movement are truly optional determined by the presence or absence of an EPP feature. Heads that lack an EPP feature cannot host a specifier. This means that the specifier of the CP, TP, NegP, and AspP can stay empty. Thus, Cable concludes that C, T, Neg, and Asp do not need to bear an EPP-feature but can. He summarises this principle as follows:

- (210) If a head *lacks* the feature EPP, then it is not possible to (internally) merge a phrase in its specifier position. (Cable 2012: 685)

This implies that movement is possible only when it is triggered by an EPP-feature and that phrasal movement can never occur freely. Cable (2012) then concludes that the EPP feature must be optional in Dholuo.

Cable (2012) is not the first to question the universality of the EPP feature. As discussed in 4.2.3, the existence of an EPP feature has also been challenged for V1 languages such as Irish, where the subject is argued to remain below T (McCloskey 1996). Turkana aligns with the Irish pattern: It is a V1 language in which the subject remains below T, indicating that the specifier of T can remain empty.

To summarise, in this section I have argued that the EPP-feature is optional in Turkana: Both T and v can carry an EPP-feature but do not have to. This accounts for the observed optionality of focus and wh-movement in the language. When the EPP feature is absent, movement to the specifier is blocked, and the position remains empty.

In this chapter, I examined the syntactic structure of Turkana, focusing on the canonical VSO word order and the landing sites of movement. I argued that the verb raises to T via head movement, while the subject remains in-situ. This results in an empty specifier of TP, consistent with analyses of other V1 languages.

I furthermore explored the landing sites of focal and wh-elements in long-distance movement. I showed that these elements do not land in the specifier of CP how it is usually assumed, but rather land in the specifier of TP. For post-verbal landing site, I demonstrated that the element moves into the vP of the higher clause, landing

below the subject in an inner specifier of vP. These findings challenge standard phase-based theories, particularly with regard to the locality of successive-cyclic movement.

In the following chapter, I address the implications of these structures for phasality and discuss the locality constraints in Turkana.

Chapter 5

Locality Constraints in Long Distance Movement

5.1 Introduction

In this chapter, I argue that the locality constraints observed in long-distance wh-movement in Turkana require a theory that goes beyond standard phase-based accounts. As shown in Chapter 3, Turkana prohibits constructions with three CPs and only one overt wh-element. Here, it does not matter whether the wh-element is in- or ex-situ. These patterns cannot be adequately explained under standard phase theory where CP and vP are phases. I show that in Turkana only CPs are phases and that PIC₂ applies.

To account for the full range of locality effects in Turkana, I further adopt the system of selective opacity developed by Keine (2019, 2020b), in which locality is determined by horizons assigned to individual probes. Under this approach, each probe is equipped with a horizon that specifies the boundary of its search space.

While the selective opacity framework provides the right architecture to account for probe-specific locality, it requires several modifications to accommodate the Turkana data. I propose that C bears an edge feature with a particularly narrow horizon (to V), which blocks certain instances of cyclic movement. This horizon-based restriction simulates the effect of vP-phases without invoking absolute phasal opacity. To capture this configuration, I reject the assumption of category inheritance, which would otherwise forbid such a horizon on C.

Under PIC₂, wh-elements do not have to move via the specifier of every CP. However, C still bears an edge feature, which remains

unchecked if no element moves to its specifier. While this does not result in ungrammaticality per se, multiple unchecked C heads can trigger so called *ganging up effects* (cf. Pater 2009; Haegeman, Jiménez-Fernández & Radford 2014; Murphy 2017, 2018a,b) lead to ungrammaticality. I will analyse these ganging up effects under the framework of Serial Harmonic Grammar a constraint-based theory originally developed for phonology (see e.g. McCarthy 2000, 2008; Kimper 2011, 2016) and later extended to syntax (e.g. Heck & Müller 2003, 2013; Murphy 2017, 2018a,b; Müller, Englisch & Opitz 2022). Critically, adopting Serial Harmonic Grammar does not require abandoning the assumptions about phases and horizons. Rather, it complements them by providing a mechanism for evaluating cumulative violations within phase-based derivations.

The chapter is structured as follows. In 5.2, I show that standard phase-based locality is too restrictive for Turkana. sec:selective-opacity introduces the selective opacity system and argues for its theoretical advantages. In 5.4, I adapt the system to Turkana, proposing language-specific horizons. In 5.5, I derive the observed data patterns through a series of case studies. 5.6 concludes the chapter.

5.2 Previous Analyses on Phases

In this section, I explore the most common theories of phase theory and why they cannot account for the long distance dependency patterns in Turkana. Since the development of Phase Theory by Chomsky (2000, 2001), it has been widely assumed that vP and CP constitute phases. This standard implementation has been adopted in a number of studies, including van Urk (2015); van Urk & Richards (2015), who provide detailed empirical support for the phasiness of vP and CP in Dinka.

However, there is a growing number of scholars who question the status of vP as a phase. For instance, Keine (2017, 2020b) argue, based on long-distance agreement and wh-licensing in Hindi-Urdu, that vP does not function as a phase. Similarly, Keine & Zeijlstra (to appear) reanalyse the Dinka data from van Urk (2015), arguing that the patterns previously taken as evidence for phasal vP behaviour are better explained by intervention effects.

Nevertheless, even more permissive versions of phase theory, as

suggested by Keine (2017, 2020b), cannot capture the long distance dependency pattern in Turkana. In what follows, I survey these previous analyses and demonstrate why none of them can account for the locality behaviour observed in Turkana long-distance movement.

5.2.1 vP and CP as Phases

In standard phase theory (Chomsky 2000, 2001), syntactic derivation proceeds in cycles known as phases. Once a phase is sent to spell out, its internal domain becomes opaque to further syntactic operations, a restriction formalised as the *Phase Impenetrability Condition* (PIC). The Phase Impenetrability Condition states that once a phase is completed, its internal structure is no longer accessible for syntactic operations from outside. Only *the edge*, the phase head and its specifier, remain visible for further computation:

- (215) *Phase Impenetrability Condition*₁ (PIC₁) (Chomsky 2000):
The domain of a head X of a phase XP is not accessible to operations outside XP; only X and its edge are accessible to such operations.

Elements can escape a phase to remain accessible for further syntactic operations by moving out of the phase via the *escape hatch*, the specifier of the phase head. Therefore, in order to still be accessible, fronted elements must move through the phase edge of every phase which is called successive-cyclic movement (Chomsky 2000, Chomsky 2001). This is schematically illustrated below:

- (216)

One of the clearest empirical implementations of a strict phase-based derivation is found in Dinka, a Western Nilotic language extensively analysed by van Urk (2015); van Urk & Richards (2015). Dinka provides compelling evidence for successive-cyclic movement through the specifiers of vP and CP. van Urk (2015); van Urk & Richards (2015) argue that elements move cyclically through the edge of every phase. We have already seen this in 4.4.2 for vP. The same pattern also holds for CPs. Dinka is a V2 language in which exactly one nominal element must precede the verb (van Urk 2015,

- 'The woman that has seen Bol.'

- b. tíŋ cǐ Bòl tǐŋ.
 woman.CS PRF.OV Bol.GEN see.NF
 (O-V-S)

‘The woman that Bol has seen.’

- c. *tíŋ Bòl cǐ tǐŋ.
 woman.CS Bol PRF.SV see.NF
 (S-O-V)

‘The woman that has seen Bol.’

- d. *tíŋ Bòl cé tǐŋ.
 woman.CS Bol PRF.SV see.NF
 (O-S-V)

‘The woman that Bol has seen.’ (Dinka, [van Urk 2015: 97](#))

When a constituent is extracted from a CP, the pre-verbal position must be empty ([van Urk 2015](#); [van Urk & Richards 2015](#)). At first glance, this seems to contradict the generalisation illustrated in (217). However, van Urk argues that the specifier must be empty because the moved element leaves a gap behind rendering the position unavailable for other elements. The following example shows that the position preceding the verb must be empty when a constituent is moved from it:

- (218) a. yeyà cúkkù luél, [__ cé kítáp yòoc]
 who PRF.1.PL say PRF book buy.TR
 b. *yeyà cúkkù luél, [kítáp (à-)cǐ yòoc]
 who PRF.1.PL say book 3.SG-PRF.NS buy.TR
 ‘Who did we say bought a book?’ (Dinka, [van Urk & Richards 2015: 125](#))

The pattern is the same for vPs (see 4.4.2). Moreover, plural DPs leave a morphological reflex behind in the specifier of vP:

- (219) yeyínà yé ké tâak [CP cǐ bôl ké tǐŋ]?
 who.PL HAB.2.SG PL think PRF.NS Bol.GEN PL see
 ‘Who all do you think Bol saw?’ (Dinka, [van Urk & Richards 2015: 128](#))

van Urk (2015, 2018); van Urk & Richards (2015) interpret the *ké*-reflexes and the empty edge of vP and CP after movement as evidence for strict successive cyclic movement through the specifier of every vP and CP in Dinka.

Although the Dinka data support successive-cyclic movement through vP and CP, the status of vP as a phase is questioned by constructions that seem to violate phase-based opacity. Consider passives and unaccusatives for example. On standard assumptions, T assigns nominative case to the subject (e.g., Chomsky 1995) but in unaccusatives and passives, the subject originates as a complement of V. If vP is a phase, this position should be inaccessible to T once vP has been spelled out—unless an escape hatch is used. Yet T can agree with and assign case to such arguments, even when they remain in-situ.

- (220) a. There arrive-s/*-Ø a cat.
b. There arrive*-s/-Ø many cats.

To account for this, Chomsky (2000) proposes that passive and unaccusative vPs are *weak* and do not constitute phases. However, Chomsky (2001) revised the PIC to make it less powerful. Under PIC₂, the complement of a phase head remains accessible until the next phase head is merged:

- (221) *Phase Impenetrability Condition*₂ (PIC₂) (Chomsky 2001):
If XP is a phase headed by X, the complement of X remains accessible for operations outside XP until the next phase head is merged. After that point, only the edge of XP remains accessible.

Müller (2004a) and Richards (2011) observe that only PIC₂ can explain agreement with low nominative objects in Icelandic quirky subject constructions. These sentences are fully transitive and involve a dative subject and a nominative object. However, the nominative DP remains low, yet T still agrees with it and assigns nominative case:

- (222) henni leiddust strákanir.
her.DAT bored.3.PL boys.NOM.PL
'She found the boys boring.' (Icelandic, Sigurðsson 1996: 1)

This revised formulation of the PIC accounts for cases like Icelandic

quirky subjects by delaying Spell-Out. However, it raises a more fundamental question whether vPs are phases at all. In the next subsection, I explore alternative proposals in which only CP qualifies as a phase.

5.2.2 CP as the Sole Phase

Although PIC₂ relaxed the strict locality conditions to account for phenomena such as quirky subjects in Icelandic, several scholars have gone further by rejecting the phasal status of vP altogether (Grano & Lasnik 2018; Keine 2017, 2020a,b; Keine & Zeijlstra to appear). This subsection examines this line of analysis, focusing on evidence from Hindi-Urdu.

Keine (2017, 2020b) argues that in Hindi-Urdu, only CP functions as a phase, but not vP. His argument is based on patterns of long-distance φ -agreement and wh-licensing.

The verb in Hindi-Urdu agrees with the structurally highest argument that is not overtly case-marked. If the subject is overtly case marked, the verb agrees with the object (Pandharipande & Kachru 1977). Crucially, this agreement still obtains even when the object remains in-situ and resists movement (Bhatt 2005; Bhatt & Keine 2017; Keine 2017, 2020b). This is illustrated in the following example. In both examples, the verb agrees with the object *biin* ('flute'). However, the object resists movement, meaning the sentence loses its idiomatic reading when the object is fronted:

- (223) a. raam-ne bhains ke aage **biin** bajaa-**yii**.
 Ram-ERG buffalo in front of **flute** play-**PERF.F.SG**
 'Ram did something futile.' (*lit.* 'Ram played the flute in front of a buffalo.')
- b. #**biin_i** raam-ne bhains ke aage *t_i* bajaa-**yii**.
flute Ram-ERG buffalo in front of play-**PERF.F.SG**
 'Ram played the flute in front of a buffalo.' (*idiomatic reading deviant*) (Hindi-Urdu Keine 2017: 178-179)

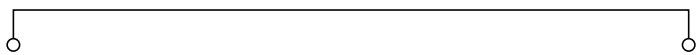
It is usually assumed that the φ -probe is located on T (Bhatt 2005; Bhatt & Keine 2017; Keine 2017, 2020b). Under PIC₂, this pattern is unproblematic: although the agreeing DP remains inside the vP domain, it is still accessible to T until the next phase head, C, is

merged.

However, Hindi-Urdu also allows long distance agreement that crosses multiple vPs (cf. Mahajan 1990; Butt 1995; Bhatt 2005; Chandra 2007; Bhatt & Keine 2017; Keine 2017, 2020b). In the following example, the matrix verb *caahii* (wanted) agrees with the embedded object *biin* ('flute').

- (224) raam-ne [bhains ke aage **biin** bajaa-nii]
 Ram-ERG buffalo in front of **flute** play-INF.F.SG
 cah-**ii**.
 want-PERF.F.SG
 'Ram wanted to something futile.' (*lit.* 'Ram wanted oot the flute in front of a buffalo.') (Hindi-Urdu Keine 2017: 179)

Bhatt (2005); Bhatt & Keine (2017); Keine (2017, 2020b) argue that the object remains in-situ in such constructions. Furthermore, Dayal (1996); Bhatt (2005); Chandra (2007) show that these non-finite clauses in these constructions contain a vP-projection but miss a CP layer. As a result, the φ -probe on T in (224) must agree with a DP that is embedded within two vPs. The structure is schematised below:

- (225)  [TP T_[*φ*] [vP v [VP V [nonfinite clause [vP v [VP V DP_[φ]]]]]]] (Keine 2020b: 277)

Such long-distance agreement is unexpected if vP is a phase, as the targeted DP would be rendered inaccessible after the first vP is completed and sent to spell out. PIC₂ also fails to explain this, as it only delays spell-out until the next phase head is merged, not the next two.

A similar pattern arises with wh-licensing. In Hindi-Urdu, wh-movement is optional, and there is no evidence for covert movement of in-situ wh-elements (Mahajan 1990; Dayal 1996, 2017; Beck 2006). When a wh-phrase appears in a VP-internal position, it remains there throughout the derivation:

- (226) raam-ne kyaa khaa-yaa thaa?
 ram.ERG what eat.PERF.M.SG be.PST.M.SG
 'What did Ram eat?' (Hindi-Urdu, Mahajan 2000: 125, retrieved from Keine 2020b)

It is generally assumed that C licenses wh-elements (Chomsky 1995). Under PIC₂, this is possible only if the wh-element is embedded in one phase. However, licensing persists even across two vPs:

- (227) tum [**kyaa** kar-naa] jaan-te
 you **what** do-INF.M.SG know-IMPERF.M.PL
 ho?
 be.PRS.2.PL
 ‘What do you know to do?’ (Hindi-Urdu, Dayal 1996: 23
 retrieved from Keine 2017)

The structure is schematised below:

- (228) [CP C_[*wh*] [TP T [vP v [VP V [nonfinite clause [vP v [VP V DP_[wh]]]]]]]]
- (Keine 2020b: 279)

Again, this is unexpected if vP were a phase: C can license the *wh*-element that is embedded in two vPs. PIC₂ cannot account for this either, since it only delays spell-out until the next phase head is merged, and not the next two phase heads.

Interestingly, CPs in Hindi-Urdu behave exactly as predicted under standard phase theory. When a CP is present, long-distance agreement and *wh*-licensing are blocked. This asymmetry strengthens the claim that CP, but not vP, acts as a phase boundary in Hindi-Urdu.

First, long-distance agreement is not possible when the embedded clause is a CP. Control verbs, which take CPs as complements (Chomsky 1981b; Landau 2001, 2013), cannot agree with the embedded object:

- (229) laṛkō-ne soc-aa/[*-ii] [CP ki Monaa-ne
 boys-ERG think-PFV.M.SG/[*-PFV.F.SG] that Mona-ERG
 ghazal gaa-yii thii].
 ghazal.F sing.PERF.F.SG be.PST.F.SG
 ‘Firoz thought that Mona sings ghazals.’ (Hindi-Urdu,
 Keine 2017: 181)

Here, the embedded CP blocks agreement between the matrix T and the object *ghazal*. This is exactly what is expected if CP is a phase and has already been spelled out when matrix T is merged.

Similarly, wh-licensing is blocked across CP boundaries. If the wh-element remains inside the embedded clause, the sentence is ungrammatical:

- (230) *siitaa-ne soc-aa [CP ki ravii-ne kis-ko
 Sita-ERG think-PERF-M-SG that Ram-ERG who-ACC
 dekh-aa].
 see-PERF-M-SG
 ‘Who did Sita think Ravi saw?’ (Hindi-Urdu, Mahajan 2000:
 319, retrieved from Keine 2017)

Again, this is expected under standard phase theory: Once the embedded CP has been spelled out, its internal constituents are no longer accessible for licensing by the matrix C head. The embedded wh-element *kisko* is already sent to spell out when the C head of the matrix clause is merged.

Keine (2017, 2020b) thus concludes that vP is not a phase in Hindi-Urdu. Going further, he argues that vP is never a phase in any language. Instead, he proposes a feature-based Horizon approach to locality that redefines when syntactic material becomes inaccessible. This alternative framework will be discussed in the next section.

However, the strongest empirical argument for treating vP as a phase comes from Dinka (van Urk 2015; van Urk & Richards 2015) where fronting of plural objects behind a morphological reflex (see 4.4.2). This reflex has been taken to indicate intermediate movement through the specifier of vP, suggesting that vP functions as a phase with an escape hatch. However, when plural subjects are fronted, the reflex is ungrammatical:

- (231) ròoor áa- cé (*ké) yîin tîŋ.
 man.PL 3.PL-PFV (*PL) you see.NFIN
 ‘The men have seen you.’ (Dinka, [van Urk 2018: 950](#))

[Keine & Zeijlstra \(to appear\)](#) challenge this interpretation and argue that this asymmetry cannot be explained by appealing to phase-based movement. Instead, they propose that *ké* is not a phase diagnostic at all, but rather a reflex of intervention: It appears when the movement of a DP crosses over another DP in the clause, creating a configuration in which an intermediate copy must be spelled out. In object fronting, the object crosses the subject (which is itself a DP), triggering the insertion of *ké*. In subject fronting, by contrast, no such DP-crossing occurs, and no reflex appears. Crucially, *ké* is also absent in unaccusatives, where no external argument intervenes, reinforcing the view that the reflex arises not from phase boundaries, but from clause-internal intervention effects.

This section has shown that vPs are not universally phases, and may in fact never function as phases. The next sub section will show that standard approaches for phasality do not account for locality constraints in Turkana.

5.2.3 Issues with Standard Approaches for Phasality in Turkana

As briefly introduced in [3.4.3](#), long-distance *wh*-movement in Turkana poses a challenge to standard assumptions about phasality. This section explores the empirical patterns in more depth and argues that conventional approaches cannot capture the locality constraints attested in Turkana.

In Chapter [3](#), I showed that *wh*-elements in Turkana can appear both in-situ and ex-situ. Importantly, there is no covert movement in *wh* in-situ constructions but I assume that the *wh*-element moves to an inner specifier of vP and that the lowest copy is deleted. Ex-situ *wh*-elements undergo syntactic movement rather than being base-generated in a high position.

Wh-elements in Turkana can be licensed or extracted across a CP boundary. This is shown in the following examples, where the *wh*-element *ŋáé* (‘who’) appears either in-situ within the embedded clause ([232a](#)) or in a fronted, ex-situ position ([232b](#)).

- (232) a. [_{CP} é-jén-ì ákìrù [_{CP} (átàmàr)
 3-know.PRS-TAM Akiru.NOM that
 é-mín-à ákàjì **ɲáé**]]
 3-love.PRS-TAM Akai.NOM **who.DEFL**
 b. [_{CP} **ɲáé** é-jén-ì ákìrù [_{CP}
who.DEFL 3-know.PRS-TAM Akiru.NOM
 (átàmàr) é-mín-à ákàjì]]
 that 3-love.PRS-TAM Akai.NOM
 ‘Who does Akiru know that Akai loves?’

However, if a second CP layer is added, sentences with only one wh-element become ungrammatical, regardless of whether the wh-element remains in-situ or undergoes movement. The sentence is only grammatical if there is an additional copy:

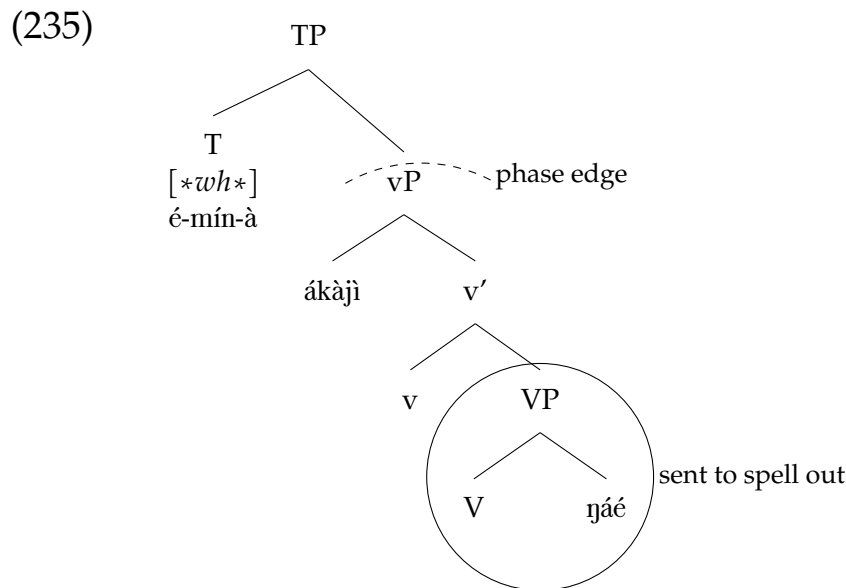
- (233) a. *[[_{CP} é-jén-ì ákìrù [_{CP} átàmàr
 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì [_{CP} átàmàr é-mín-à
 3-think-TAM Akai.NOM that 3-love.PRS-TAM
 ádzìkòn **ɲáé**]]]?
 Adjikon.NOM **who.DEFL**
 b. *[[_{CP} **ɲáé** é-jén-ì ákìrù [_{CP} átàmàr
who.DEFL 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì [_{CP} átàmàr é-mín-à
 3-think-TAM Akai.NOM that 3-love.PRS-TAM
 ádzìkòn]]]?
 Adjikon.NOM
 c. [_{CP} **ɲáé** é-jén-ì ákìrù [_{CP} átàmàr
who.DEFL 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì [_{CP} átàmàr é-mín-à
 3-think-TAM Akai.NOM that 3-love.PRS-TAM
 ádzìkòn **ɲáé**]]]?
 Adjikon.NOM boxed **who.DEFL**
 ‘Who does Akiru know that Akai thinks that Adjikon loves?’

This pattern is puzzling: if vP and CP were phases, licensing wh-elements in-situ should not be possible when the wh-element is

in an embedded CP. On the other hand, the *wh*-element should be able to escape each phase by moving cyclically to the front. In fact, the positions where *wh*-elements surface, the pre- and post-verbal positions, are consistent with the expected phase edges. The following example shows where the additional copy can appear when the *wh*-element is fronted:

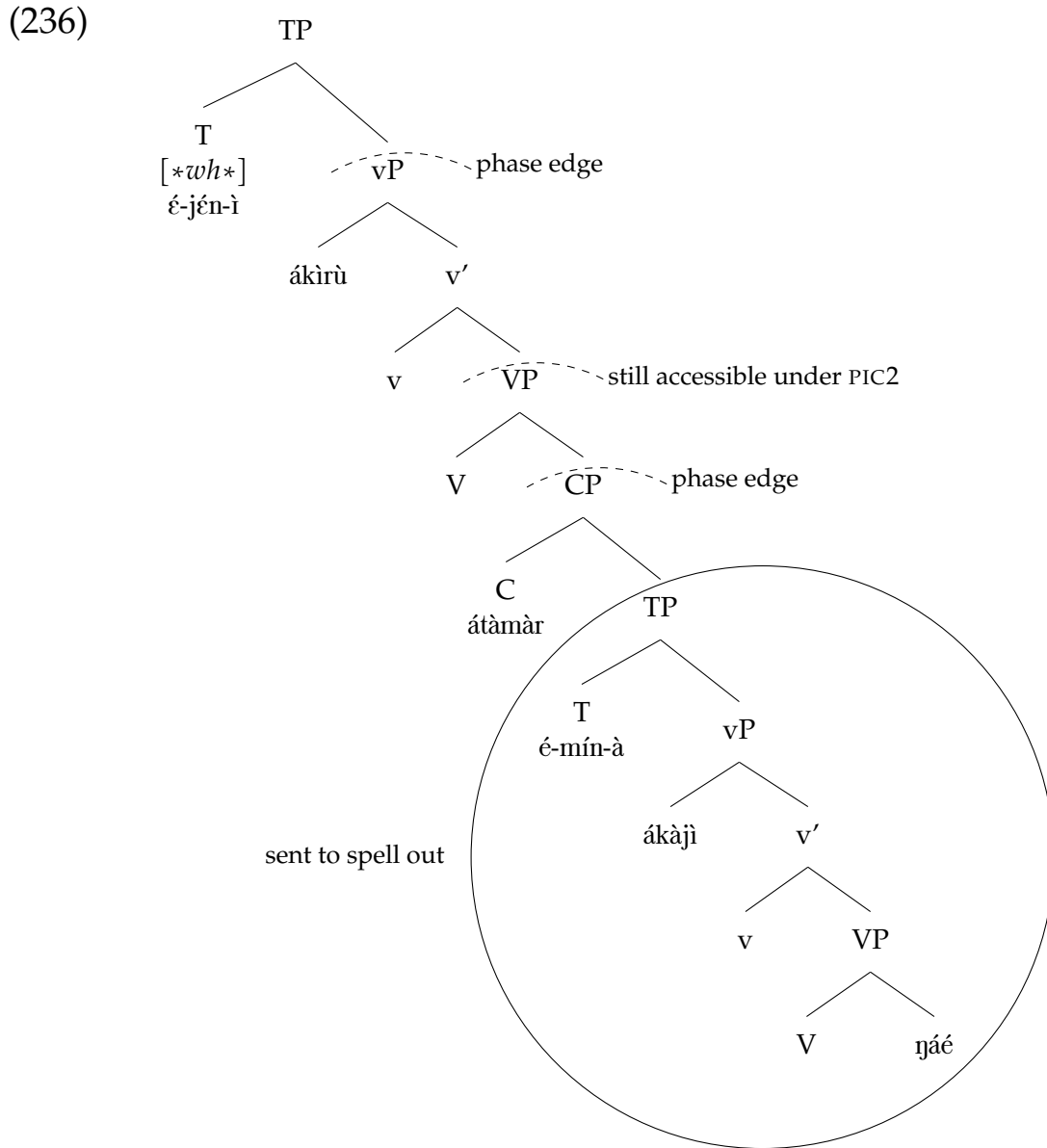
- (234) [CP **ŋáé** who.DEFL é-jén-ì ákìrù [CP átàmàr ✓
 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì ✓ [CP átàmàr ✓ é-mín-à
 3-think-TAM Akai.NOM that 3-love.PRS-TAM
 ádzìkòn ✓]]]?
 Adjikon.NOM
 ‘Who does Akiru know that Akai thinks that Adjikon loves?’

The positions of the copies leads to assume that Turkana behaves like [van Urk \(2015\)](#); [van Urk & Richards \(2015\)](#) suggested for Dinka: vP and CP are phases and *wh*-elements move cyclically through the edge of every vP and CP. However, the in-situ data suggest otherwise. *Wh*-elements can be licensed when they stay in-situ. In [4.3.2](#), I showed that the *wh*-probe is located on T. If vP were a phase, T would not be able to license a *wh*-element that stays in-situ as a complement of VP:



However, PIC₂ could potentially still explain that pattern as it has been done for quirky subjects in Icelandic. The *wh*-element would be accessible to T until the next phase head, C, is merged. However,

this cannot explain the ungrammaticality of examples involving two CPs, where the *wh*-element is embedded in the lower CP and still must be licensed by the matrix T. PIC₂ delays spell-out until the next phase head is merged, and not the next two phase heads:



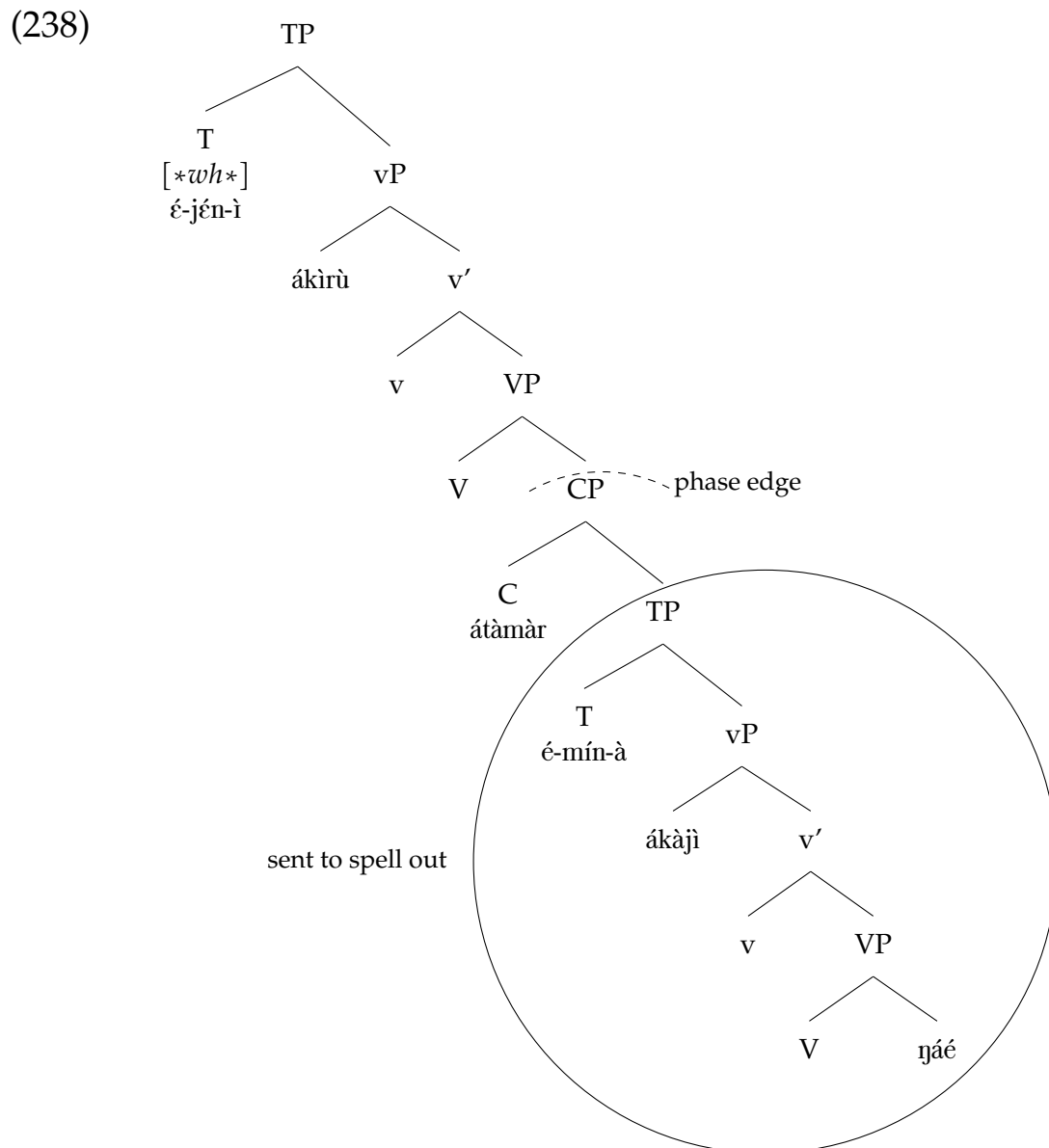
As shown above, Keine (2017, 2020a,b) suggests that vPs are not phases but only CPs are. The single CP data with the *wh*-element then poses no issue. However, if CPs would be phases we would expect Turkana behave like Hindi-Urdu where *wh* in-situ in one CP is grammatical but ungrammatical with two CPs:

- (237) a. raam-ne kyaa khaa-yaa thaa?
 ram.ERG what eat.PERF.M.SG be.PST.M.SG
 ‘What did Ram eat?’ (Hindi-Urdu, Mahajan 2000: 125,

retrieved from [Keine 2020b](#))

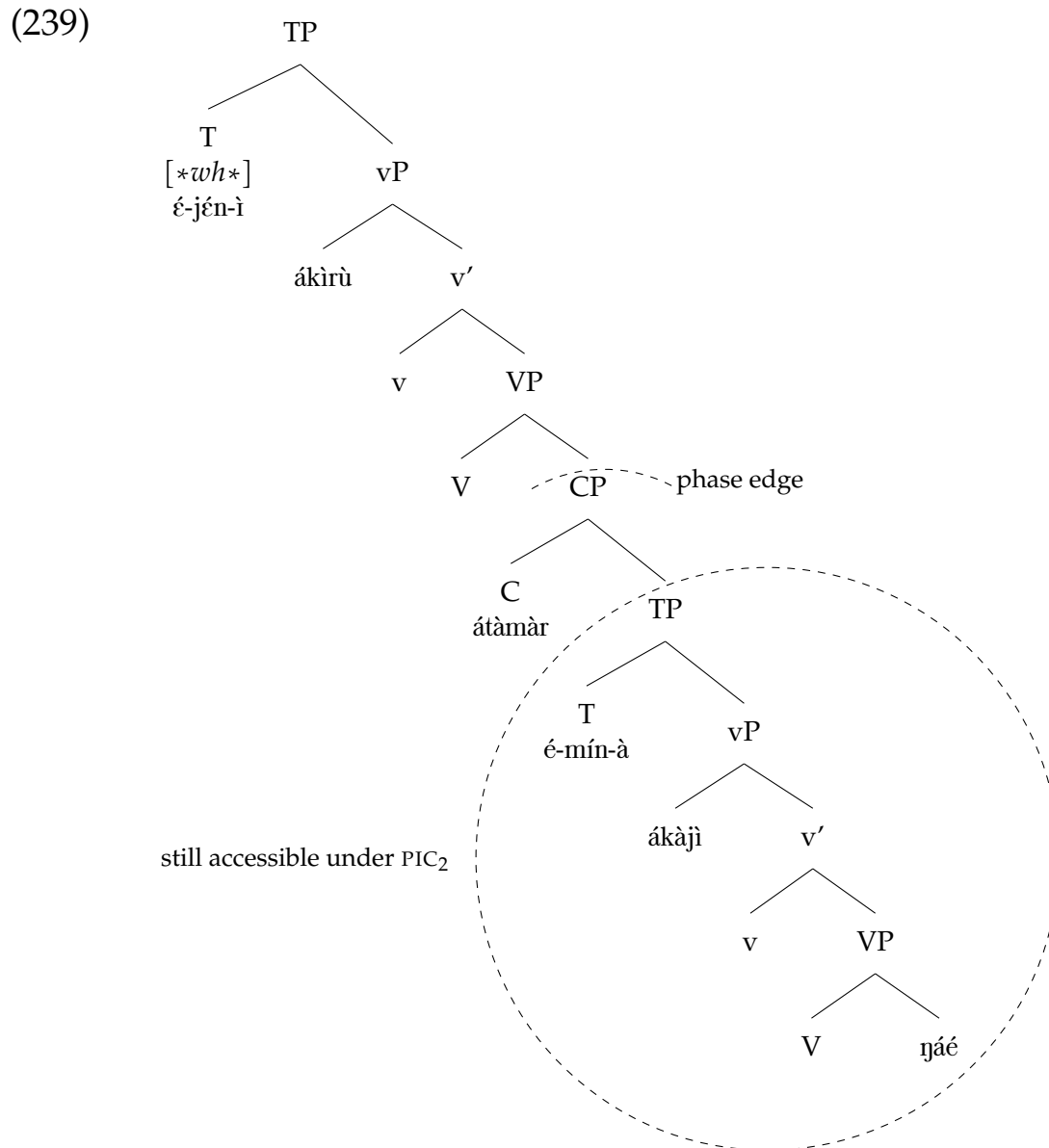
- b. *siitaa-ne soc-aa [CP ki ravii-ne
 Sita-ERG think-PERF-M-SG that Ram-ERG
kis-ko dekh-aa]?
who-ACC see-PERF-M-SG
 'Who did Sita think ravi saw?' (Mahajan 2000: 319,
 retrieved from Keine 2017)

However, this is not the case. As shown above, wh-elements can stay in-situ in an embedded clause. Thus, Keine's (2017; 2020a; 2020b) approach can also not capture the pattern in Turkana. Under his approach, the wh-element in-situ should not be accessible for the wh-probe on the matrix T:



This could lead to the conclusion that there are no restrictions in

Turkana when it comes to *wh*-licensing. This is not the case. If an additional CP layer is added, the *wh*-probe in the matrix clause cannot license the *wh*-element in-situ anymore. To capture this, one could assume that only CPs are phases in Turkana and PIC₂ takes place. This would explain why the matrix T can license a *wh*-element in-situ when it is embedded in one CP but not in two CPs:



Interestingly, the sentence remains ungrammatical if the *wh*-element moves to the pre-verbal domain. This is unexpected because according to phase theory, the *wh*-element should be able to move successive-cyclically through the edge of every phase and therefore not violate any locality constraints. Instead, there needs to be an additional *wh*-element when the *wh*-element moves via two CPs. Thus,

phasality alone cannot capture the locality constraints in Turkana by itself.

To summarise, in both ex-situ and in-situ contexts, wh-elements behave alike with respect to locality: they can move or be licensed across a single CP boundary, but not across more than one CP. Then an additional wh-element is obligatory. Phasality alone does not capture the locality behaviour of wh-elements in Turkana.

In the next section, I turn to the selective opacity framework proposed by [Keine \(2019, 2020b\)](#) that addresses locality by giving every probe a horizon where the probe halts. I will argue that this system captures the Turkana data better than traditional phase-based accounts.

5.3 Selective Opacity

[Keine \(2019, 2020b\)](#) observes that phasality cannot account for all locality phenomena such as A vs $\bar{A}$ -movement where one type of extraction is allowed while another is blocked:

- (240) a. Who_i do you think [_{CP} t_i eats oatmeal for breakfast]?
 b. *Who_i seems [_{CP} t_i eats oatmeal for breakfast]? ([Keine 2019: 14](#))

In both examples, the wh-element has to move via the specifier of CP, which is considered an $\bar{A}$ -position. The difference between these two constructions has traditionally been explained by the ban of improper movement ([Chomsky 1973, 1977, 1981a](#)) which prohibits movement from an $\bar{A}$ -position to an A-position.

The selective opacity approach aims to account for these extraction asymmetries without appealing to stipulative constraints such as improper movement. Instead, [Keine \(2019, 2020b\)](#) proposes that locality should be relative and not absolute like phases. He argues that a syntactic domain can permit extraction of one type of movement while blocking another:

- (241) *Selective opacity*
 A syntactic domain Δ is selectively opaque for α -extraction if Δ prohibits α -extraction but allows β -extraction out of it, where α and β are different types of extraction. ([Keine 2019:](#)

13)

To implement this, Keine (2019, 2020b) argues that each probe is associated with a horizon: a boundary beyond which the probe can no longer operate. These horizons are probe and language specific, allowing for fine-grained distinctions in locality.

This section explores the selective opacity framework in more detail. Since phasehood alone fails to capture the locality phenomena in Turkana, this approach will be crucial for the analysis developed in the next section.

5.3.1 Horizons

The central point of the selective opacity approach is the concept of *horizons*. Keine (2019, 2020b) proposes that each probe (wh-probe, φ -probe etc.) has a horizon. A horizon is a structural boundary beyond which the probe cannot search.

(242) *Horizons*

If a category label X is a horizon for probe π (notated as " $\pi \dashv\!\parallel X$ ") then a π -initiated search terminates at a node of category X . All elements dominated by XP are therefore outside of π ' search space. (Keine 2019: 36)

Unlike phases, which impose uniform boundaries for all syntactic operations, horizons define locality domains relative to the specific probe. While phases block all operations beyond the phase edge, Keine's (2019; 2020b) horizon approach allows certain probes to access a domain while restricting others. This makes horizons inherently probe-specific and sensitive to the type of syntactic dependency involved. For example, if a probe π has C as its horizon ($[\pi] \dashv\!\parallel C$), the categorial head of C will terminate the search of π . This means the CP is not accessible for π .

For example, consider again the contrast between A- and $\bar{A}$ -movement in English:

- (243) a. Who_{*i*} do you think [_{CP} t_{*i*} eats oatmeal for breakfast]?
 b. *Who_{*i*} seems [_{CP} t_{*i*} eats oatmeal for breakfast]? (Keine 2019: 14)

Here, the difference arises because the A-probe (the EPP probe on T)

has C as its horizon ($[A] \dashv\parallel C$), and thus cannot access material inside CP. In contrast, the $\bar{A}$ -probe on C has a horizon that can look into the CP ($[\bar{A}] \dashv\parallel C$) and can successfully establish a dependency with the *wh*-element in the specifier of CP. Horizons therefore provide a principled account of why certain extractions are blocked while others succeed, without invoking stipulative constraints such as the ban on improper movement. Furthermore, horizons account for crosslinguistic variation. Unlike English, many Bantu languages, such as Lusaamia, allow A-movement from within CPs (Carstens 2011; Diercks 2012; Halpert 2015):

- (244) Efula yi-bonekhana i-na-kwa muchiri.
 9rain 9SA-appear 9SA-FUT-fall tomorrow
 'It seems that it will rain tomorrow.'
 (lit: Rain seems will fall tomorrow.) (Lusaamia, Carstens 2011: 737)

The improper movement account does not explain such patterns, as it would predict A-movement to be uniformly blocked across languages. The horizon approach, by contrast, can capture this variation: While in English the A-probe (EPP on T) has C as its horizon ($[A] \dashv\parallel C$), in Lusaamia the A-probe does not ($[A] \dashv\parallel C$), allowing it to search into CPs.

5.3.2 Constraints on Horizons

The concept of horizons makes it possible for different probes to have different search spaces. But horizons are not completely free. Keine (2019, 2020b) identifies several generalisations that appear to constrain how horizons behave crosslinguistically. The section below outlines these claims.

One of the observations concerns the relationship between the structural height of a probe and the size of its search space. Across languages, probes located higher in the structure uniformly have larger search domains than lower probes (cf. Sternefeld 1992; Williams 2003, 2011; Abels 2007; Abels et al. 2009; Müller 2014a,b). In English, for example, the A-probe on T is unable to search into CPs ($[A] \dashv\parallel C$), whereas the structurally higher $\bar{A}$ -probe on C can access material inside CPs ($[\bar{A}] \dashv\parallel C$). Keine (2019, 2020b) argues that this is not an accident, but a general property of how horizons work.

This correlation between structural height and locality domain is formulated as the *Height-Locality Connection*:

- (245) *Height-Locality Connection*
The higher the structural position of a probe π , the more kinds of structure π can search into.

A second generalisation [Keine \(2019, 2020b\)](#) makes concerns the relation between the size of a clause and its opacity to a given operation. If a clause of a certain structural size is opaque for some operation, then all clauses that are structurally larger are likewise opaque. This means that if a given structural size (e.g. TP) is opaque to some operation, then structurally larger clauses (e.g. CP) are likewise opaque to this operation. This generalisation is referred to as *Upward Entailment*:

- (246) *Upward Entailment*
If a clause of a certain structural size is opaque to an operation, then clauses that are structurally larger are likewise opaque to this operation. ([Keine 2019: 37](#))

in the case of extraposition in English. Both nonfinite TP clauses and finite CP clauses are opaque for this operation:

- (247) *[John₁ is believed [_{TP} t_1 to be certain t_2] by everybody that [that Fred is crazy]_i]. ([Baltin 1978: 144](#))

[Keine \(2019, 2020b\)](#) proposes that both generalisations follow from how category features are inherited in extended projections (like CP > TP > vP > VP). Each layer in an extended projection inherits the categorial features of the projections below it. This means that a CP consists of the categorial features of CP, TP, vP, and VP: CP = [C, T, v, V]:

- (248) *Category Inheritance*
Given an extended projection $\Phi = \langle \Pi_n > \Pi_{n-1} > \dots > \Pi_1 \rangle$ where Π_x 's are phrases, the categorial features of Π_m are inherited up to Π_{m+1} . ([Keine 2019: 38](#))

Category inheritance directly leads to the *Horizon Inheritance Theorem*. This means that once a horizon is set at a certain level, no larger structure can remain transparent to that probe. Once a probe is

blocked at a certain level, that blocking effect continues for larger structures. For instance, if the horizon is set at TP, the probe cannot see into anything larger like CP, because the category features that define the horizon at TP are also present in CP (via inheritance):

(249) *Horizon Inheritance Theorem*

Given a probe π and an extended projection $\Phi = \langle \Pi_n > \Pi_{n-1} > \dots > \Pi_1 \rangle$, if $\Pi_m \in \Phi$ is a horizon for π , then all projections $\Pi_{m+1}, \dots, \Pi_n$ are likewise horizons for π (due to category inheritance). (Keine 2019: 39)

On the other hand, this system also restricts probes to search too local which (Keine 2019, 2020b) calls *vacuous probes*. This system also excludes the possibility of vacuous probes. A probe merged at a given level cannot have horizons at a structurally lower position in the same extended projection, because the search domain of such a probe would be contained within its own merge site. For example, C cannot have T, v, or V as its horizon. This restriction is formalised as the Height-Locality Theorem:

(250) *Height-Locality Theorem*

Given an extended projection $\Phi = \langle \Pi_n > \Pi_{n-1} > \dots > \Pi_1 \rangle$, for any non vacuous probe π ,

a. *Height $\rightarrow$ locality entailment*

If π is located on Π_m , then a projection $\in \{\Pi_{m-1} \dots \Pi_1\}$ cannot be a horizon for π .

b. *Locality $\rightarrow$ height entailment*

If π has Π_m as a horizon then π cannot be located on a projection $\in \{\Pi_n \dots \Pi_{m+1}\}$.

In sum, the system of horizons is tightly constrained by structural inheritance. This leaves little room for mixed horizon configurations within a single clause. As will become clear in the analysis of Turkana, this rigidity raises questions about whether these axioms aren't too strict. Before, I will apply this approach to analyse the Turkana data, I will show how horizons and phases interact in Keine's (2019; 2020b) approach.

5.3.3 Horizons and Phases

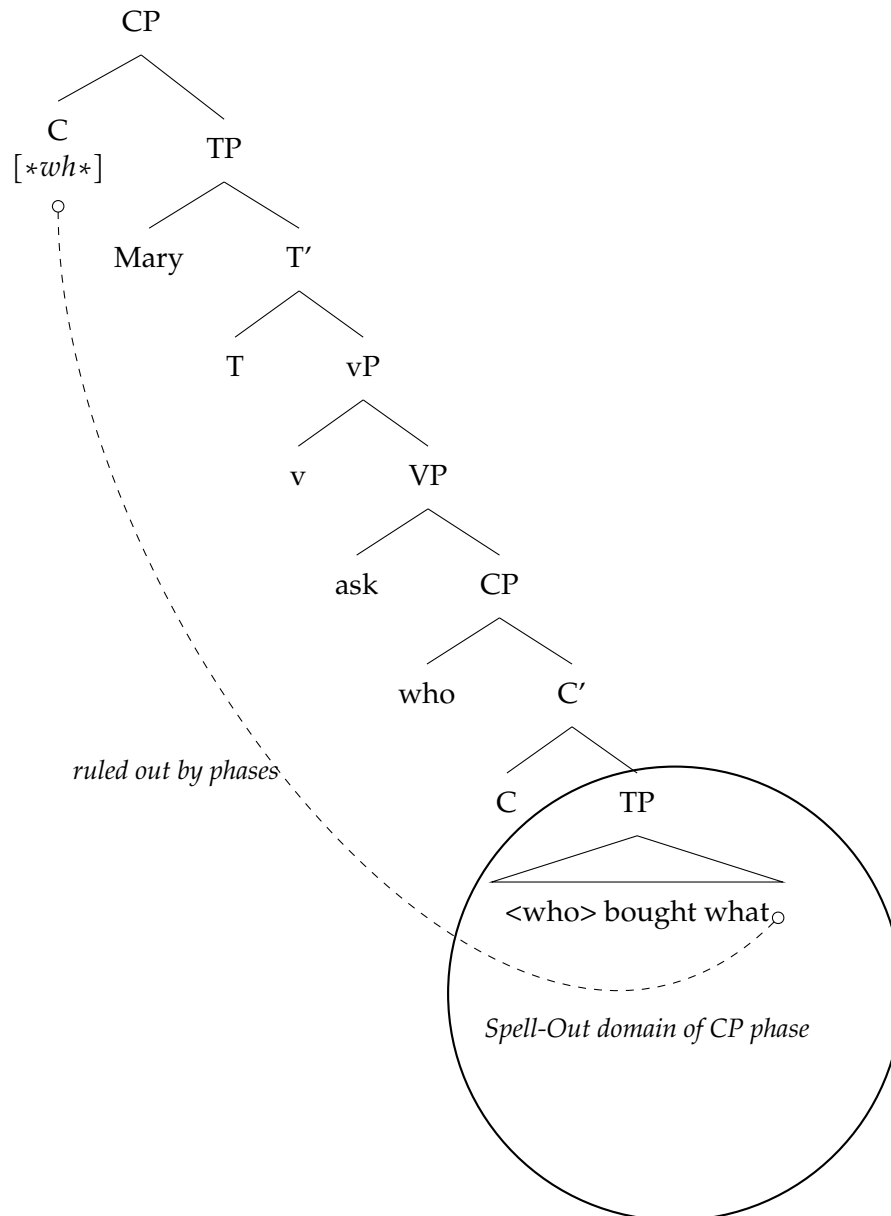
While Keine (2019, 2020b) argues that phase theory fails to account for a number of locality phenomena due to its assumption of absolute, uniform domains, he does not discard the notion of phases altogether. Instead, he incorporates them into his account of selective opacity and argues that horizons and phases interact by addressing different aspects of locality. In this system, phases and horizons serve distinct but complementary roles. Horizons determine whether a domain is accessible to a particular probe. Phases, in contrast, regulate how extraction proceeds structurally: they force movement to pass through the phase edge in a successive-cyclic fashion.

Horizons cannot derive successive-cyclic movement, as they either block access at a given domain entirely (e.g. $\pi \dashv \parallel$ CP) or permit unrestricted probing ($\pi \dashv \parallel \emptyset$). Neither outcome explains why intermediate movement through CP edges is mandatory. Consider the following example from English which is ungrammatical because the wh-element _{what} fails to move via the intermediate CP edge, and is thus inaccessible to the matrix wh-probe:

(251) *What_i did Mary ask who bought t_i ? (Keine 2020b: 208)

The ungrammaticality cannot be explained by horizons. Instead, it follows from a violation of the PIC. The embedded CP has been spelled out and its complement *what* is inaccessible to the *wh*-probe. The *wh*-element must escape via the CP edge, but this position is already occupied. The derivation fails, as shown by Keine (2020b: 208):

(252)



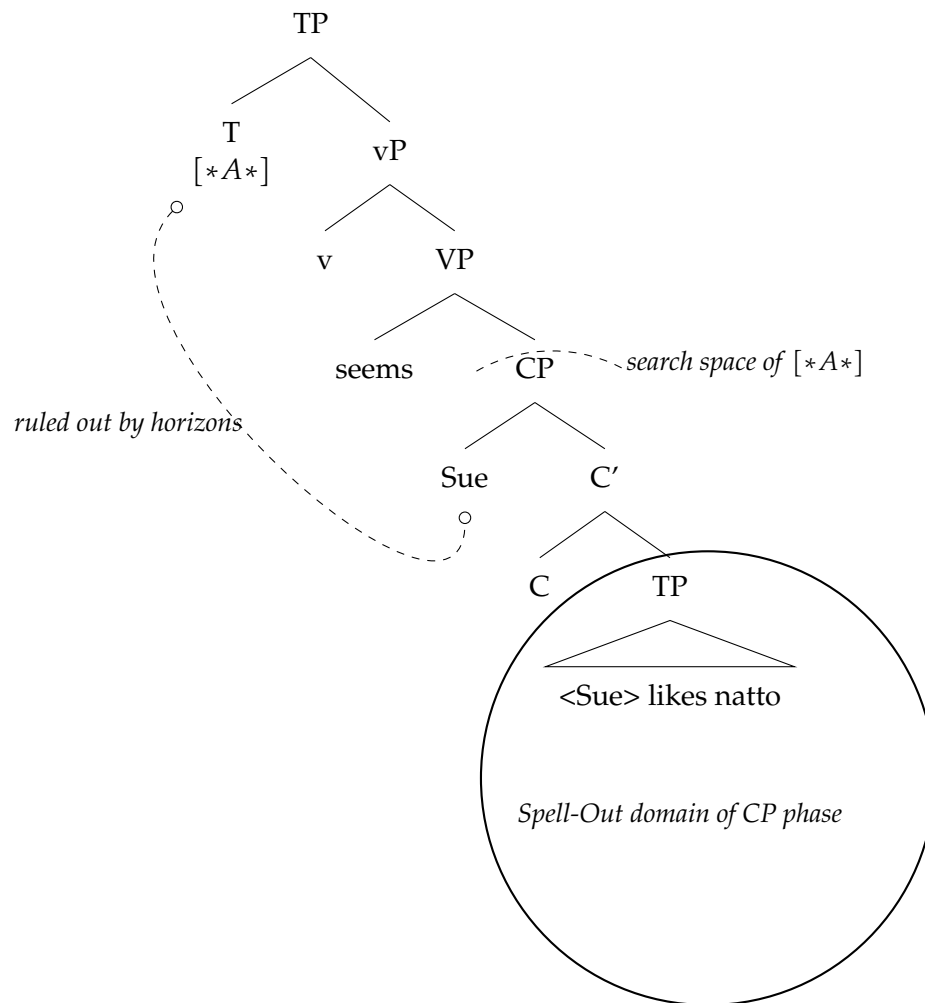
On the other hand, phases cannot explain the ungrammaticality of A-movement out of a finite clause like in the following example:

(253) *Sue seems (that) likes natto. (Keine 2020b: 208)

Phases allow access to the specifier of CP but the horizon of A-probe

($[*A*] \dashv\parallel$ CP) blocks agreement as (Keine 2020b: 208) showed in the following tree:

(254)

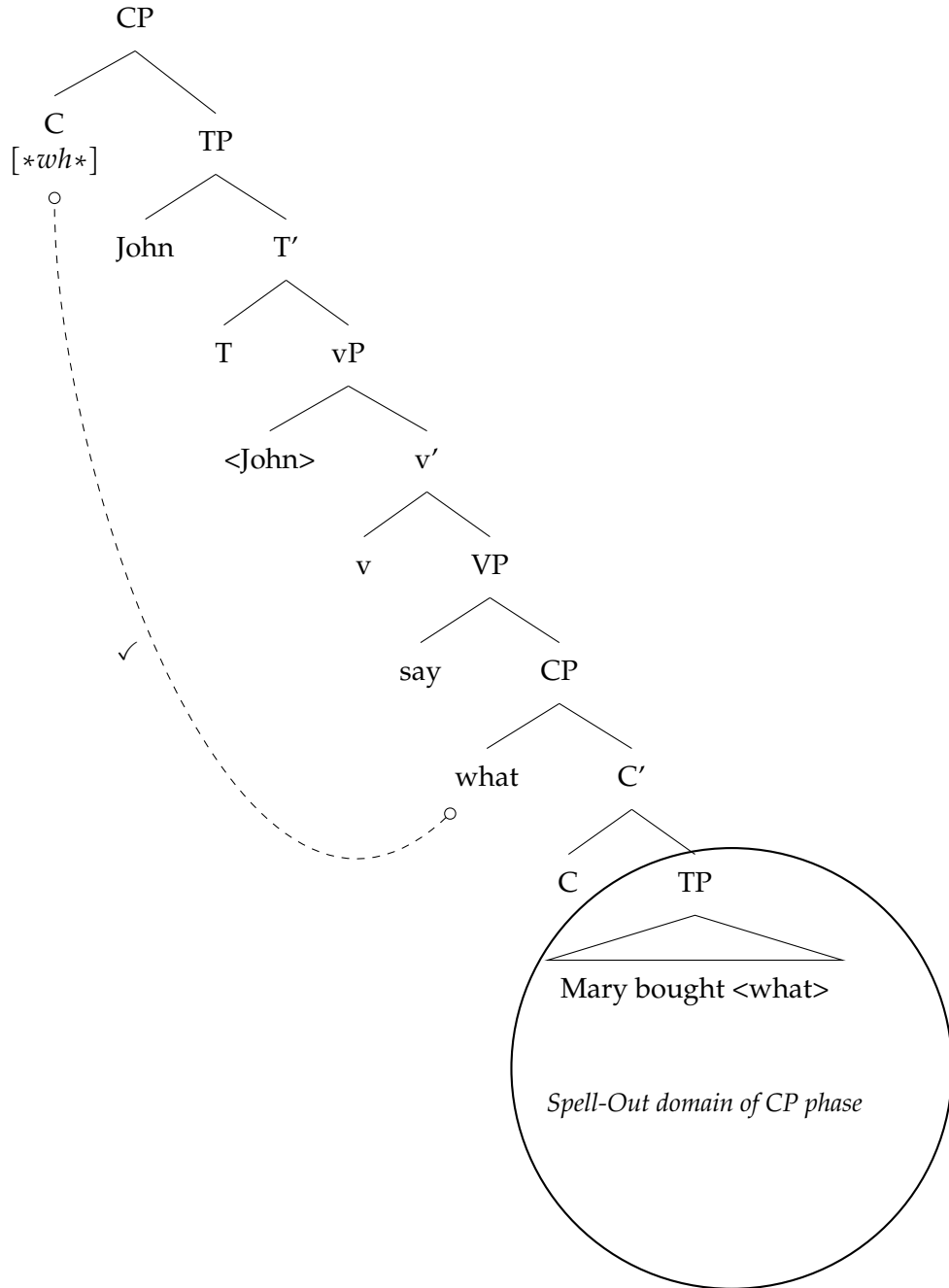


Only when both phases and horizons are obeyed, is the sentence well formed like in the following sentence:

(255) What did John say that Mary bought? (Keine 2020b: 209)

The $\bar{A}$ -probe can see into the next CP ($[*wh*] \dashv\parallel \emptyset$) and successive cyclic movement is not blocked as Keine (2020b: 209) shows in the following tree:

(256)



In sum, phases and horizons coexist in Keine's (2019; 2020b) system: horizons determine the probe's reach, and phases determine the structural requirements for movement through opaque domains.

However, Keine (2019, 2020b) argue that vPs can never be phases in his system. I have already shown in 5.2.2 that Keine rejects vPs as phases. However, vP phases are incompatible with the horizon-based system. Keine states that the problem emerges when an element originates in an A-position inside a vP, for instance as the subject of an unaccusative verb, moves to the phase edge, the specifier of vP which is an $\bar{A}$ -position, and then attempts to move to an

A-position, such as the specifier of TP. If vP is a phase, this constitutes an instance of improper movement: A to $\bar{A}$ to A. Under standard phase theory, the original A-position inside the vP is sent to spell out and therefore inaccessible to the probe on T. The probe on T only sees the element in its $\bar{A}$ -position in the specifier of vP and cannot see its A-origin. As Keine points out, there is no mechanism in this architecture to diagnose that this is improper movement, since the $\bar{A}$ -trace at the phase edge is the only accessible one. In this way, clause-internal phases like vP are structurally incompatible with the horizon-based model.

To summarise, in this section I have presented the selective opacity approach by Keine (2019, 2020b). In this approach, probes are equipped with a horizon where the probes stops searching. Horizons allow for probe-specific locality effects and account for crosslinguistic variation in extraction patterns. In the next section, I will show how a modified approach can capture the Turkana data.

5.4 A Modified Selective Opacity Approach for Turkana

I showed in 5.2.3 that a standard phasal approach does not capture the long distance dependency data in Turkana. In the previous section, I presented the selective opacity framework by Keine (2019, 2020b) which introduces probe-specific horizons. In this section, I apply this approach to the Turkana data presented in chapter 3. I argue, however, that Keine's original approach must be modified to account for the empirical patterns in Turkana

In this section, I begin by discussing phasehood in Turkana and argue that only CPs qualify as phases, and that the relevant locality condition is PIC₂. I then turn to the role of horizons in Turkana, showing how they interact with the attested movement patterns. I will derive the relevant long-distance dependencies.

5.4.1 Phasehood in Turkana

As noted above, a standard phase-based account cannot derive the full range of long-distance dependencies observed in Turkana. In this subsection, I define the phasehood properties of Turkana and argue that only CPs function as phases. Furthermore, I argue that

the patterns in Turkana are more compatible with PIC₂.

At first glance, the Turkana data appear to support an analysis in which both vP and CP are phases, as proposed by [van Urk \(2015\)](#); [van Urk & Richards \(2015\)](#) for Dinka. Turkana exhibits overt wh-movement through a specifier of vP (see [4.4.2](#)). However, I argue that this movement is not phase-driven, but rather triggered by an optional EPP feature on v.

Keine (2020b) proposes the following diagnostics to determine whether vP is a phase or not. First, the obligatoriness requirement which states that movement through the specifier of vP must be obligatory. Second, the extraction requirement which states that movement to the specifier of vP must take place in order to be extracted out of the vP and not for independently observable vP-internal requirements.

Neither requirement is met in Turkana. Movement through the specifier of vP is optional and not obligatory as shown in the following example. Hence, the obligatoriness requirement is not met.

- (257) [CP ɲáé é-jén-ì ákirù [CP átàmàr
 who.DEFL 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì (ɲáé) [CP átàmàr
 3-think-TAM Akai.NOM (who.DEFL) that
 é-mín-à ádʒìkòn ɲáé]]]?
 3-love.PRS-TAM Adjikon.NOM who.DEFL
 ‘Who does Akiru know that Akai thinks that Adjikon loves?’

As shown in 4.5, movement to the post-verbal position is optional and motivated by an optional EPP feature on v. Thus, the extraction requirement is also not met, since movement occurs due to a vP-internal requirement, not to enable extraction. This is what [Keine \(2020b\)](#) argued for Dinka: Movement to the specifier of vP happens because of a language specific EPP-feature and not because of phasality. Thus, I conclude that vPs are not phases in Turkana.

Now that it is established that vP is not a phase, let's turn to the nature of CP in Turkana. I have already shown in 5.2.3 that the locality conditions in Turkana are less restrictive. Compare Turkana and Hindi-Urdu. Both languages allow wh-elements to stay in-situ without covert movement. Wh-licensing into the lower CP is not allowed in Hindi-Urdu, since it crosses a phase boundary:

Turning now to CP, I argue that CP is a phase in Turkana. However, the locality conditions in Turkana are less restrictive than in languages like Hindi-Urdu. Both languages allow *wh*-elements to remain in-situ, but differ in how far these elements can be licensed: while Hindi-Urdu blocks licensing across a single CP boundary, Turkana permits it, as the following examples show:

- (258) *siitaa-ne soc-aa [CP ki ravii-ne **kis-ko**
Sita-ERG think-PERF-M-SG that Ram-ERG **who-ACC**
dekh-aa].
see-PERF-M-SG
'Who did Sita think Ravi saw?' (Hindi-Urdu, Mahajan 2000: 319, retrieved from Keine 2017)

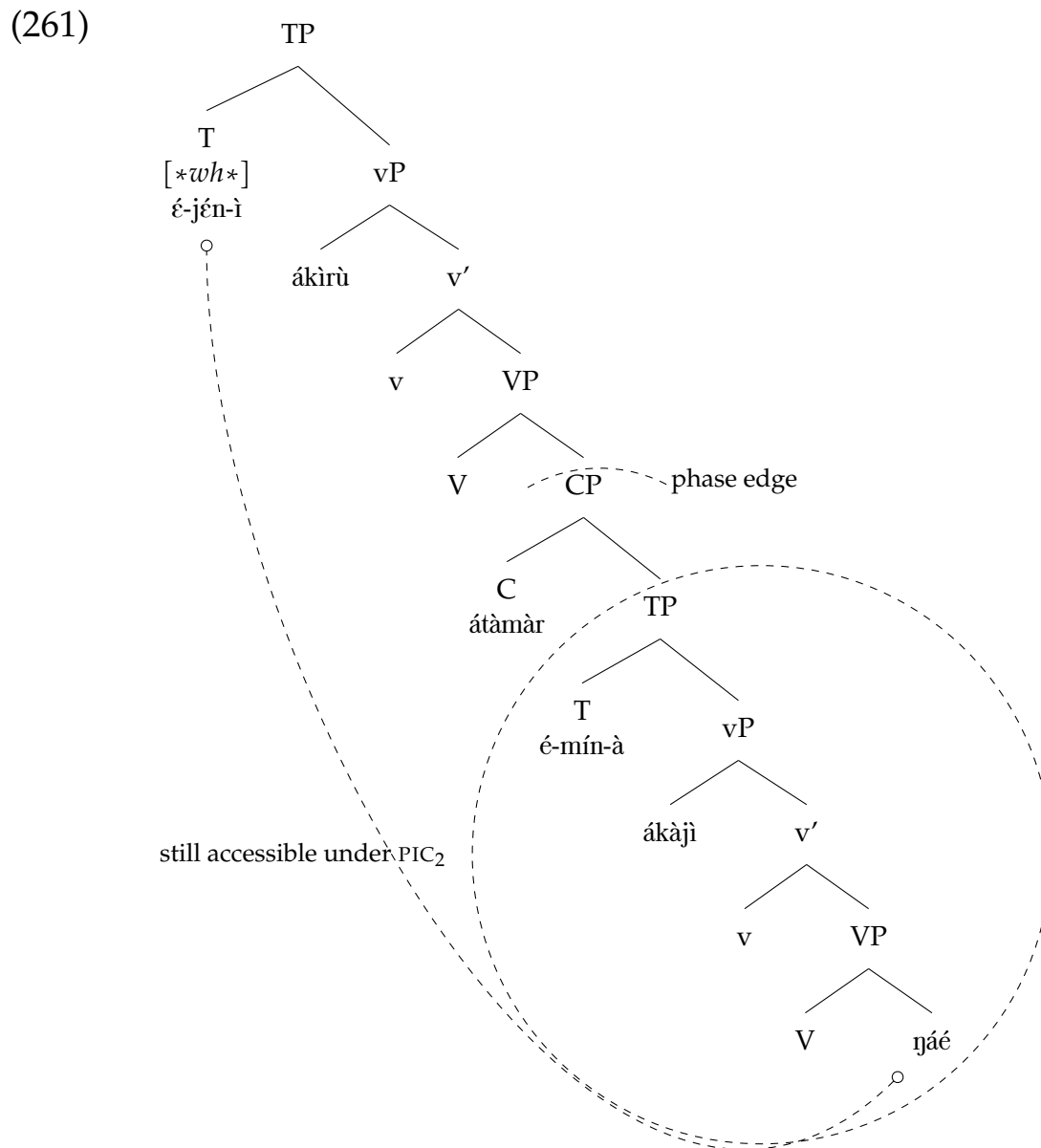
- (259) é-jén-ì ákìrù [CP (átàmàr) é-mín-à
3-know.PRS-TAM Akiru.NOM that 3-love.PRS-TAM
ákàjì **ḡáé**]?
Akai.NOM **who.DEFL**
'Who does Akiru know that Akai loves?'

Only when an additional CP layer is added, this sentence is ungrammatical:

- (260) *[CP é-jén-ì ákìrù [CP átàmàr í-tám-ì
3-know.PRS-TAM Akiru.NOM that 3-think-TAM
ákàjì [CP átàmàr é-mín-à ádzìkòn
Akai.NOM that 3-love.PRS-TAM Adjikon.NOM
ḡáé]]]?
who.DEFL
'Who does Akiru know that Akai thinks that Adjikon loves?'

This contrast suggests that Turkana CPs are subject to a more permissive version of the PIC, specifically PIC₂, which delays spell-out until the next phase head is merged. In (259), the *wh*-element remains accessible to the probe on matrix T because the embedded CP phase has not yet been completed at the time the probe is merged. Spell-out is delayed until the next phase head, namely, the higher C, is introduced. As a result, the probe on T can still search into the embedded TP and establish an Agree relation with the *wh*-element. This contrasts with PIC₁, under which the CP complement would

be spelled out as soon as the lower C head is merged, rendering the wh-element inaccessible. The following tree shows the derivation of (259) under PIC₂:



In contrast, in (264), the addition of an additional CP renders the wh-element in-situ inaccessible under PIC₂. Thus, the sentence is ungrammatical.

However, phasehood alone cannot account for all locality effects

in Turkana. If CP phases were the sole factor constraining dependencies, (262) should be grammatical: the *wh*-element is fronted and should therefore be accessible to the *wh*-probe on matrix T. Nonetheless, the sentence is ungrammatical, suggesting that an additional mechanism is needed to restrict the derivation:

- (262) [CP ɲáé é-jén-ì ákìrù [CP átàmàr
who.DEFL 3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàjì [CP átàmàr é-mín-à
3-think-TAM Akai.NOM that 3-love.PRS-TAM
ádʒìkòn]]]?
Adjikon.NOM
‘Who does Akiru know that Akai thinks that Adjikon loves?’

Thus, while phasehood (under PIC₂) explains certain locality effects, it is not sufficient to capture all the relevant generalisations in Turkana. In the next subsection, I introduce horizons as a probe-specific mechanism to complement phase-based locality.

5.4.2 Horizons in Turkana

In the previous subsection, I have shown that phasality alone cannot capture all locality effects in Turkana. To account for the locality patterns in Turkana, I adopt a system in which probes on *v*, T, and C are equipped with specific horizons, following the framework proposed by Keine (2019, 2020b). This subsection discusses the horizons in Turkana.

As established in 4.3.2, the *wh*-/ focus-probe is located on T. This probe can license in-situ *wh*-elements located within an embedded CP, which suggests that its horizon is unrestricted. I therefore assume that the horizon of the *wh*-probe is $\emptyset$ ($[*wh*] \dashv \emptyset$).

- (263) [CP é-jén-ì ákìrù [CP (átàmàr)
3-know.PRS-TAM Akiru.NOM that
é-mín-à ákàjì ɲáé]]
3-love.PRS-TAM Akai.NOM who.DEFL
‘Who does Akiru know that Akai loves?’

Crucially, however, horizons are still subject to the constraints imposed by phase theory and PIC₂. If the *wh*-element has already been

spelled out, the probe cannot be checked and the derivation crashes. This predicts the ungrammaticality of the following example, where the *wh*-probe on the matrix T cannot access the *wh*-element since it has already been sent to spell-out when the matrix T is merged:

- (264) *_{[CP} é-jén-ì ákìrù _{[CP} átàmàr í-tám-ì
 3-know.PRS-TAM Akiru.NOM that 3-think-TAM
 ákàjì _{[CP} átàmàr é-mín-à ádzìkòn
 Akai.NOM that 3-love.PRS-TAM Adjikon.NOM
ɲáé]]]?
who.DEFL
 ‘Who does Akiru know that Akai thinks that Adjikon loves?’

I showed in 4.5 that T and v can carry an optional EPP feature, [*•D•*], which triggers movement to their respective specifiers. Since this probe can target elements across a CP boundary obeying PIC₂, I assume that the horizon of this horizon is also unrestricted ([*•D•*] → ∅). Consider the example below, in which the matrix T carries an [*•D•*]-probe. The *wh*-element is attracted to the specifier of the matrix clause. This derivation succeeds because PIC₂ allows the probe to search into the complement of the lower CP head, as long as no additional phase head is merged:

- (265) _{[CP} **ɲáé_i** é-jén-ì ákìrù _{[CP} (átàmàr)
who.DEFL 3-know.PRS-TAM Akiru.NOM that
 é-mín-à ákàjì *t_i*]]]?
 3-love.PRS-TAM Akai.NOM
 ‘Who does Akiru know that Akai loves?’

However, if the *wh*-element is already sent to spell out, when the [*•D•*]-probe is merged, the probe cannot be checked and the derivation crashes. This is what happens in the following configuration with three CPs and the [*•D•*]-probe on the matrix T. The [*•D•*]-probe cannot attract the *wh*-element in-situ because it has already been sent to spell-out and thus the derivation crashes:

- (266) *_{[CP} **ɲáé_i** é-jén-ì ákìrù _{[CP} átàmàr
who.DEFL 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì _{[CP} átàmàr é-mín-à
 3-think-TAM Akai.NOM that 3-love.PRS-TAM

ádʒíkòn t_i]]]?

Adjikon.NOM

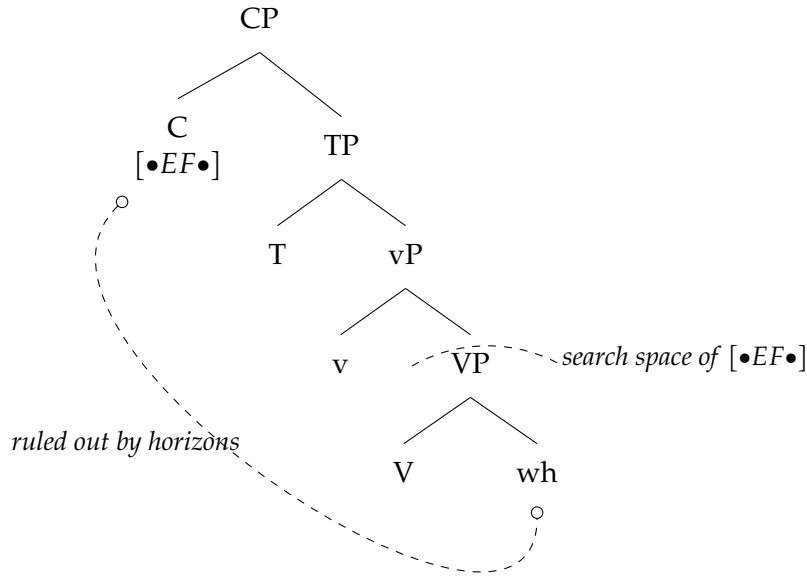
‘Who does Akiru know that Akai thinks that Adjikon loves?’

Cyclic movement through the edge of CP might seem like a potential repair strategy in such cases. However, this option is blocked by the nature of the edge probe on C. Following Chomsky (2001, 2006, 2007), I assume that C carries an edge feature $[EF]$, which attracts an element in order to escape the phase. In Turkana, $[\bullet EF \bullet]$ has a highly restricted horizon: $V ([\bullet EF \bullet] \dashv V)$. Under Keine (2019, 2020b), such a configuration is prohibited because of the category inheritance. Under the category inheritance, C inherits the category features of TP, vP, and VP. Any horizon lower than C would result in a vacuous probe, a configuration in which the probe’s search space is entirely contained within its own projection.

In order to derive the locality effects observed in Turkana, I assume that the language does not exhibit category inheritance in the sense of Keine (2019, 2020b). Specifically, I take C to be able to host an edge probe with a horizon in the same CP, namely $V ([\bullet EF \bullet] \dashv V)$.

A comparable locality restriction has been implemented in standard phase theory where vP is assumed to be a phase (Chomsky 2000, 2001; Legate 2003; van Urk 2015). Under this view, only the edge of vP remains accessible for further syntactic operations, while the complement of v is sent to spell-out. However, I have shown that vP does not function as a phase in Turkana. Instead, I propose that this locality effect arises from the horizon of the edge feature on C. By assigning V as the horizon ($[\bullet EF \bullet] \dashv V$), the same boundary is established: material within the VP-phase is inaccessible unless it moves to a higher position. Crucially, this restriction is probe-specific and applies only to cyclic movement triggered by the $[\bullet EF \bullet]$ -probe, in contrast to the absolute opacity imposed by phasehood. This allows the system to simulate vP phase like effects without assuming that vP is a phase.

(267)



A consequence of rejecting category inheritance is that the generalisations proposed by Keine (2019, 2020b), such as the height locality condition and upward entailment, no longer hold. This is not problematic in the case of Turkana, since the edge probe on C has a more limited search space than probes on T and v which are located lower in the structure.

Furthermore, I assume the derivation does not crash if the [•EF•]-probe is not checked. The probe is active but nothing happens if it does not agree (cf. Preminger (2014)). The probe enables movement through the phase edge, but the derivation does not crash if the probe fails to attract a goal. Following Chomsky (2000, 2001), I argue that an edge feature is purely there to trigger movement to the phase edge so that an element can continue its movement out of the current phase. Thus, movement triggered by the [•EF•]-probe can never be a final landing site.

To conclude, I assume the following probes with their horizons:

Probe	Location	Horizon	Role
[*wh*]	T	¬ ∅	wh-licensing
[•D•]	T/v	¬ ∅	triggers overt movement
[•EF•]	C	¬ V	enables successive cyclic movement

Table 5.1: Horizons in Turkana

With these assumptions in place, I now turn to the derivation of

long-distance wh-dependencies in Turkana.

5.5 Deriving Long-Distance Movement

Having outlined the phase structure and horizon system of Turkana, I now turn to the derivation of long-distance dependencies. The following cases illustrate how the proposed system accounts for the patterns with three CPs. Crucially, the interaction between probe-specific horizons, PIC_2 , and the optional EPP features on T and v predicts the observed distribution.

5.5.1 Case 1: Ungrammatical Wh-Element In-Situ

In this case, the wh-element remains in-situ within the most deeply embedded CP and is not targeted by any intermediate movement operation. As a result, it is sent to spell-out before the matrix wh-probe on T is merged. Since the $[*wh*]$ probe cannot access material that has already been sent to spell out, the derivation fails.

- (268) a. $*[_{CP} \text{ é-jén-ì} \quad \text{ákìrù} \quad [_{CP} \text{ átàmàr}$
 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì $[_{CP} \text{ átàmàr é-mín-à}$
 3-think-TAM Akai.NOM that 3-love.PRS-TAM
 ádžìkòn $\boxed{\text{ṇáé}}$ $]]]$?
 Adjikon.NOM $\boxed{\text{who.DEFL}}$
 ‘Who does Akiru know that Akai thinks that Adjikon
 loves?’
- b. $*[_{CP} C \quad T_{[*wh*]} \dots v \dots$
 wh-element is already sent to spell out $\frac{1}{2}$
 $\dots [_{CP} C_{[\bullet EF \bullet]} \dashv \parallel V \dots T \dots v \dots [_{CP} C_{[\bullet EF \bullet]} \dashv \parallel V \dots T \dots v \dots wh]]]$

5.5.2 Case 2: Ungrammatical Wh-Movement

In this case, the wh-element moves from its base position to a higher position within the most deeply embedded CP but remains inaccessible to the matrix $[*wh*]$ -probe.

- (269) a. $*[_{CP} \text{é-jén-ì} \quad \text{ákìrù} \quad [_{CP} \text{átàmàr}$
 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì $[_{CP} \text{átàmàr} \boxed{\text{ḡáé}}$
 3-think-TAM Akai.NOM that $\boxed{\text{who.DEFL}}$
 é-mín-à ádzìkòn]]]?
 3-love.PRS-TAM Adjikon.NOM
 ‘Who does Akiru know that Akai thinks that Adjikon
 loves?’

- b. $*[_{CP} C \quad T_{[*wh*]} \dots v \dots$
 wh-element is already sent to spell out ⚡
 ... $[_{CP} C_{[\bullet EF \bullet]} \dashv V \dots T \dots v \dots [_{CP} C_{[\bullet EF \bullet]} \dashv V \dots wh_i \quad T_{[\bullet D \bullet]} \dots v \dots wh_i]$

At the same time, movement from the in-situ position to the matrix CP is also ruled out. It cannot move there directly because of phasal-ity but it can also not move cyclically because of the horizon of the edge feature. The wh-element is located in the VP of the embedded CP and the horizon of edge feature cannot look into the VP. Thus, the wh-element cannot move out of the embedded CP.

- (270) a. $*[_{CP} \boxed{\text{ḡáé}} \quad \text{é-jén-ì} \quad \text{ákìrù} \quad [_{CP} \text{átàmàr}$
 $\boxed{\text{who.DEFL}}$ 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì $[_{CP} \text{átàmàr} \text{é-mín-à}$
 3-think-TAM Akai.NOM that 3-love.PRS-TAM
 ádzìkòn]]]?
 Adjikon.NOM
 ‘Who does Akiru know that Akai thinks that Adjikon
 loves?’

- b. $*[_{CP} C \quad wh_i \quad T_{[*wh*, \bullet D \bullet]} \dots v \dots$
 ⚡
 ... $[_{CP} C_{[\bullet EF \bullet]} \dashv V \dots T \dots v \dots [_{CP} C_{[\bullet EF \bullet]} \dashv V \dots T \dots v \dots wh_i]]]$
 ⚡

5.5.3 Case 3: Grammatical Wh-Movement

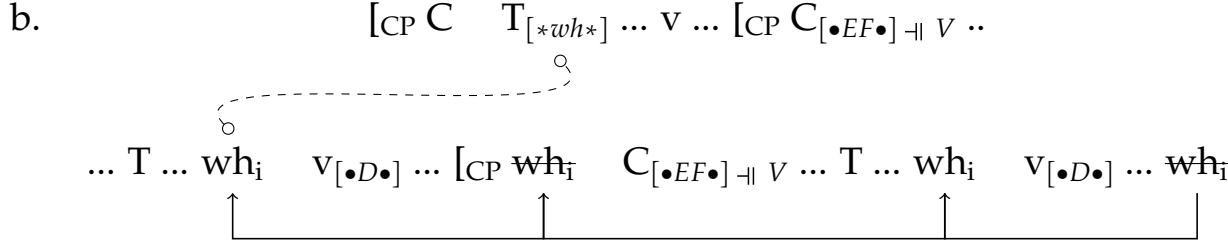
However, the sentence becomes grammatical when the *wh*-element moves within the lowest CP because it is attracted by an EPP-feature. It is then within the horizon of the edge feature on C. Once the *wh*-element has moved to C in the lowest CP, it is visible to matrix T under PIC₂.

- (271) a. [CP **ŋáé** **who.DEFL** é-jén-ì 3-know.PRS-TAM Akìrù Akiru.NOM [CP átàmàr that í-tám-ì 3-think-TAM Akàjì Akai.NOM [CP átàmàr é-mín-à 3-love.PRS-TAM ádzìkòn Adjikon.NOM **ŋáé** **who.DEFL**]]]?
 'Who does Akiru know that Akai thinks that Adjikon loves?'

- b. [CP C *wh_i* T_[*wh*, •D•] ... V ... [CP *wh_i* C_[•EF•] V ... T ... *wh_i* V_[•D•] ... *wh_i*]]]
-

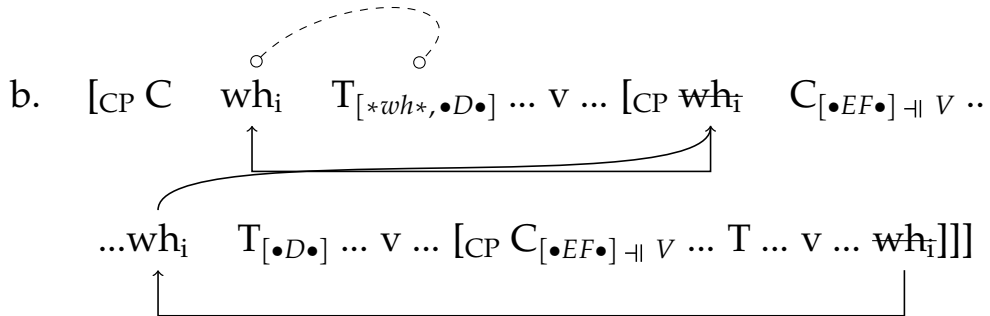
In Turkana, matrix T can license a *wh*-element that has moved only to the intermediate CP. This is expected under PIC₂, which allows probes to access material in the complement of a lower phase head until the next phase head is merged.

- (272) a. [CP é-jén-ì 3-know.PRS-TAM Akìrù Akiru.NOM [CP átàmàr that í-tám-ì 3-think-TAM Akàjì Akai.NOM **ŋáé** **who.DEFL** [CP átàmàr é-mín-à 3-love.PRS-TAM ádzìkòn Adjikon.NOM **ŋáé** **who.DEFL**]]]?
 'Who does Akiru know that Akai thinks that Adjikon loves?'



The *wh*-element can also move directly to the intermediate CP without violating phasality because of PIC_2 . From there it can move to matrix CP:

- (273) a. [CP **ɲáé** **who.DEFL** $\acute{e}$ -jén-ì $\acute{a}$ kìrù [CP $\acute{a}$ tàmàr
 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì $\acute{a}$ kàjì **ɲáé** [CP $\acute{a}$ tàmàr
 3-think-TAM Akai.NOM **who.DEFL** that
 $\acute{e}$ -mín-à $\acute{a}$ džìkòn]]]?
 3-love.PRS-TAM Adjikon.NOM
 ‘Who does Akiru know that Akai thinks that Adjikon
 loves?’



This system also allows one *wh*-element per CP. In this case, there is an EPP-feature, either on *v* or *T*, in each CP, that attracts the *wh*-element.

5.5.4 Case 4: Ungrammaticality from Ganging-Up Effects

Given the phase- and horizon-based assumptions outlined above, the ungrammaticality of the following sentence is unexpected. The *wh*-probe on matrix *T* can see the *wh*-element and movement to the next higher CP is permitted under PIC_2 .

- (274) *_{[CP é-jén-ì ákìrù [CP átàmàr **ṇǎé**]}
 3-know.PRS-TAM Akiru.NOM that **who.DEFL**
 í-tám-ì ákàjì [CP átàmàr é-mín-à
 3-think-TAM Akai.NOM that 3-love.PRS-TAM
 ádžìkòn]]]?
 Adjikon.NOM
 ‘Who does Akiru know that Akai thinks that Adjikon loves?’

As shown in the previous sections, movement from the in-situ position to the intermediate CP is independently grammatical in Turkana, and so is agreement between matrix T and a *wh*-element located in the intermediate CP. However, their combination results in ungrammaticality. This pattern can be analysed as a cumulative or gang effect (cf. Pater 2009; Haegeman et al. 2014; Murphy 2017, 2018a,b), in which two grammatical processes become ungrammatical when combined.

Such effects have been analysed within the framework of Serial Harmonic Grammar (SHG), a constraint-based theory originally developed for phonology (see e.g. McCarthy 2000, 2008; Kimper 2011, 2016) and later extended to syntax (e.g. Heck & Müller 2003, 2013; Murphy 2017, 2018a,b; Müller et al. 2022). In SHG, constraints are not ranked, as in classic Optimality Theory (Prince & Smolensky 1993/2004; McCarthy & Prince 1995), but are instead assigned numerical weights. Competing derivational candidates accumulate harmony scores based on how often they violate these constraints. The candidate with the best overall score (the lowest weighted sum of violations) is selected as optimal. This framework can predict cumulative effects, where two operations that are each independently tolerated result in ungrammaticality when combined. The following example illustrates this. Even though constraint B and C are weighted lower than constraint A, in cumulation they outweigh constraint A. Thus, candidate 2 becomes the optimal output:

(275)	Input	A	B	C	$\mathcal{H}$
		w = 6	w = 3	w = 2	
a.	Candidate 1	-1			-6
 b.	Candidate 2		-1	-1	-5

A core idea in SHG is cyclic optimisation. Following Chomsky

(1993), syntactic structures are built step by step from the bottom up. According to cyclic optimisation, at each derivational step, the grammar looks at the structure so far and compares different possible operations that could apply next (Heck & Müller 2013). The best most optimal candidate is then the input for the next optimisation until no further optimisation is possible.

Importantly, this approach is compatible with my existing assumptions about phases and horizons: the phase-based locality domains and horizon-defined accessibility conditions continue to structure the derivation, while SHG provides a graded explanation for cumulative constraint violations within those domains. Only derivations that respect PIC₂ and horizons are evaluated for well-formedness under SHG.

In what follows, I propose that the ungrammaticality in (274) results from such a cumulative effect, modelled via two constraints. I stated earlier that a failure of checking the [$\bullet EF\bullet$]-probe does not lead to the derivation crashing yet it is a constraint in Turkana.

- (276) AGREE (Murphy 2018b: 23)
No unchecked/unvalued probe features.

The second constraint is EGDE which penalises using a phase edge as a final landing site. Movement to a phase edge is only licensed as an intermediate step.

- (277) EDGE
An element moved to an edge position must continue its movement to a higher position.

In step one¹⁴, the derivation consists of the lowest and the intermediate CP. Because of the restricted horizon on C, none of the elements can move to the specifier of C and check the probe. As a result, all three candidates have the same harmony scores and can but used as the input in the next step. This is in line with the observed optionality in Turkana.

- (278) *Step one:*

¹⁴for simplicity reasons, I do not distinguish between the pre- and post-verbal landing sites here

	$C_{[\bullet EF \bullet]} \dots [CP C_{[\bullet EF \bullet]} \dots wh]$	AGREE w = 2	EDGE w = 1	$\mathcal{H}$
☞ a.	$C_{[\bullet EF \bullet]} \dots [CP C_{[\bullet EF \bullet]} \dots V wh]$	-2	0	-4
☞ b.	$C_{[\bullet EF \bullet]} \dots [CP C_{[\bullet EF \bullet]} \dots wh V \cancel{wh}]$	-2	0	-4
☞ c.	$C_{[\bullet EF \bullet]} \dots wh \dots [CP C_{[\bullet EF \bullet]} \dots V \cancel{wh}]$	-2	0	-4

As discussed earlier, *wh*-movement in Turkana is optional. Thus, several inputs are allowed for the next step of the cyclic optimisation. However, I will focus on the crucial input for (274): candidate c. Here the cumulative effect becomes obvious. Violating the AGREE constraint twice, makes the output less optimal than the candidate that violates this constraint only once even though it leaves the *wh*-element stranded at an edge. This step rules out (274) as an optimal output.

(279) *Step two, input candidate c*

	$[CP T_{[*wh*]} \dots [CP C_{[\bullet EF \bullet]} \dots wh \dots [CP C_{[\bullet EF \bullet]} \dots V \cancel{wh}]]]$	AGREE w = 2	EDGE w = 1	$\mathcal{H}$
a.	$[CP T_{[*wh*]} \dots [CP C_{[\bullet EF \bullet]} \dots wh \dots [CP C_{[\bullet EF \bullet]} \dots V \cancel{wh}]]]$	-2	0	-4
☞ b.	$[CP T_{[*wh*]} \dots [CP wh C_{[\bullet EF \bullet]} \dots wh \dots [CP C_{[\bullet EF \bullet]} \dots V \cancel{wh}]]]$	-1	-1	-3
c.	$[CP wh T_{[*wh*]} \dots [CP C_{[\bullet EF \bullet]} \dots wh \dots [CP C_{[\bullet EF \bullet]} \dots V \cancel{wh}]]]$	-2	-0	-4

In the final derivational step, the optimal output is the candidate b which results in a construction of three CPs and two copies.

(280) *Step three:*

	$[CP T_{[*wh*]} \dots [CP C_{[\bullet EF \bullet]} \dots wh \dots [CP C_{[\bullet EF \bullet]} \dots V \cancel{wh}]]]$	AGREE w = 2	EDGE w = 1	$\mathcal{H}$
a.	$[CP T_{[*wh*]} \dots [CP wh C_{[\bullet EF \bullet]} \dots wh \dots [CP C_{[\bullet EF \bullet]} \dots V \cancel{wh}]]]$	-1	-1	-3
☞ b.	$[CP wh T_{[*wh*]} \dots [CP \cancel{wh} C_{[\bullet EF \bullet]} \dots wh \dots [CP C_{[\bullet EF \bullet]} \dots V \cancel{wh}]]]$	-1	-0	-2

These results show that the ungrammaticality of (274) does not arise from a single constraint violation, but from the interaction of multiple violations. A Serial Harmonic Grammar analysis captures these patterns.

5.6 Conclusion

This chapter has shown that the locality patterns observed in Turkana cannot be captured by standard phase theory. Thus, I argued for Turkana that only CPs are phases and that PIC₂ takes place. However,

phase theory alone cannot capture all locality phenomena in Turkana.

To account for these patterns, I have proposed a modified version of the selective opacity framework by Keine (2019, 2020b), in which probes are equipped with individual horizons. I proposed that T and v in Turkana can carry probes with unrestricted horizons, while C carries an edge feature whose horizon is limited to V. This configuration explains why cyclic movement is sometimes blocked, even when the phase edge is available.

A key assumption in my analysis is that Turkana does not exhibit category inheritance. This diverges from Keine's proposal but is necessary to account for the restricted behaviour of the edge feature on C. This horizon imitates the locality restrictions of vP being a phase while not having the absolute locality restrictions that comes with phasehood.

Chapter 6

Conclusion and Outlook

In this dissertation, I investigated the locality restrictions of long distance dependencies in Turkana, an under-researched Eastern Nilotic language. The goal was to bring new empirical data and thus new insight into the theoretical discussion of the syntax of locality.

In chapter 2, I gave a grammar sketch of Turkana. I provided basic information on the language and showed how understudied the language is. I presented novel data on an applicative marker that has not been previously discussed in the literature:

- (281) a. í-pír-ì ákìrù
 3-stir.PRS-TAM Akiru.NOM
 é-è-gètʃ.
 with-M-wooden.whisk.INS
 b. í-pír-í-à ákìrù
 3-stir.PRS-TAM-APPL Akiru.NOM
 é-gètʃ.
 M-wooden.whisk.DEFL
 ‘Akiru is stirring with a wooden whisk.’

Furthermore, I demonstrated that the canonical word order in Turkana is VSO but that Turkana has an optional pre-verbal focus position which has not been investigated in detail before in the literature:

- (282) a. é-múdʒ-í ájòŋ ákírín. (canonical
 1-eat.PRS-TAM I.NOM meat.DEFL
 word order)
 ‘I am eating meat.’

- b. ákírĩŋ é-múdʒ-í ájòŋ.
 meat.DEFL 1-eat.PRS-TAM I.NOM
 (pre-verbal focus)

‘I am eating MEAT.’

I discussed the properties of the pre-verbal focus position such as that only nominals are allowed in this position, that nouns lose their case and get default case, and that not more than one element is allowed in this position which has already been observed for other Nilotic languages (van Urk 2015 for Dinka, Bossi & Diercks 2019; Driemel & Kouneli 2022 for Kipsigis). I furthermore showed that this pre-verbal position has mixed A- and $\bar{A}$ -properties like Dinka (van Urk 2015).

In chapter 3, I examined focus and wh-movement in Turkana. I showed that the language allows both wh-in-situ and wh-ex-situ constructions:

- (283) a. é-mín-à ákàjì ŋáé?
 3-like-TAM Akai.NOM who.DEFL
 b. ŋáé é-mín-à ákàjì?
who.DEFL 3-like-TAM Akai.NOM
 ‘Who does Akai like?’

In this chapter, I showed that wh in-situ does not undergo covert movement but it is truly in-situ. In contrast, wh-ex-situ is a result of movement and not because the wh-element is externally merged high in the syntactic structure.

I further investigated long distance wh-movement: In sentences containing more than two CPs, at least two overt wh-elements are obligatory, and they may appear in either the pre-verbal or post-verbal position within each CP which challenges current ideas of phasality:

- (284) a. *_{CP} é-jén-ì ákírù _{CP} átàmàr
 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì _{CP} átàmàr é-mín-à
 3-think-TAM Akai.NOM that 3-love.PRS-TAM
 ádʒìkòn ŋáé]]]?
 Adjikon.NOM who.DEFL

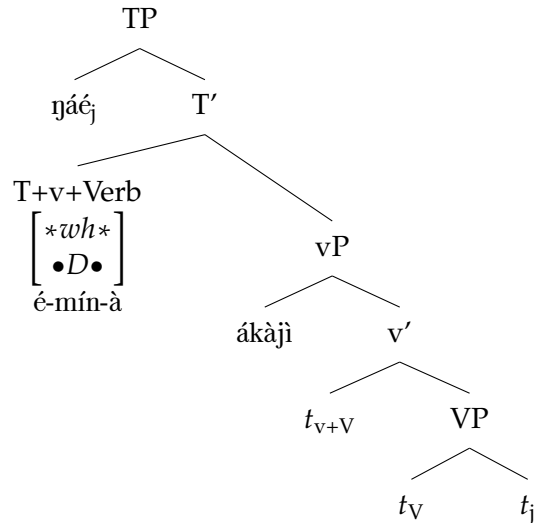
- b. *_[CP] ɲáé é-jén-ì ákìrù _[CP] átàmàr
who.DEFL 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ákàjì _[CP] átàmàr é-mín-à
 3-think-TAM Akai.NOM that 3-love.PRS-TAM
 ádzìkòn]]]?
 Adjikon.NOM
- c. _[CP] ɲáé é-jén-ì ákìrù _[CP] átàmàr
who.DEFL 3-know.PRS-TAM Akiru.NOM that
✓ í-tám-ì ákàjì ✓ _[CP] átàmàr ✓
 3-think-TAM Akai.NOM that
 é-mín-à ádzìkòn ✓]]]?
 3-love.PRS-TAM Adjikon.NOM
 ‘Who does Akiru know that Akai thinks that Adjikon
 loves?’

In chapter 4, I analysed the syntax of the basic word order of Turkana and the landing sites of the pre- and post-verbal *wh*- or focal element. I argued that the subject remains in-situ in the specifier of *vP* and that the verb raises via head movement to *T*. I further argued that the *wh*/focus-probe is located on *T*. For the post-verbal position, I proposed that it is an inner specifier of *v*. Furthermore, I argued that both *v* and *T* can have an optional EPP-feature, accounting for the observed optionality in movement:

- (285) ɲáé é-mín-à ákàjì?
who.DEFL 3-like-TAM Akai.NOM
 ‘Who does Akai like?’

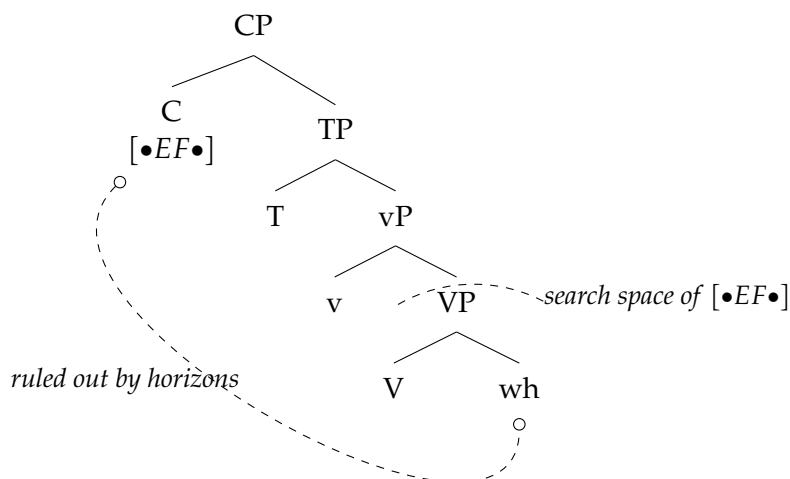
The *wh*-element is the object of the sentence. It is attracted by the *wh*-feature on *T*. The subject stays in-situ in *vP*:

(286)



Chapter 5 explored the long-distance dependencies in Turkana from the perspective of current theories of locality. I showed that standard assumptions about phasality (Chomsky 2000, 2001) cannot account for the Turkana data. I argued that only CPs function as phases in the language, and that phasehood follows PIC₂, resulting in a relatively weak version of phasality. However, I adopt Keine's (2019; 2020b) horizon approach and made it more restrictive by allowing horizons being very small. I proposed that T and v in Turkana can carry probes with unrestricted horizons, while C carries an edge feature whose horizon is limited to V. This horizon imitates the locality restrictions of vP being a phase while not having the absolute locality restrictions that come with phasehood. Crucially, this restriction is probe-specific and applies only to cyclic movement triggered by the $[\bullet EF\bullet]$ -probe, in contrast to the absolute opacity imposed by phasehood. This allows the system to simulate vP phase like effects without assuming that vP is a phase.

(287)



While phase theory has often treated locality as a uniform property across languages, typically vP and CP being phases, recent proposals have begun to question this assumption. [Keine \(2019, 2020b\)](#), for instance, argued that probes have different horizons meaning that they differ in the size of their search domains. The horizons of a probe can vary from language to language.

The new Turkana data further challenge the assumption about locality being uniform. I have shown that only CPs function as phases in Turkana, and that phasality obeys PIC₂. This suggests that not only horizons, but also the very architecture of phasal domains, are subject to cross-linguistic variation. Turkana thus offers compelling evidence that the phase system is less rigid and more diverse than traditionally assumed. These findings highlight the importance of investigating a broader range of languages beyond the Indo-European languages to get a full understanding of locality restrictions.

A promising direction for future research lies in comparative work on other Nilotic languages, which may yield additional insight into locality constraints. Languages that are mutually intelligible with Turkana, Toposa, or partially intelligible, Karimojong, Jie, and Nyangatom, offer a valuable opportunity for exploring microvariation. These languages offer a useful testing ground for whether the phase properties observed in Turkana are stable across the subgroup or subject to microvariation.

The pre-verbal domain in Turkana displays some striking properties which it seems to share with other Nilotic languages (cf. [van](#)

Urk 2015 for Dinka and Driemel & Kouneli 2022; Bossi 2023b; Kouneli 2024 for Kipsigis). nouns systematically lose morphological case when moved into this domain. This case loss has so far received only limited attention in the theoretical literature (e.g., Baker 2015, van Urk 2015, Kouneli 2024).

Moreover, the pre-verbal position is restricted to nominals. When adjuncts, such as locatives, instruments, or temporals, are fronted, the verb obligatorily bears an applicative marker. This suggests that the pre-verbal domain functions as an argument position even though it is a focus position. This is consistent with the mixed A- & $\bar{A}$ -properties of that language and with my analysis which claims that the landing site of the pre-verbal position is the specifier of TP. However, a detailed analysis of the argument structure in Turkana remains an open topic.

I hope that this dissertation encourages further research into the syntactic patterns of Nilotic languages.

Appendix A

A.1 Data with an object wh-element

3 CPs with one wh-element are always forbidden:

- (288) a. *[_{CP} é-jén-ì ákìrù [_{CP} átàmàr
3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàyì [_{CP} átàmàr é-mín-à
3-think-TAM Akai.NOM that 3-love.PRS-TAM
ádžìkòn **ṣáé**]]]?
Adjikon.NOM **who.DEFL**
- b. *[_{CP} é-jén-ì ákìrù [_{CP} átàmàr
3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàyì [_{CP} átàmàr **ṣáé**
3-think-TAM Akai.NOM that **who.DEFL**
é-mín-à ádžìkòn]]]?
3-love.PRS-TAM Adjikon.NOM
- c. *[_{CP} é-jén-ì ákìrù [_{CP} átàmàr
3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàyì **ṣáé** [_{CP} átàmàr
3-think-TAM Akai.NOM **who.DEFL** that
é-mín-à ádžìkòn]]]?
3-love.PRS-TAM Adjikon.NOM
- d. *[_{CP} é-jén-ì ákìrù [_{CP} átàmàr **ṣáé**
3-know.PRS-TAM Akiru.NOM that **who.DEFL**
í-tám-ì ákàyì [_{CP} átàmàr é-mín-à
3-think-TAM Akai.NOM that 3-love.PRS-TAM
ádžìkòn]]]?
Adjikon.NOM

- e. *_{[CP} é-jén-ì ákìrù ɲáé _{[CP} átàmàr
3-know.PRS-TAM Akiru.NOM who.DEFL that
í-tám-ì ákàyì _{[CP} átàmàr é-mín-à
3-think-TAM Akai.NOM that 3-love.PRS-TAM
ádžìkòn]]]?
Adjikon.NOM
- f. *_{[CP} ɲáé é-jén-ì ákìrù _{[CP} átàmàr
who.DEFL 3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàyì _{[CP} átàmàr é-mín-à
3-think-TAM Akai.NOM that 3-love.PRS-TAM
ádžìkòn]]]?
Adjikon.NOM
'Who does Akiru know that Akai thinks that Adjikon
loves?'

The following examples illustrate all grammatical cases where one of the overt copies remains in-situ:

- (289) a. _{[CP} é-jén-ì ákìrù _{[CP} átàmàr
3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàyì ɲáé _{[CP} átàmàr
3-think-TAM Akai.NOM who.DEFL that
é-mín-à ádžìkòn ɲáé]]]?
3-love.PRS-TAM Adjikon.NOM who.DEFL
- b. _{[CP} é-jén-ì ákìrù _{[CP} átàmàr ɲáé
3-know.PRS-TAM Akiru.NOM that who.DEFL
í-tám-ì ákàyì _{[CP} átàmàr é-mín-à
3-think-TAM Akai.NOM that 3-love.PRS-TAM
ádžìkòn ɲáé]]]?
Adjikon.NOM who.DEFL
- c. _{[CP} é-jén-ì ákìrù ɲáé _{[CP} átàmàr
3-know.PRS-TAM Akiru.NOM who.DEFL that
í-tám-ì ákàyì _{[CP} átàmàr é-mín-à
3-think-TAM Akai.NOM that 3-love.PRS-TAM
ádžìkòn ɲáé]]]?
Adjikon.NOM who.DEFL

- d. [_{CP} **ɲáé** **who.DEFL** é-jén-ì ákìrù [_{CP} átàmàr
3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàyì [_{CP} átàmàr é-mín-à
3-think-TAM Akai.NOM that 3-love.PRS-TAM
ádžìkòn **ɲáé**]]]?
Adjikon.NOM **who.DEFL**
'Who does Akiru know that Akai thinks that Adjikon
loves?'

The following examples show all grammatical examples with two overt copies, both ex-situ:

- (290) a. [_{CP} é-jén-ì ákìrù [_{CP} átàmàr
3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàyì **ɲáé** [_{CP} átàmàr
3-think-TAM Akai.NOM **who.DEFL** that
ɲáé é-mín-à ádžìkòn]]]?
who.DEFL 3-love.PRS-TAM Adjikon.NOM
- b. [_{CP} é-jén-ì ákìrù [_{CP} átàmàr **ɲáé**
3-know.PRS-TAM Akiru.NOM that **who.DEFL**
í-tám-ì ákàyì [_{CP} átàmàr **ɲáé**
3-think-TAM Akai.NOM that **who.DEFL**
é-mín-à ádžìkòn]]]?
3-love.PRS-TAM Adjikon.NOM
- c. [_{CP} é-jén-ì ákìrù **ɲáé** [_{CP} átàmàr
3-know.PRS-TAM Akiru.NOM **who.DEFL** that
í-tám-ì ákàyì [_{CP} átàmàr **ɲáé**
3-think-TAM Akai.NOM that **who.DEFL**
é-mín-à ádžìkòn]]]?
3-love.PRS-TAM Adjikon.NOM
- d. [_{CP} **ɲáé** **who.DEFL** é-jén-ì ákìrù [_{CP} átàmàr
3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàyì [_{CP} átàmàr **ɲáé**
3-think-TAM Akai.NOM that **who.DEFL**
é-mín-à ádžìkòn]]]?
3-love.PRS-TAM Adjikon.NOM

- e. [_{CP} é-jén-ì ákìrù ηάέ [_{CP} átàmàr
3-know.PRS-TAM Akiru.NOM who.DEFL that
í-tám-ì ákàyì ηάέ [_{CP} átàmàr
3-think-TAM Akai.NOM who.DEFL that
é-mín-à ádzìkòn]]]?
3-love.PRS-TAM Adjikon.NOM
- f. [_{CP} ηάέ é-jén-ì ákìrù [_{CP} átàmàr
who.DEFL 3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàyì ηάέ [_{CP} átàmàr
3-think-TAM Akai.NOM who.DEFL that
é-mín-à ádzìkòn]]]?
3-love.PRS-TAM Adjikon.NOM
- g. [_{CP} é-jén-ì ákìrù ηάέ [_{CP} átàmàr
3-know.PRS-TAM Akiru.NOM who.DEFL that
ηάέ í-tám-ì ákàyì [_{CP} átàmàr
who.DEFL 3-think-TAM Akai.NOM that
é-mín-à ádzìkòn]]]?
3-love.PRS-TAM Adjikon.NOM
- h. [_{CP} ηάέ é-jén-ì ákìrù [_{CP} átàmàr
who.DEFL 3-know.PRS-TAM Akiru.NOM that
ηάέ í-tám-ì ákàyì [_{CP} átàmàr
who.DEFL 3-think-TAM Akai.NOM that
é-mín-à ádzìkòn]]]?
3-love.PRS-TAM Adjikon.NOM
'Who does Akiru know that Akai thinks that Adjikon
loves?'

The following examples show that 2 overt copies in the same CP are always ungrammatical no matter whether they are in the lowest, intermediate or highest CP:

- (291) a. * [_{CP} é-jén-ì ákìrù [_{CP} átàmàr
3-know.PRS-TAM Akiru.NOM that
í-tám-ì ákàyì [_{CP} átàmàr ηάέ
3-think-TAM Akai.NOM that who.DEFL
é-mín-à ádzìkòn ηάέ]]]?
3-love.PRS-TAM Adjikon.NOM who.DEFL

- b. *_{[CP é-jén-ì ákìrù [CP átàmàr ɲáé]}
 3-know.PRS-TAM Akiru.NOM that _{who.DEFL}
 í-tám-ì ákàjì _{ɲáé} _[CP átàmàr]
 3-think-TAM Akai.NOM _{who.DEFL} that
 é-mín-à ádzìkòn]]]?
 3-love.PRS-TAM Adjikon.NOM
- c. *_[CP ɲáé] é-jén-ì ákìrù
_{who.DEFL} 3-know.PRS-TAM Akiru.NOM
_{ɲáé} _{[CP átàmàr í-tám-ì ákàjì [CP]}
_{who.DEFL} that 3-think-TAM Akai.NOM
 átàmàr é-mín-à ádzìkòn]]]?
 that 3-love.PRS-TAM Adjikon.NOM
 ‘Who does Akiru know that Akai thinks that Adjikon loves?’

The following examples shows all grammatical examples with 3 CPs and 3 wh-elements

- (292) a. _[CP é-jén-ì ákìrù ɲáé] _[CP átàmàr]
 3-know.PRS-TAM Akiru _{who.DEFL} that
 í-tám-ì ákàjì _{ɲáé} _[CP átàmàr]
 3-think.PRS-TAM Akai.NOM _{who.DEFL} that
 é-mín-à ádzìkòn _{ɲáé}]]]?
 3-love.PRS-TAM Adjikon.NOM _{who.DEFL}
- b. _[CP é-jén-ì ákìrù ɲáé] _[CP átàmàr]
 3-know.PRS-TAM Akiru _{who.DEFL} that
 í-tám-ì ákàjì _[CP átàmàr ɲáé]
 3-think.PRS-TAM Akai.NOM that _{who.DEFL}
 é-mín-à ádzìkòn _{ɲáé}]]]?
 3-love.PRS-TAM Adjikon.NOM _{who.DEFL}
- c. _[CP é-jén-ì ákìrù ɲáé] _[CP átàmàr]
 3-know.PRS-TAM Akiru _{who.DEFL} that
_{ɲáé} í-tám-ì ákàjì _[CP átàmàr]
_{who.DEFL} 3-think.PRS-TAM Akai.NOM that
 é-mín-à ádzìkòn _{ɲáé}]]]?
 3-love.PRS-TAM Adjikon.NOM _{who.DEFL}

- d. [CP é-jén-ì ákìrù **ɲáé** [CP átàmàr
3-know.PRS-TAM Akiru **who.DEFL** that
ɲáé í-tám-ì ákàjì [CP átàmàr
who.DEFL 3-think.PRS-TAM Akai.NOM that
ɲáé é-mín-à ádžìkòn]]]?
who.DEFL 3-love.PRS-TAM Adjikon.NOM
- e. [CP **ɲáé** é-jén-ì ákìrù [CP átàmàr
who.DEFL 3-know.PRS-TAM Akiru that
í-tám-ì ákàjì **ɲáé** [CP átàmàr
3-think.PRS-TAM Akai.NOM **who.DEFL** that
é-mín-à ádžìkòn **ɲáé**]]]?
3-love.PRS-TAM Adjikon.NOM **who.DEFL**
- f. [CP **ɲáé** é-jén-ì ákìrù [CP átàmàr
who.DEFL 3-know.PRS-TAM Akiru that
í-tám-ì ákàjì **ɲáé** [CP átàmàr
3-think.PRS-TAM Akai.NOM **who.DEFL** that
ɲáé é-mín-à ádžìkòn]]]?
who.DEFL 3-love.PRS-TAM Adjikon.NOM
- g. [CP **ɲáé** é-jén-ì ákìrù [CP átàmàr
who.DEFL 3-know.PRS-TAM Akiru that
ɲáé í-tám-ì ákàjì [CP átàmàr
who.DEFL 3-think.PRS-TAM Akai.NOM that
é-mín-à ádžìkòn **ɲáé**]]]?
3-love.PRS-TAM Adjikon.NOM **who.DEFL**
- h. [CP **ɲáé** é-jén-ì ákìrù [CP átàmàr
who.DEFL 3-know.PRS-TAM Akiru that
ɲáé í-tám-ì ákàjì [CP átàmàr
who.DEFL 3-think.PRS-TAM Akai.NOM that
ɲáé é-mín-à ádžìkòn]]]?
who.DEFL 3-love.PRS-TAM Adjikon.NOM
'Who does Akiru know that Akai thinks that Adjikon
loves?'

A.2 Data with a subject wh-element

Three CPs with one wh-element are always forbidden:

- (293) a. *_{CP} í-tám-ì ákìrù _{CP} átàmàr
 3-think.PRS-TAM Akiru.NOM that
 é-jén-ì ádzìkòn _{CP} átàmàr
 3-know.PRS-TAM Adjikon.NOM that
 í-múdɜ-ì ɲáè á-kírín]]]?
 3-eat.PRS-TAM who.NOM F-meat.DEFL
- b. *_{CP} textipáí-tám-ì ákìrù _{CP} átàmàr
 3-think.PRS-TAM Akiru.NOM that
 é-jén-ì ádzìkòn _{CP} átàmàr ɲáé
 3-know.PRS-TAM Adjikon.NOM that who.DEFL
 í-múdɜ-ì á-kírín]]]?
 3-eat.PRS-TAM F-meat.DEFL
- c. *_{CP} í-tám-ì ákìrù _{CP} átàmàr
 3-think.PRS-TAM Akiru.NOM that
 é-jén-ì ádzìkòn ɲáé _{CP} átàmàr
 3-know.PRS-TAM Adjikon.NOM who.DEFL that
 í-múdɜ-ì á-kírín]]]?
 3-eat.PRS-TAM F-meat.DEFL
- d. *_{CP} í-tám-ì ákìrù _{CP} átàmàr
 3-think.PRS-TAM Akiru.NOM that
 é-jén-ì ɲáé ádzìkòn _{CP} átàmàr
 3-know.PRS-TAM who.DEFL Adjikon.NOM that
 í-múdɜ-ì á-kírín]]]?
 3-eat.PRS-TAM F-meat.DEFL
- e. *_{CP} í-tám-ì ákìrù _{CP} átàmàr ɲáé
 3-think.PRS-TAM Akiru.NOM that who.DEFL
 é-jén-ì ádzìkòn _{CP} átàmàr
 3-know.PRS-TAM Adjikon.NOM that
 í-múdɜ-ì á-kírín]]]?
 3-eat.PRS-TAM F-meat.DEFL

- f. *<sub>[CP í-tám-ì ákìrù ɲáé _{[CP átàmàr}
 3-think.PRS-TAM Akiru.NOM who.DEFL that
 é-jén-ì ádzìkòn _{[CP átàmàr}
 3-know.PRS-TAM Adjikon.NOM that
 í-múdz-ì á-kírín]]]?
 3-eat.PRS-TAM F-meat.DEFL</sub>
- g. *<sub>[CP ɲáé í-tám-ì ákìrù _{[CP átàmàr}
who.DEFL 3-think.PRS-TAM Akiru.NOM that
 é-jén-ì ádzìkòn _{[CP átàmàr}
 3-know.PRS-TAM Adjikon.NOM that
 í-múdz-ì á-kírín]]]?
 3-eat.PRS-TAM F-meat.DEFL
 ‘Who does Akiru think that Adjikon knows that is eating
 (the) meat?’</sub>

The following examples illustrate all grammatical cases where one of the overt copies remains in-situ:

- a. <sub>[CP í-tám-ì ákìrù _{[CP átàmàr}
 3-think.PRS-TAM Akiru.NOM that
 é-jén-ì ádzìkòn ɲáé _{[CP átàmàr}
 3-know.PRS-TAM Adjikon.NOM who.DEFL that
 í-múdz-ì ɲàè á-kírín]]]?
 3-eat.PRS-TAM who.NOM F-meat.DEFL</sub>
- b. <sub>[CP í-tám-ì ákìrù _{[CP átàmàr ɲáé}
 3-think.PRS-TAM Akiru.NOM that who.DEFL
 é-jén-ì ádzìkòn _{[CP átàmàr í-múdz-ì}
 3-know.PRS-TAM Adjikon.NOM that 3-eat.PRS-TAM
ɲàè á-kírín]]]?
who.NOM F-meat.DEFL</sub>
- c. <sub>[CP í-tám-ì ákìrù ɲáé _{[CP átàmàr}
 3-think.PRS-TAM Akiru.NOM who.DEFL that
 é-jén-ì ádzìkòn _{[CP átàmàr í-múdz-ì}
 3-know.PRS-TAM Adjikon.NOM that 3-eat.PRS-TAM
ɲàè á-kírín]]]?
who.NOM F-meat.DEFL</sub>

- d. $[\text{CP } \boxed{\eta\acute{a}\acute{e}} \text{ } \acute{í}\text{-tám-}\acute{i} \text{ } \acute{á}\text{kìrù} \text{ } [\text{CP } \acute{á}\text{tàmàr}$
 $\boxed{\text{who.DEFL}} \text{ 3-think.PRS-TAM Akiru.NOM that}$
 $\acute{é}\text{-jén-}\acute{i} \text{ } \acute{á}\text{dzìkòn} \text{ } [\text{CP } \acute{á}\text{tàmàr } \acute{í}\text{-múdz-}\acute{i}$
 $\text{3-know.PRS-TAM Adjikon.NOM that 3-eat.PRS-TAM}$
 $\boxed{\eta\acute{a}\acute{e}} \text{ } \acute{á}\text{-kírín} \text{ }]]]?$
 $\boxed{\text{who.NOM}} \text{ F-meat.DEFL}$
 ‘Who does Akiru think that Adjikon knows that is eating
 (the) meat?’

The following examples show all grammatical examples with two overt copies, both ex-situ:

- (294) a. $[\text{CP } \acute{í}\text{-tám-}\acute{i} \text{ } \acute{á}\text{kìrù} \text{ } [\text{CP } \acute{á}\text{tàmàr}$
 $\text{3-think.PRS-TAM Akiru.NOM that}$
 $\acute{é}\text{-jén-}\acute{i} \text{ } \acute{á}\text{dzìkòn} \text{ } \boxed{\eta\acute{a}\acute{e}} \text{ } [\text{CP } \acute{á}\text{tàmàr}$
 $\text{3-know.PRS-TAM Adjikon.NOM } \boxed{\text{who.DEFL}} \text{ that}$
 $\boxed{\eta\acute{a}\acute{e}} \text{ } \acute{í}\text{-múdz-}\acute{i} \text{ } \acute{á}\text{-kírín} \text{ }]]]?$
 $\boxed{\text{who.DEFL}} \text{ 3-eat.PRS-TAM F-meat.DEFL}$
- b. $[\text{CP } \acute{í}\text{-tám-}\acute{i} \text{ } \acute{á}\text{kìrù} \text{ } [\text{CP } \acute{á}\text{tàmàr}$
 $\text{3-think.PRS-TAM Akiru.NOM that}$
 $\boxed{\eta\acute{a}\acute{e}} \text{ } \acute{é}\text{-jén-}\acute{i} \text{ } \acute{á}\text{dzìkòn}$
 $\boxed{\text{who.DEFL}} \text{ 3-know.PRS-TAM Adjikon.NOM}$
 $[\text{CP } \acute{á}\text{tàmàr } \boxed{\eta\acute{a}\acute{e}} \text{ } \acute{í}\text{-múdz-}\acute{i} \text{ } \acute{á}\text{-kírín} \text{ }]]]?$
 $\text{that } \boxed{\text{who.DEFL}} \text{ 3-eat.PRS-TAM F-meat.DEFL}$
- c. $[\text{CP } \acute{í}\text{-tám-}\acute{i} \text{ } \acute{á}\text{kìrù} \text{ } \boxed{\eta\acute{a}\acute{e}} \text{ } [\text{CP } \acute{á}\text{tàmàr}$
 $\text{3-think.PRS-TAM Akiru.NOM } \boxed{\text{who.DEFL}} \text{ that}$
 $\acute{é}\text{-jén-}\acute{i} \text{ } \acute{á}\text{dzìkòn} \text{ } [\text{CP } \acute{á}\text{tàmàr } \boxed{\eta\acute{a}\acute{e}}$
 $\text{3-know.PRS-TAM Adjikon.NOM that } \boxed{\text{who.DEFL}}$
 $\acute{í}\text{-múdz-}\acute{i} \text{ } \acute{á}\text{-kírín} \text{ }]]]?$
 $\text{3-eat.PRS-TAM F-meat.DEFL}$
- d. $[\text{CP } \boxed{\eta\acute{a}\acute{e}} \text{ } \acute{í}\text{-tám-}\acute{i} \text{ } \acute{á}\text{kìrù} \text{ } [\text{CP } \acute{á}\text{tàmàr}$
 $\boxed{\text{who.DEFL}} \text{ 3-think.PRS-TAM Akiru.NOM that}$
 $\acute{é}\text{-jén-}\acute{i} \text{ } \acute{á}\text{dzìkòn} \text{ } [\text{CP } \acute{á}\text{tàmàr } \boxed{\eta\acute{a}\acute{e}}$
 $\text{3-know.PRS-TAM Adjikon.NOM that } \boxed{\text{who.DEFL}}$
 $\acute{í}\text{-múdz-}\acute{i} \text{ } \acute{á}\text{-kírín} \text{ }]]]?$
 $\text{3-eat.PRS-TAM F-meat.DEFL}$

- e. [_{CP} í-tám-ì ákìrù **ḡáé** [_{CP} átàmàr
3-think.PRS-TAM Akiru.NOM **who.DEFL** that
é-jén-ì ádzìkòn **ḡáé** [_{CP} átàmàr
3-know.PRS-TAM Adjikon.NOM **who.DEFL** that
í-múdz-ì á-kírín]]]?
3-eat.PRS-TAM F-meat.DEFL
- f. [_{CP} **ḡáé** í-tám-ì ákìrù [_{CP} átàmàr
who.DEFL 3-think.PRS-TAM Akiru.NOM that
é-jén-ì ádzìkòn **ḡáé** [_{CP} átàmàr
3-know.PRS-TAM Adjikon.NOM **who.DEFL** that
í-múdz-ì á-kírín]]]?
3-eat.PRS-TAM F-meat.DEFL
- g. [_{CP} í-tám-ì ákìrù **ḡáé** [_{CP} átàmàr
3-think.PRS-TAM Akiru.NOM **who.DEFL** that
ḡáé é-jén-ì ádzìkòn [_{CP} átàmàr
who.DEFL 3-know.PRS-TAM Adjikon.NOM that
í-múdz-ì á-kírín]]]?
3-eat.PRS-TAM F-meat.DEFL
- h. [_{CP} **ḡáé** í-tám-ì ákìrù [_{CP} átàmàr
who.DEFL 3-think.PRS-TAM Akiru.NOM that
ḡáé é-jén-ì ádzìkòn [_{CP}
who.DEFL 3-know.PRS-TAM Adjikon.NOM that
átàmàr í-múdz-ì á-kírín]]]?
3-eat.PRS-TAM F-meat.DEFL
'Who does Akiru think that Adjikon knows that is eating
(the) meat?'

The following examples show that 2 overt copies in the same CP are always ungrammatical no matter whether they are in the lowest, intermediate or highest CP:

- (295) a. * [_{CP} í-tám-ì ákìrù [_{CP} átàmàr
3-think.PRS-TAM Akiru.NOM that
é-jén-ì ádzìkòn [_{CP} átàmàr **ḡáé**
3-know.PRS-TAM Adjikon.NOM that **who.DEFL**
í-múdz-ì **ḡáé** á-kírín]]]?
3-eat.PRS-TAM **who.NOM** F-meat.DEFL

‘Who does Akiru think that Adjikon knows that is eating (the) meat?’

- b. *<sub>[CP í-tám-ì ákìrù [CP átàmàr ɲáé
3-think.PRS-TAM Akiru.NOM that who.DEFL
é-jén-ì ádzìkòn ɲáé [CP átàmàr
3-know.PRS-TAM Adjikon.NOM who.DEFL that
í-múdz-ì á-kírín]]]?</sub>
3-eat.PRS-TAM F-meat.DEFL

‘Who does Akiru think that Adjikon knows that is eating (the) meat?’

- c. *<sub>[CP ɲáé í-tám-ì ákìrù
who.DEFL 3-think.PRS-TAM Akiru.NOM
ɲáé [CP átàmàr é-jén-ì ádzìkòn
who.DEFL that 3-know.PRS-TAM Adjikon.NOM
[CP átàmàr í-múdz-ì á-kírín]]]?</sub>
that 3-eat.PRS-TAM F-meat.DEFL

‘Who does Akiru think that Adjikon knows that is eating (the) meat?’

The following examples shows all grammatical examples with 3 CPs and 3 wh-elements

- (296) a. <sub>[CP í-tám-ì ákìrù ɲáé [CP átàmàr
3-think.PRS-TAM Akiru.NOM who.DEFL that
é-jén-ì ádzìkòn ɲáé [CP átàmàr
3-know.PRS-TAM Adjikon.NOM who.DEFL that
í-múdz-ì ɲáè á-kírín]]]?</sub>
3-eat.PRS-TAM who.NOM F-meat.DEFL
- b. <sub>[CP ɲáé í-tám-ì ákìrù [CP átàmàr
who.DEFL 3-think.PRS-TAM Akiru.NOM that
é-jén-ì ádzìkòn ɲáé [CP átàmàr
3-know.PRS-TAM Adjikon.NOM who.DEFL that
í-múdz-ì ɲáè á-kírín]]]?</sub>
3-eat.PRS-TAM who.NOM F-meat.DEFL
- c. <sub>[CP í-tám-ì ákìrù ɲáé [CP átàmàr
3-think.PRS-TAM Akiru.NOM who.DEFL that</sub>

- ɲáé é-jén-ì ádzìkòn [CP átàmàr
 who.DEFL 3-know.PRS-TAM Adjikon.NOM that
 í-múdz-ì ɲàè á-kírín]]]?
 3-eat.PRS-TAM who.NOM F-meat.DEFL
- d. [CP ɲáé í-tám-ì ákìrù [CP átàmàr
 who.DEFL 3-think.PRS-TAM Akiru.NOM that
 ɲáé é-jén-ì ádzìkòn [CP átàmàr
 who.DEFL 3-know.PRS-TAM Adjikon.NOM that
 í-múdz-ì ɲàè á-kírín]]]?
 3-eat.PRS-TAM who.NOM F-meat.DEFL
- e. [CP í-tám-ì ákìrù ɲáé [CP átàmàr
 3-think.PRS-TAM Akiru.NOM who.DEFL that
 é-jén-ì ádzìkòn ɲáé [CP átàmàr
 3-know.PRS-TAM Adjikon.NOM who.DEFL that
 ɲáé í-múdz-ì á-kírín]]]?
 who.DEFL 3-eat.PRS-TAM F-meat.DEFL
- f. [CP ɲáé í-tám-ì ákìrù [CP átàmàr
 who.DEFL 3-think.PRS-TAM Akiru.NOM that
 é-jén-ì ádzìkòn ɲáé [CP átàmàr
 3-know.PRS-TAM Adjikon.NOM who.DEFL that
 ɲáé í-múdz-ì á-kírín]]]?
 ɲáé 3-eat.PRS-TAM F-meat.DEFL
- g. [CP í-tám-ì ákìrù ɲáé [CP átàmàr
 3-think.PRS-TAM Akiru.NOM who.DEFL that
 ɲáé é-jén-ì ádzìkòn [CP átàmàr
 who.DEFL 3-know.PRS-TAM Adjikon.NOM that
 ɲáé í-múdz-ì á-kírín]]]?
 who.DEFL 3-eat.PRS-TAM F-meat.DEFL
- h. [CP ɲáé í-tám-ì ákìrù [CP átàmàr
 who.DEFL 3-think.PRS-TAM Akiru.NOM that
 ɲáé é-jén-ì ádzìkòn [CP átàmàr
 who.DEFL 3-know.PRS-TAM Adjikon.NOM that
 ɲáé í-múdz-ì á-kírín]]]?
 who.DEFL 3-eat.PRS-TAM F-meat.DEFL
 ‘Who does Akiru think that Adjikon knows that is eating

(the) meat?’

A.3 Data with D-linked elements

Repetition of the same d-linked element is ungrammatical

- (297) a. *_{[CP é-jén-ì ákìrù [CP átàmàr}
 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ádzìkòn _{[CP átàmàr}
 3-think.PRS-TAM Adjikon.NOM that
 è-kilé álí é-mín-à
 M-man.DEFL which.M.DEFL 3-love.PRS-TAM
 ákàjì **è-kilé álí**]]]?
 Akai.NOM **M-man.DEFL which.M.DEFL**
- b. *_{[CP é-jén-ì ákìrù [CP átàmàr}
 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ádzìkòn
 3-think.PRS-TAM Adjikon.NOM
 è-kilé álí _{[CP átàmàr}
 M-man.DEFL which.M.DEFL that
 é-mín-à ákàjì
 3-love.PRS-TAM Akai.NOM
 è-kilé álí]]]?
 M-man.DEFL which.M.DEFL
- c. *_{[CP é-jén-ì ákìrù [CP átàmàr}
 3-know.PRS-TAM Akiru.NOM that
 è-kilé álí í-tám-ì
 M-man.DEFL which.M.DEFL 3-think.PRS-TAM
 ádzìkòn _{[CP átàmàr é-mín-à ákàjì}
 Adjikon.NOM that 3-love.PRS-TAM Akai.NOM
 è-kilé álí]]]?
 M-man.DEFL which.M.DEFL
- d. *_{[CP é-jén-ì ákìrù}
 3-know.PRS-TAM Akiru.NOM
 è-kilé álí _{[CP átàmàr}
 M-man.DEFL which.M.DEFL that

í-tám-ì ádzìkòn [CP átàmàr
3-think.PRS-TAM Adjikon.NOM that

é-mín-à ákàjì
3-love.PRS-TAM Akai.NOM

è-kílé álí]]]?

M-man.DEFL which.M.DEFL

e. *[CP **è-kílé álí]]]?** é-jén-ì
M-man.DEFL which.M.DEFL 3-know.PRS-TAM

ákìrù [CP átàmàr í-tám-ì ádzìkòn
Akiru.NOM that 3-think.PRS-TAM Adjikon.NOM

[CP átàmàr é-mín-à ákàjì
that 3-love.PRS-TAM Akai.NOM

è-kílé álí]]]?

M-man.DEFL which.M.DEFL

‘Which man does Akiru know that Adjikon think that Akai loves?’

(298) *[CP **álú-tùṅá]]]?** é-jén-ì ákàjì
which.PL-people.DEFL 3-know.PRS-TAM Akai.NOM

[CP átàmàr í-tám-ì ádzìkòn [CP átàmàr
that 3-think.PRS-TAM Adjikon.NOM that

álú-tùṅá í-múdž-é-tè á-kírín]]]?

which.PL-people.DEFL 3-eat.PRS-TAM-PL meat.DEFL

‘Which people does Akai know that Adjikon thinks are eating meat?’

D-linked element with a wh-element as the lower copy is allowed

(299) [CP

álú-tùṅá / ṅí-tùṅá álú]]]?
which.PL-people.DEFL / PL-people.DEFL which.PL.DEFL

é-jén-ì ákàjì [CP átàmàr í-tám-ì
3-know.PRS-TAM Akai.NOM that 3-think.PRS-TAM

ádzìkòn [CP átàmàr **tá-ṅáé / kètjí**

Adjikon.NOM that **PL-who.DEFL / 3.PL.DEFL**

í-múdž-é-tè á-kírín]]]?

3-eat.PRS-TAM-PL meat.DEFL

‘Which people does Akai know that Adjikon thinks are eating meat?’

Wh-element being the higher copy is allowed

- (300) a. [_{CP} **tá-ŋáé** **PL-who.DEFL** é-jén-ì 3-know.PRS-TAM Akai.NOM
 átàmàr í-tám-ì ádzìkòn [_{CP} átàmàr
 that 3-think.PRS-TAM Adjikon.NOM that
 í-múdʒ-é-tè **álú-tùṅà** á-kírín
 3-eat.PRS-TAM-PL **which.PL-people.NOM** meat.DEFL
]]]?

- b. [_{CP} **tá-ŋáé** **PL-who.DEFL** é-jén-ì 3-know.PRS-TAM Akai.NOM
 átàmàr í-tám-ì ádzìkòn [_{CP} átàmàr
 that 3-think.PRS-TAM Adjikon.NOM that
álú-tùṅá í-múdʒ-é-tè á-kírín
which.PL-people.DEFL 3-eat.PRS-TAM-PL meat.DEFL
]]]?

- c. [_{CP} é-jén-ì ákàjì [_{CP} átàmàr
PL-who.DEFL 3-know.PRS-TAM Akai.NOM
tá-ŋáé í-tám-ì ádzìkòn [_{CP} átàmàr
 that 3-think.PRS-TAM Adjikon.NOM that
álú-tùṅá í-múdʒ-é-tè á-kírín
which.PL-people.DEFL 3-eat.PRS-TAM-PL meat.DEFL
]]]?

‘Which people does Akai know that Adjikon thinks are eating meat?’

- (301) [_{CP} **(ká)táŋáé** **PL-who.DEFL** é-jén-ì ákàjì [_{CP} átàmàr
 í-tám-ì ádzìkòn [_{CP} átàmàr é-mín-à
 3-think.PRS-TAM Adjikon.NOM that 3-like.PRS-TAM
 ákirù **ŋí-tùṅá** **álù**]]]?
 Akiru.NOM **PL-people.DEFL which.DEFL.PL**

‘Which people does Akai know that Adjikon thinks that Akiru likes?’ (*there is no difference between tangae and katangae*)

- (302) [CP (ka)tá-ŋáé é-jén-ì ákàjì [CP átàmàr
 PL-who.DEFL 3-know.PRS-TAM Akai.NOM that
 í-tám-ì ádzìkòn [CP átàmàr
 3-think.PRS-TAM Adjikon.NOM that
 álú-túná í-múdz-é-tè á-kírín]]]?
 which.PL-people.DEFL 3-eat.PRS-TAM-PL meat.DEFL
 ‘Which people does Akai know that Adjikon thinks are eat-
 ing meat.’ (*there is no difference between tangae and katangae*)

- (303) a. [CP ŋáé é-jén-ì ákìrù [CP átàmàr
 who.DEFL 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ádzìkòn [CP átàmàr
 3-think.PRS-TAM Adjikon.NOM that
 é-mín-à ákàjì
 3-love.PRS-TAM Akai.NOM
 è-kilé álí]]]?
 M-man.DEFL which.M.DEFL
 b. [CP ŋáé é-jén-ì ákìrù [CP átàmàr
 who.DEFL 3-know.PRS-TAM Akiru.NOM that
 í-tám-ì ádzìkòn [CP átàmàr
 3-think.PRS-TAM Adjikon.NOM that
 è-kilé álí é-mín-à
 M-man.DEFL which.M.DEFL 3-love.PRS-TAM
 ákàjì]]]?
 Akai.NOM
 c. [CP ŋáé é-jén-ì ákìrù [CP átàmàr
 who.DEFL 3-know.PRS-TAM Akiru.NOM that
 è-kilé álí í-tám-ì
 M-man.DEFL which.M.DEFL 3-think.PRS-TAM
 ádzìkòn [CP átàmàr é-mín-à ákàjì
 Adjikon.NOM that 3-love.PRS-TAM Akai.NOM
]]]?

‘Which man does Akiru know that Adjikon think that Akai loves?’

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